TimeMorph: Dual-view LLM-based Few-Shot Time Series Classification

Shizhuo Deng, *Member, IEEE*, Yonghui Qin, Dongyue Chen, Ye Yuan, Xinde Li, *Senior Member, IEEE*, and Tong Jia, *Member, IEEE*

Abstract—Few-shot time-series classification (TSC) is challenging due to scarce labeled sequences, where each class provides only a few examples. Reliable prediction requires capturing both temporal dynamics and waveform morphology, reflecting how humans interpret sequential signals by tracking numeric trends and recognizing overall shape patterns. Recent pretrained large language models exhibit strong potential for few-shot learning due to their pretrained knowledge and reasoning capabilities. However, directly applying them remains nontrivial: continuous time-series features are not naturally aligned with language embeddings, and predictions must be restricted to a fixed set of target classes. Therefore, we propose TimeMorph, a dual-view framework that extracts temporal dynamics via a temporal branch and waveform morphology via a visual pseudo-image branch. To bridge time-series features and language embeddings, we employ a stage-wise transfer strategy to project the fused tokens into a pretrained causal LLM. During inference, TimeMorph replaces open-ended text generation with closed-set class-token scoring over the valid label set, and further incorporates semantic label priors when meaningful class names are available. Extensive few-shot experiments on the UCR archive and additional external datasets show that TimeMorph consistently outperforms strong task-specific, Transformer-based, and LLM/foundation-model baselines. Further analyses validate the effectiveness of dual-view modeling, transfer pretraining, and the class-token interface.

Index Terms—Few-shot learning, time-series classification, large language models, multimodal learning, closed-set classification.

1 Introduction

Time series classification (TSC) is challenging when labeled sequences are scarce in healthcare monitoring, industrial diagnosis, and related domains [1]. Reliable TSC requires capturing both temporal dynamics and waveform morphology. This reflects how humans interpret sequential data: they track numerical evolution over time while recognizing characteristic shape patterns [2], [3]. Some classes are mainly distinguished by temporal dynamics, such as local trends and long-range evolution, whereas others are more naturally characterized by local subsequence shapes, raw waveform patterns, or recurring morphological structures [4], [5]. Inspired by this dual-view observation of human, it believes that effective few-shot TSC models would benefit from extracting and combining these temporal and morphological features. However, existing approaches often focus primarily on numerical trends while overlooking the intuitive visual structure of time-series waveforms [6]–[8].

Traditional TSC methods, including convolutional networks [9], Transformers [10], and prototype-based models [7], can achieve strong performance when sufficient labeled data are available. In few-shot settings, however, they often overfit to the limited support set and generalize poorly [11]. Pretrained large language models (LLMs) offer a promising direction for data-scarce classification. Recent studies adapt LLMs to time-series forecasting, classification, and anomaly detection through prompting, sequence reprogramming, or cross-modal alignment, showing encouraging generalization potential [12]–[15]. Meanwhile, the in-context learning and prompt-based adaptation abilities of LLMs have demonstrated strong task adaptability in few-shot and zero-shot settings [16], [17]. Nevertheless, directly applying LLMs to TSC remains non-trivial, since time series are continuous, domain-specific, and not naturally expressed in language. [18].

To make pretrained LLMs effective for few-shot TSC with dual-view time-series representations, three key challenges must be addressed: (i) the model should preserve both temporal dynamics and morphology-oriented cues. (ii) the fused continuous time-series features must be aligned with the input space of a pretrained LLM. (iii) predictions should be restricted to valid candidate classes rather than produced by unconstrained text generation. The latter may generate invalid labels and introduce undesirable biases from tokenization or label length. This issue is closely related to the broader problem of enforcing valid outputs in language-model decoding, which has motivated constrained decoding methods for sequence generation and structured prediction

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Shizhuo Deng and Dongyue Chen are with the College of Information Science and Engineering, Key Laboratory of Data Analytics and Optimization for Smart Industry, Northeastern University, Shenyang 110819, China, and also with the Foshan Graduate School of Innovation, Northeastern University, Foshan 528311, China (e-mail: dengshizhuo@mail.neu.edu.cn; chendongyue@ise.neu.edu.cn).
Yonghui Qin and Tong Jia are with the College of Information Science and Engineering, Northeastern University, Shenyang 110819, China (e-mail: 2400885@stu.neu.edu.cn; jiatong@ise.neu.edu.cn).
Xinde Li is with the Faculty of Robot Science and Engineering, Northeastern University, Shenyang 110819, China (e-mail: lixinde@mail.neu.edu.cn).
Ye Yuan is with [Department/School], [University/Institution], [City] [Postal Code], China (e-mail: [email]).

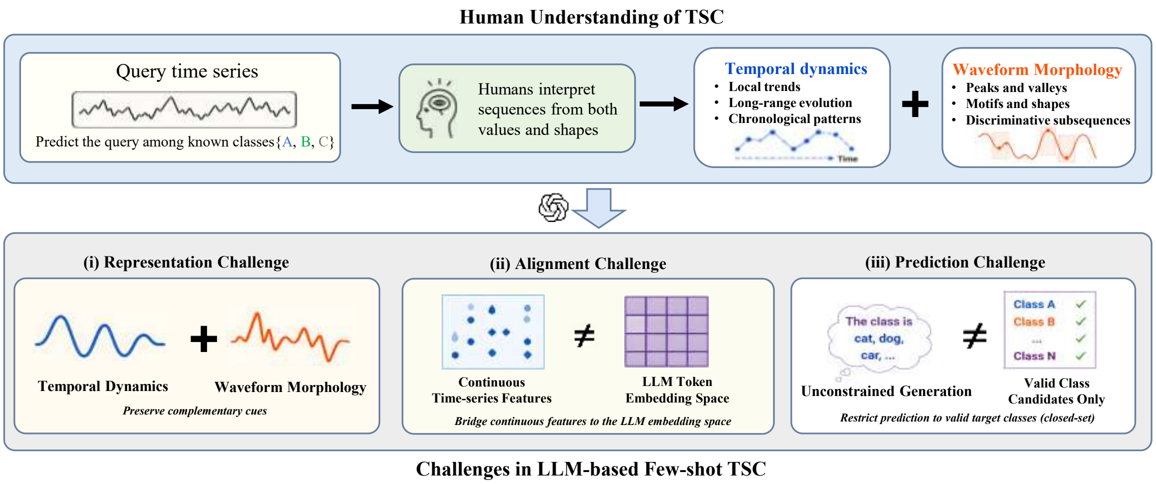

Fig. 1. Motivation of TimeMorph. Effective few-shot TSC requires both temporal dynamics and waveform morphology, while adapting LLMs to this setting introduces representation, alignment, and closed-set prediction challenges.

[19], [20].

We propose **TimeMorph**, a dual-view generative classification framework for few-shot TSC. For each input sequence, TimeMorph constructs two complementary views: a temporal branch that models ordered local relations and long-range dynamics, and a morphological branch that converts the waveform into a morphological map, enabling a frozen visual encoder to extract shape-level structures. Through a stage-wise cross-modal alignment strategy, the resulting temporal and morphological tokens are projected into the embedding space of a frozen causal language model, mitigating the mismatch between time-series representations and language embeddings. For downstream prediction, TimeMorph replaces unconstrained text generation with supervised closed-set class-token scoring over valid target labels. When semantically meaningful class names are available, label verbalizers can further initialize the class tokens, injecting useful semantic priors while avoiding the instability of multi-token generation.

Our main contributions are summarized as follows:

- We propose a **dual-view time-series modeling paradigm** for few-shot TSC that jointly models temporal dynamics and waveform morphology through complementary views.
- We introduce a **closed-set class-token scoring** interface for LLM-based TSC. It replaces autoregressive label generation with supervised scoring over valid classes, ensuring legal predictions, stabilizes optimization, and effectively leverages semantic label priors.
- We design a **stage-wise cross-modal pretraining strategy** that aligns continuous time-series tokens with the embedding space of a pretrained causal LLM, enabling the model to leverage foundation-model priors without numeric-to-text tokenization.

2 Related Work

Few-shot time-series classification. Few-shot time-series classification (TSC) aims to recognize target classes from only a few labeled sequences. Existing methods typically improve data efficiency by introducing inductive biases from meta-learning, metric learning, or prototype-based decision rules. Narwariya et al. [21] meta-train residual networks across heterogeneous UCR tasks so that the model can rapidly adapt to new few-shot TSC problems. TapNet [22] learns a task-adaptive projection space and compares query samples with class reference vectors, while Prototypical Networks [23] provide a general metric-learning formulation in which classification is performed by measuring distances to class prototypes. Shapelet-based few-shot models, such as Dual Prototypical Shapelet Networks (DPSN) [8], further introduce interpretable local subsequences as discriminative evidence. More recently, COSCO [24] improves few-shot multivariate TSC by coupling prototypical learning with sharpness-aware optimization. These methods provide strong task-specific or metric-learning priors, but they are mainly discriminative models over numerical sequences. They usually do not exploit pretrained language or multimodal foundation-model priors, nor do they explicitly combine chronological dynamics with morphology-oriented waveform evidence.

LLMs for time-series modeling. Recent studies have explored adapting pretrained large language models (LLMs) to time-series data through numerical serialization, prompting, reprogramming, and embedding-space alignment. LLM-Time [12] serializes numerical values as text and casts forecasting as next-token prediction. GPT4TS/One-Fits-All [13] adapts frozen pretrained language or vision Transformers to multiple time-series tasks, suggesting that pretrained sequence models can provide transferable representations. TEST [15] aligns time-series embeddings with the LLM embedding space using instance-wise, feature-wise, and text-prototype-aligned objectives, whereas Time-LLM [14] reprograms time-series patches into a frozen LLM through text prototypes and prompt-as-prefix conditioning. LLM4TS [25] further studies staged alignment and downstream fine-tuning for data-efficient forecasting. OpenTSLM [26] integrates time series as a native modality for multimodal reasoning through soft-prompt and cross-attention interfaces.

For few-shot classification, LLMFew [11] uses a patch-wise temporal convolutional encoder and LoRA-tuned LLM decoder for few-shot multivariate TSC.

Despite these advances, directly applying LLMs to time-series classification remains non-trivial. Recent analyses show that LLM backbones do not automatically improve time-series models when the modality interface is weak or when the task head is mismatched [27], [28]. This motivates models that carefully design both the time-series-to-LLM interface and the prediction mechanism. TimeMorph differs from prior LLM-based time-series methods by aligning dual-view temporal and morphological tokens to a causal LLM and by replacing open-ended label generation with closed-set class-token scoring over valid target labels.

Time-series foundation models. In parallel with adapting textual LLMs to time series, recent work has developed time-series foundation models pretrained directly on large collections of numerical sequences. Forecasting-oriented models such as Chronos [29], TimesFM [30], and Moirai [31] demonstrate that large-scale pretraining can provide strong cross-domain priors for zero-shot or few-shot forecasting. Chronos tokenizes scaled time-series values into a fixed vocabulary and trains language-model architectures with a next-token objective, while TimesFM and Moirai design large Transformer-based architectures for universal forecasting. Beyond forecasting, MOMENT [32] constructs a large Time-series Pile and studies limited-supervision transfer across forecasting, classification, anomaly detection, and imputation. UniTS [33] further unifies predictive and generative time-series tasks through task tokenization and a shared Transformer backbone. These studies show that foundation-model pretraining is promising for time-series analysis under limited supervision. However, most time-series foundation models either focus primarily on forecasting or rely on task-specific adaptation heads for classification. They also do not explicitly address the combination of morphology-oriented waveform evidence, LLM embedding-space alignment, and closed-set label-token scoring. TimeMorph is complementary to this line of work: rather than building a standalone time-series foundation backbone, it adapts a pretrained causal LLM to few-shot TSC through a dual-view time-series interface and a constrained class-token prediction space.

Temporal and morphology-aware representations. Time-series labels can depend on both temporal dynamics and waveform morphology. Sequence-domain methods model this evidence through discriminative subsequences, convolutional kernels, or local patches. Shapelet methods extract class-discriminative local patterns [4], [8]; convolutional architectures and random convolutional kernels capture multi-scale local structures efficiently [6], [9]; and patch-based Transformers preserve local-to-global temporal dependencies by treating subsequences as tokens [34]. A complementary direction maps one-dimensional signals to two-dimensional structures, such as Gramian angular fields and Markov transition fields, enabling image models to capture global shape and transition patterns [35]. Recent visualization-oriented studies further support the value of exposing time-series structure through visual forms. VL-Time [36] shows that visualization-modeled time-series inputs can make multimodal LLMs more effective time-series reasoners, and TiViT [37] demonstrates that hidden representations of frozen pretrained vision Transformers can be useful for time-series classification. TimeMorph follows this broader insight but differs in two aspects. First, the morphology branch does not simply render a line plot or a generic image encoding; it constructs a sliding-window morphology map that emphasizes local waveform shapes and cross-window structural transitions. Second, the visual morphology tokens are fused with chronological temporal tokens before being aligned to a causal LLM, allowing the final classifier to exploit both ordered temporal dynamics and shape-level morphological evidence.

Generative classification and label-space constraints. Text-to-text models cast classification as conditional generation, where labels are produced as target tokens [38]. This interface is flexible, but it can be poorly matched to closed-set classification under scarce supervision. Free-form decoding may generate invalid labels, and likelihood-based label selection can be sensitive to answer priors, surface-form competition, tokenization, and verbalizer choices [39], [40]. Constrained decoding restricts the output space to valid lexical or structural forms [41], [42], while prompt-based classification uses label words or verbalizers to connect semantic classes with vocabulary tokens [43], [44]. TimeMorph takes a more direct closed-set route. It introduces dataset-specific single-token class identifiers and scores only valid classes, avoiding autoregressive label generation and reducing tokenization-induced label bias. When class names are informative, label verbalizers initialize and regularize the corresponding class-token rows, injecting semantic priors without relying on unstable multi-token free-form outputs.

3 Method

We propose **TimeMorph**, a dual-view LLM-based framework for few-shot time-series classification (TSC). TimeMorph adapts a pretrained causal language model to continuous time-series inputs without serializing raw observations into text. Given an input sequence, it extracts complementary temporal-dynamic and waveform-morphology patterns, projects the resulting time-series tokens into the LLM embedding space, and performs prediction by closed-set scoring over dataset-specific class tokens.

3.1 Problem Setup

3.1.1 Univariate Time-Series Classification

Let D be a target classification dataset with label set $\mathcal{Y}_D = \{0, \ldots, C-1\}$. Each example consists of a univariate time series $x \in \mathbb{R}^T$ and a class label $y \in \mathcal{Y}_D$. Given a query sequence x, the goal of TSC is to infer its label from the temporal signal. We consider closed-set classification over the full label space: all C classes are available during few-shot fine-tuning and evaluation, and predictions are restricted to the valid label set $\mathcal{Y}_D$.

3.1.2 Few-Shot Learning Setup

In the K-shot setting, the support set $\mathcal{S}_D^K$ contains at most K labeled examples per class:

$$\mathcal{S}_D^K = \{(x_i, y_i)\}_{i=1}^{n_D}, \qquad |\{i : y_i = c\}| \leq K, \quad c \in \mathcal{Y}_D. \quad (1)$$

The model is fine-tuned on $\mathcal{S}_D^K$ and evaluated on query examples from the official test split.

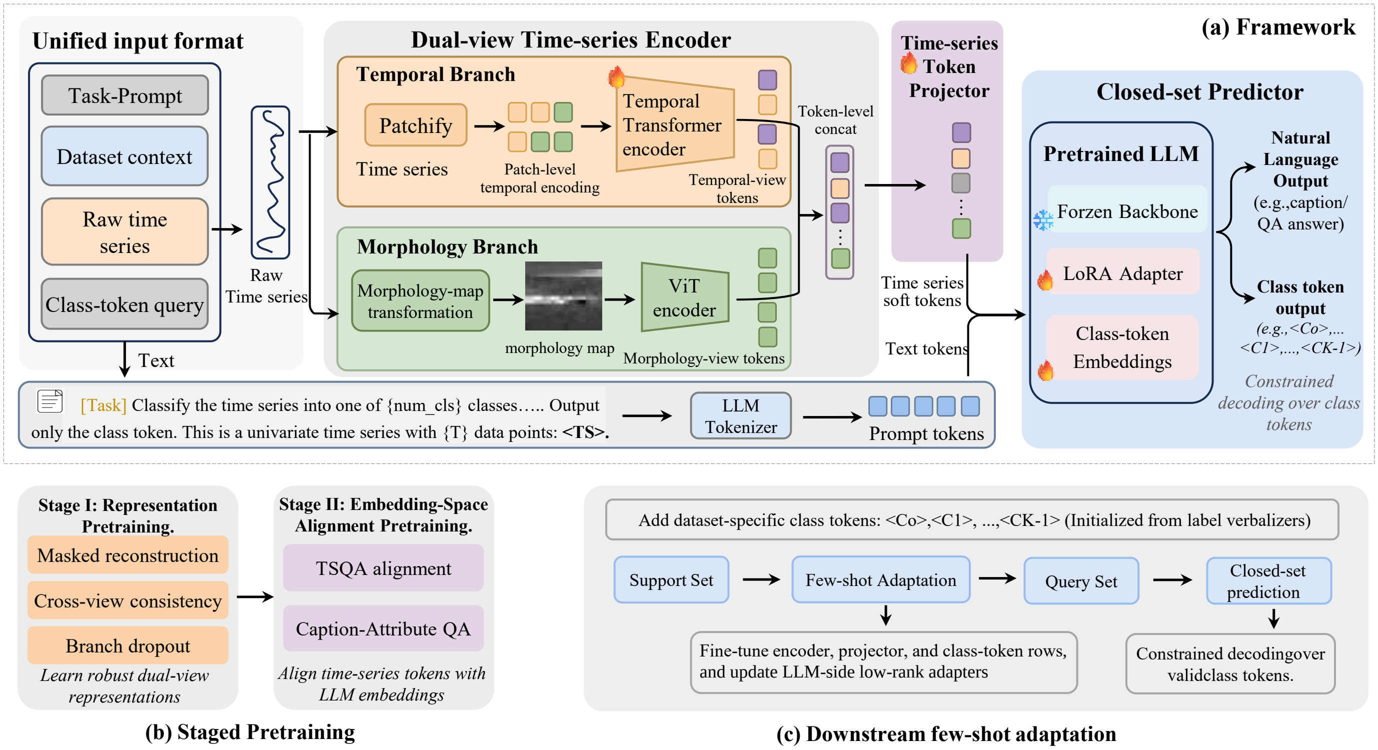

Fig. 2. Overview of TimeMorph. (a) TimeMorph encodes a raw univariate time series with temporal and morphology branches, fuses the resulting tokens, and projects them into the embedding space of a pretrained causal LLM for closed-set prediction. (b) Staged pretraining learns robust dual-view representations and aligns time-series tokens with LLM embeddings under language supervision. (c) During few-shot fine-tuning, dataset-specific class tokens are added and predictions are made by scoring only valid class tokens.

3.2 Framework

The framework consists of three components: a dual-view encoder, an embedding-space projection interface, and a closed-set class-token scoring rule. The dual-view encoder G_θ extracts complementary representations using a temporal-view encoder and a morphology-view encoder. The projection interface P_ψ maps the fused time-series tokens to the hidden dimension of a pretrained causal LLM. The class-token scoring rule converts the LLM output logits into predictions over the valid labels of the target dataset. All components are optimized through the pretraining and fine-tuning schedule described in Section 3.2.4.

3.2.1 Dual-View Time-Series Encoding

Class-discriminative information in time series can arise from heterogeneous cues. Some classes are primarily distinguished by temporal dynamics, such as local trends, phase shifts, amplitude variations, or long-range evolution. Others are more naturally characterized by waveform morphology, including peaks, valleys, cycles, motifs, and discriminative subsequences. TimeMorph therefore encodes each input sequence with two complementary views, processed by a temporal branch and a morphology branch, respectively. Figure 3 illustrates this design by comparing the raw input, ordered temporal patches, and the corresponding morphology-map view.

Temporal branch: The temporal view preserves the chronological structure of the original sequence. Let x^{pad} denote the sequence after right-side replicate padding. Given patch length ℓ and stride s, we extract overlapping temporal patches as

$$\begin{aligned} R_i &= x^{\mathrm{pad}}_{(i-1)s+1:(i-1)s+\ell}, \qquad i = 1, \ldots, N_t, \\ N_t &= \left\lfloor \frac{T + p_t - \ell}{s} \right\rfloor + 1, \end{aligned} \tag{2}$$

where p_t is the padding length required to form complete patches. Each patch is linearly projected by W_t and augmented with a learnable positional embedding π_i that encodes the index of the i-th temporal patch. The resulting sequence is processed by a temporal Transformer:

$$Z^t(x) = G_t\left([W_t R_1 + \pi_1; \ldots; W_t R_{N_t} + \pi_{N_t}]\right). \tag{3}$$

The output $Z^t(x)$ is a sequence of temporal tokens encoding ordered patch-level dynamics of the waveform.

Morphology branch: The morphology view exposes waveform morphology by converting the same one-dimensional sequence into a two-dimensional sliding-window morphology map. We first normalize x with robust per-sequence statistics:

$$\bar{x}_t = \frac{x_t - \mathrm{median}(x)}{\mathrm{IQR}(x) + \epsilon}. \tag{4}$$

The normalized sequence is then unfolded into overlapping local windows. Let $q = \max(1, \lfloor \sqrt{T} \rfloor)$ be the window length and $\delta = \max(1, \lfloor \rho q \rfloor)$ be the stride. We construct the morphology map by normalizing the unfolded matrix,

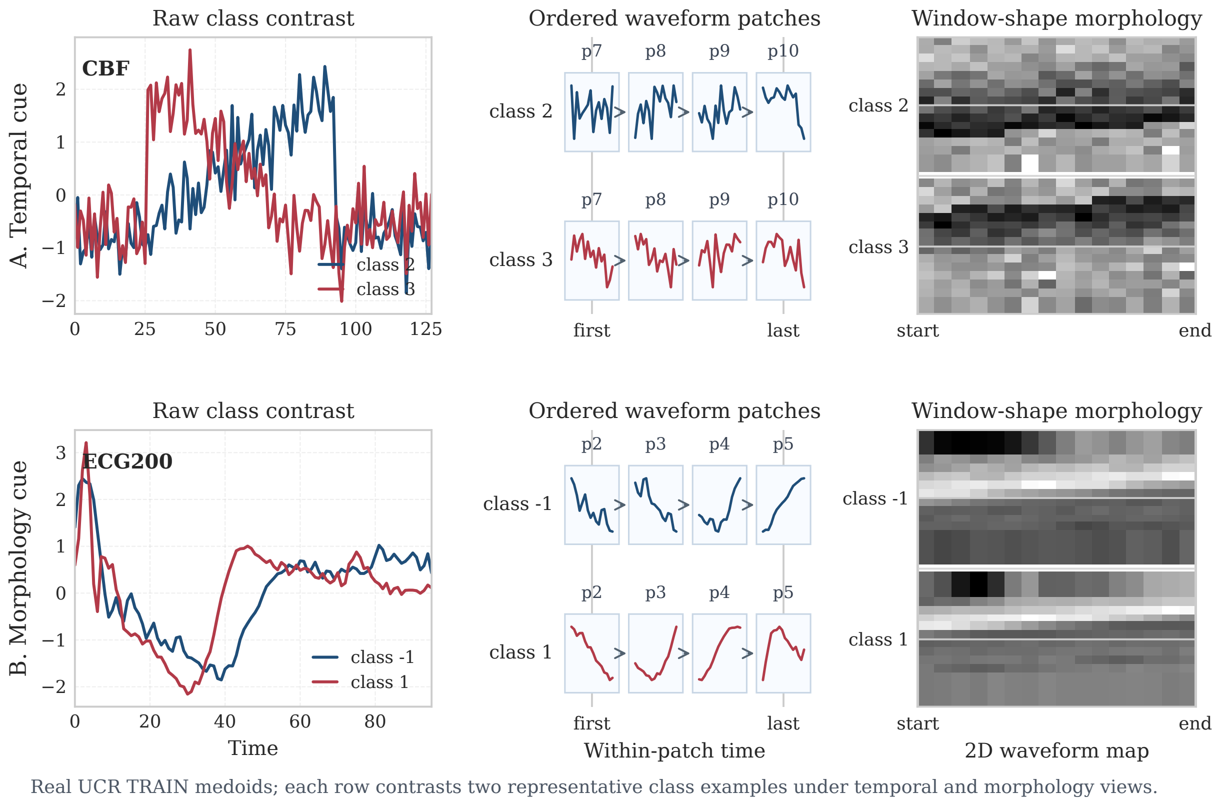

Fig. 3. Qualitative motivation for the dual-view encoder. For representative UCR tasks, we visualize real training examples nearest to their class centroids, together with their raw sequences, ordered temporal patches, and morphology-map representations. The temporal view preserves ordered patch-level evolution, whereas the morphology view exposes local subsequence shapes and cross-window structural transitions through a two-dimensional waveform map.

resizing it to the input resolution of the vision backbone, and replicating it across RGB channels:

$$I(x) = \mathrm{RGB}\left(\mathrm{Resize}\left(\Gamma_{\mathrm{img}}\left(\mathrm{Unfold}_{q,\delta}(\bar{x})\right)\right)\right), \tag{5}$$

where $\Gamma_{\mathrm{img}}(\cdot)$ denotes per-map normalization and contrast adjustment. Each row of $I(x)$ corresponds to a local waveform window, so the map preserves local subsequence shapes while making cross-window structural transitions visible to a vision encoder. This construction exposes repeated local shapes and cross-window transitions rather than treating the time series as an arbitrary natural image. The two axes of the map have explicit time-series meanings: the within-row direction preserves the shape of each local window, whereas the across-row direction follows the progression of overlapping windows and therefore records how local shapes evolve along the sequence. The median/IQR normalization in Eq. (4) reduces the effect of isolated spikes and scale outliers, which is important when only a few labeled examples are available for target adaptation. The choice $q = \lfloor\sqrt{T}\rfloor$ provides a length-adaptive window size that remains small enough for short sequences while still capturing local morphology in longer sequences. Thus, the morphology map is used as a structured waveform view with local-shape and transition semantics, rather than as an unconstrained image rendering of the signal. A frozen vision backbone G_m extracts morphology tokens:

$$Z^m(x) = G_m(I(x)). \tag{6}$$

We discard the vision class token and use only the patch tokens. The vision backbone remains frozen; trainable parameters are introduced in the temporal branch, projection modules, class-token rows, and LLM adapters.

Dual-view fusion: The temporal and morphology tokens are projected to a common dimension and concatenated:

$$\begin{aligned} \tilde{Z}^t(x) = Q_t(Z^t(x)), \qquad \tilde{Z}^m(x) = Q_m(Z^m(x)), \\ Z_x = \left[\tilde{Z}^t(x); \tilde{Z}^m(x)\right]. \end{aligned} \tag{7}$$

The fused sequence $Z_x \in \mathbb{R}^{N_x \times d_e}$ serves as the dual-view time-series token sequence passed to the LLM interface.

3.2.2 Time-Series Token Projection

The dual-view encoder outputs continuous time-series tokens, whereas a causal LLM consumes embeddings associated with discrete text tokens. TimeMorph bridges this modality gap through an input-side projection interface, which maps the fused time-series tokens Z_x into the LLM embedding space without serializing numerical observations as text.

Let P_ψ be a trainable projection module from encoder dimension d_e to the LLM hidden dimension, $\tau(\cdot)$ be the tokenizer, and E_{lm} be the LLM input embedding matrix. For a prompt $p = (p^{\mathrm{pre}}, p^{\mathrm{post}})$, the conditioned input embeddings are

$$H(x,p) = \left[E_{\mathrm{lm}}(\tau(p^{\mathrm{pre}})); P_\psi(Z_x); E_{\mathrm{lm}}(\tau(p^{\mathrm{post}}))\right]. \tag{8}$$

The projected time-series tokens are therefore inserted into the same embedding sequence as text tokens and can condition the causal LLM through its standard forward pass. When auxiliary textual descriptions are available, they are appended to the prompt and encoded by the same tokenizer. The objectives used to learn this interface are introduced together with the staged training schedule in Section 3.2.4.

3.2.3 Closed-Set Class-Token Scoring

Few-shot TSC requires prediction over a fixed set of valid labels rather than open-ended text generation. For each target dataset D, TimeMorph assigns a compact set of single-token class identifiers:

$$\mathcal{C}_D = \{\langle \text{cls}_0 \rangle, \dots, \langle \text{cls}_{C-1} \rangle\}, \qquad \mathcal{V}_D = \{v_0, \dots, v_{C-1}\}, \tag{9}$$

where v_c is the vocabulary index of class token $\langle \text{cls}_c \rangle$. We implement these identifiers as newly added special tokens, so each class corresponds to exactly one vocabulary index. Let p_D denote the classification prompt for dataset D. After the LLM conditions on $H(x, p_D)$, let $h_\Theta(x, p_D)$ be the final hidden state at the last prompt position. With output embedding matrix W_{out}, the score for class c is the next-token logit assigned to its class token:

$$s_c(x) = [W_{\text{out}} h_\Theta(x, p_D)]_{v_c}. \tag{10}$$

Prediction is restricted to the valid class-token set:

$$\hat{y} = \arg \max_{c \in \{0, \dots, C-1\}} s_c(x). \tag{11}$$

The supervised objective is the cross entropy over only the valid class tokens:

$$\mathcal{L}_{\text{ct}} = -\log \frac{\exp s_y(x)}{\sum_{c=0}^{C-1} \exp s_c(x)}. \tag{12}$$

This design keeps the autoregressive vocabulary head but converts downstream classification into a single-forward-pass closed-set comparison over C legal labels. Among vocabulary parameters, only the input embeddings and output rows of the newly introduced class tokens are trainable; all other vocabulary rows remain fixed. Compared with generating textual labels, this interface avoids multi-token likelihood comparison, label-length effects, and invalid free-form outputs. It also differs from replacing the LLM with an independent linear classifier: the class identifiers remain part of the LLM vocabulary/output-head space, so anonymous class tokens and semantic-prior class tokens can be handled by the same scoring rule. The resulting objective directly calibrates the valid class-token rows under support-set supervision while retaining the pretrained language backbone as the conditional feature processor.

Label-semantic priors: For anonymous-label benchmarks, the class tokens are learned as dataset-specific identifiers. When class names are semantically meaningful, TimeMorph uses label verbalizers to initialize and regularize the same class-token rows. For class c, let $\mathcal{R}_c$ denote a set of template-consistent verbalizers derived from its canonical label name. We form a semantic prototype by averaging the LLM input embeddings of all verbalizer tokens:

$$p_c = \frac{1}{|\mathcal{R}_c|} \sum_{r \in \mathcal{R}_c} \frac{1}{|\tau(r)|} \sum_{u \in \tau(r)} E_{\text{lm}}(u). \tag{13}$$

The prototype is rescaled to match the average norm of vocabulary rows and is used to initialize the input and output rows of the corresponding class token. During fine-tuning, we regularize these rows toward a detached prototype $\tilde{p}_c$:

$$\mathcal{L}_{\text{sem}} = \frac{1}{C} \sum_{c=0}^{C-1} \left[1 - \cos(e_{v_c}, \tilde{p}_c) + 1 - \cos(w_{v_c}, \tilde{p}_c)\right], \tag{14}$$

where e_{v_c} and w_{v_c} are the input and output rows of class token v_c. The supervised target loss is

$$\mathcal{L}_{\text{adapt}} = \mathcal{L}_{\text{ct}} + \lambda_{\text{sem}} \mathcal{L}_{\text{sem}}. \tag{15}$$

For datasets without meaningful label names, we set $\lambda_{\text{sem}} = 0$.

3.2.4 Staged Pretraining and Few-Shot Fine-Tuning

TimeMorph first learns reusable dual-view time-series representations, then aligns projected time-series tokens with a frozen causal LLM, and finally adapts to each target few-shot label space. Stage I performs representation pretraining on unlabeled raw series. Stage II performs language-alignment pretraining with time-series–language supervision. The final Stage-II checkpoint initializes all target-dataset fine-tuning runs reported in this paper. This staged route keeps the few-shot support set focused on closed-set calibration rather than representation learning and modality alignment.

Stage I: Representation Pretraining: Stage I pretrains the dual-view encoder without class labels. It uses unlabeled raw sequences from TSQA-train [45], M4-train [46], and the official TRAIN splits of a fixed 98-dataset UCR pretraining pool [47]. Only raw sequences from these TRAIN splits are used, with no access to UCR TEST sequences or class labels during pretraining. All input series are normalized by the data loader before augmentation.

The objective combines masked temporal patch reconstruction with an invariance–variance–covariance regularizer over augmented temporal, morphology, and fused representations:

$$\mathcal{L}_{\text{rep}} = \lambda_{\text{rec}} \mathcal{L}_{\text{rec}} + \sum_{b \in \{t, m, f\}} \lambda_b \mathcal{L}_{\text{ivc}}^b, \tag{16}$$

where t, m, and f denote the temporal, morphological, and fused branches, respectively. For each augmented view, a single contiguous segment with ratio 0.4 is set to 0.0 before temporal encoding, and $\mathcal{L}_{\text{rec}}$ reconstructs the normalized temporal patches extracted from the same view before this reconstruction mask is applied. Augmentations consist of Gaussian jitter with standard deviation 0.03, sequence-wise scaling sampled from $[0.8, 1.2]$, and contiguous time masking with ratio 0.1.

The IVC regularizer uses mean-pooled temporal, morphology, and fused tokens after branch-specific SSL heads. Given two augmented projections $z_a, z_b \in \mathbb{R}^{B \times d}$ for a branch, let $s_j(z) = \sqrt{\mathrm{Var}(z_{:j}) + \epsilon}$, $\tilde{z} = z - \frac{1}{B}\mathbf{1}\mathbf{1}^\top z$, and $C(z) = \tilde{z}^\top \tilde{z}/(B-1)$. We use

$$\begin{aligned}
\mathcal{L}_{\text{ivc}} &= \alpha R + \beta V + \gamma O, \\
R &= \frac{1}{Bd}\|z_a - z_b\|_F^2, \\
V &= \sum_{u\in\{a,b\}} \frac{1}{d}\sum_{j=1}^{d} \max(0, 1 - s_j(z_u)), \\
O &= \sum_{u\in\{a,b\}} \frac{1}{d}\sum_{i\neq j} C(z_u)_{ij}^2,
\end{aligned} \tag{17}$$

with $(\alpha, \beta, \gamma) = (25, 25, 1)$ and $\epsilon = 10^{-4}$. The Stage-I loss weights are $\lambda_{\text{rec}} = 1.0$, $\lambda_t = 0.25$, $\lambda_m = 0.25$, and $\lambda_f = 0.5$. Branch dropout with probability 0.1 regularizes the fused representation against single-view dependence.

Stage II: Language-Alignment Pretraining: Stage II attaches the projector P_ψ and the pretrained causal LLM to the Stage-I dual-view encoder. Let $\mathcal{D}_{\text{lang}}$ denote the language-alignment corpus, which contains TSQA question-answering examples from ChatTime [45], M4 captioning examples [46], and synthetic attribute supervision derived from TSQA and M4 raw series. Training samples are mixed with TSQA/M4/synthetic-attribute weights $1:1:2$. The synthetic labels are computed from normalized TSQA/M4 raw series using rule-based temporal attributes, including trend, periodicity, volatility, spikes, and abrupt level changes, and are verbalized as captions or yes/no attribute prompts. The projected tokens $P_\psi(Z_x)$ are inserted into the prompt embedding sequence as in Eq. (8). The alignment objective is the answer-token language-modeling loss:

$$\mathcal{L}_{\text{align}} = \mathbb{E}_{(x,p,a)\sim\mathcal{D}_{\text{lang}}}\left[-\log p_\Theta(a \mid H(x,p))\right], \tag{18}$$

where the loss is applied only to answer tokens. This stage optimizes a time-series-to-language interface under the frozen LLM next-token objective. Training runs for 20 epochs: the first two epochs update only the projector with the dual-view encoder fixed, and the remaining epochs jointly optimize the encoder and projector. The encoder and projector learning rates are 5×10^{-5} and 1×10^{-4}, respectively. The LLM backbone remains frozen throughout Stage II. The final Stage-II checkpoint supports target-dataset fine-tuning on UCR, MIT-BIH, and CinC2017. Because the loss is applied only to answer tokens, the language objective supervises the textual response conditioned on projected time-series tokens, rather than converting the raw numerical sequence into text. Freezing the LLM backbone concentrates the optimization on the time-series encoder, projector, and modality interface, preserving the pretrained language model while learning how continuous time-series evidence should condition it.

Language-Alignment Data Forms: Stage II uses heterogeneous time-series–language supervision, but all examples are normalized into the same tuple form (x, p, a), where x denotes a raw time series, p denotes a natural-language or template-based prompt, and a denotes the supervised textual target. Table 1 lists the resulting data forms for TSQA question answering, M4-derived captions, and synthetic attribute supervision. The raw series x is always encoded by the dual-view time-series encoder and projected into the LLM embedding space; it is not serialized into numerical text.

Thus, the three data sources share the same answer-token language-modeling objective in Eq. (18), while covering complementary forms of temporal semantics.

Target-Dataset Fine-Tuning: For each target dataset D, TimeMorph fine-tunes a small trainable subset of the model on the sampled support set. The warm-up phase updates the dual-view encoder, the projection module, and the dataset-specific class-token rows using Eq. (15). This phase calibrates the projected time-series tokens and the closed-set decision space before updating any LLM adapters. For datasets with meaningful class names, label verbalizers initialize the class-token rows and Eq. (14) regularizes them during fine-tuning.

The joint phase additionally updates low-rank adapters on the LLM side, while the pretrained LLM weights remain frozen. At inference time, TimeMorph performs a single forward pass for each query sequence and ranks the valid class tokens using Eq. (10). The stage-wise optimization schedule is summarized in Table 2.

4 EXPERIMENTS

4.1 Experimental Setup

4.1.1 Datasets and Few-Shot Protocol

We evaluate TimeMorph on all 128 datasets in the UCR Archive 2018 under a fixed full-label-space few-shot protocol. For each dataset D and $K \in \{1, 2, 5, 10\}$, all target classes are included. We sample up to K labeled examples per class from the official TRAIN split as the support set and evaluate on all examples in the official TEST split. UCR results are averaged over five independent runs and then macro-averaged over datasets. We further evaluate semantic-label transfer on MIT-BIH and CinC2017 with $K \in \{10, 20, 30\}$ over ten independent runs, using the same full-label-space protocol. These two external datasets contain meaningful class names, so TimeMorph enables label-name priors during fine-tuning.

4.1.2 Baselines

We compare TimeMorph with representative few-shot and time-series classification baselines, including ResNet, TapNet, COSCO-ResNet, PatchTST, DLinear, TimesNet, Autoformer, Crossformer, FEDformer, Informer, and GPT4TS. For UCR, `Avg` averages the four shot-specific mean accuracies, and `AvgRank` averages dataset-level ranks across shots. Standard deviations and full per-dataset tables are reported in Appendix A. Appendix B.4 lists the training recipes and hyperparameters for the reported baseline families.

4.1.3 Implementation Details

TimeMorph uses `Llama-3.2-1B` as the causal language backbone and `dinov2-base` as the frozen morphology encoder. The temporal branch uses patch length 16, stride 8, and hidden size 128. For each target dataset, we perform 5 warm-up fine-tuning epochs followed by 55 joint fine-tuning epochs. Language-side fine-tuning uses LoRA with rank 16 and alpha 32. Table 3 summarizes the model and downstream implementation settings; the stage-wise optimization schedule is given in Table 2.

The operator Γ_{img} in Eq. 5 applies per-sample min–max normalization over the morphology grid and a fixed contrast adjustment $u \mapsto u^{0.8}$. We use DINOv2 patch tokens from the configured intermediate layer and exclude the class token.

TABLE 1
Unified data forms used in Stage-II language-alignment pretraining. Each instance is represented as (x, p, a), where x is a raw time series, p is a natural-language or template-based prompt, and a is the supervised textual target. Raw series are encoded by the time-series encoder and are not serialized as text.

Source	**Series input** x	**Prompt** p	**Target** a
TSQA	Raw TSQA training series	Question-answering prompt, e.g., asking the model to select the temporal trend, identify an anomaly, or answer a time-series reasoning question.	Short answer or option text, e.g., *constant trend*, *upward trend*, or the corresponding QA answer.
M4-derived captions	Raw M4 training series	Captioning prompt, e.g., "Describe the dominant temporal pattern of this series."	Caption text describing temporal patterns, including trend, seasonality, volatility, spikes, or level shifts.
Synthetic attribute supervision	TSQA/M4-derived raw series	Attribute prompt, e.g., "Which temporal attributes are present in this series?"	Structured textual attributes, e.g., *trend: upward; periodicity: weak; local pattern: spike; noise: high.*

TABLE 2
Stage-wise training summary for TimeMorph.

Stage	**Data**	**Objective**	**Trainable modules**	**Frozen modules**	**Epoch / LR**
Stage I	TSQA raw, M4 raw, and UCR TRAIN raw series	Masked patch reconstruction + IVC regularization	Temporal branch, branch projectors, SSL/reconstruction heads	DINOv2; LLM not used	8 epochs; LR 2×10^{-4} temporal, 1×10^{-4} heads
Stage II	TSQA QA, M4 captions, and synthetic attributes	Answer-token language-modeling loss	Projector for the first 2 epochs; encoder + projector afterwards	LLM backbone and DINOv2	20 epochs; LR 5×10^{-5} encoder, 1×10^{-4} projector
Target warm-up	Target support set	Class-token cross-entropy + optional semantic row regularization	Encoder, projector, class-token rows	Base LLM; LoRA inactive	5 epochs; LR 2×10^{-4} encoder, 1×10^{-4} projector, 2×10^{-4} class-token rows
Target joint	Target support set	Class-token cross-entropy + optional semantic row regularization	Encoder, projector, class-token rows, LoRA adapters	Base LLM weights	55 epochs; LR 2×10^{-4} encoder, 1×10^{-4} projector/LoRA, 2×10^{-4} class-token rows

4.1.4 Protocol Fairness and Leakage Control

All methods use the same sampled support sets within each run and are evaluated on the same query examples. For UCR, support examples are sampled only from the official TRAIN split, while evaluation always uses the full official TEST split. Stage I is trained on raw TSQA-train series, raw M4-train series, and unlabeled official TRAIN-split sequences from a fixed 98-dataset UCR pretraining pool. Stage II uses TSQA question-answering examples, M4 captioning examples, and synthetic attribute supervision derived from TSQA and M4 raw series. Official TEST examples and UCR class labels are excluded from pretraining; target supervision enters only through the sampled support set during few-shot fine-tuning. All target evaluations start from the final Stage-II checkpoint. For MIT-BIH and CinC2017, label-name priors initialize and regularize class-token rows during fine-tuning. The baseline families share the same support/query protocol, but their checkpoint-selection rules and epoch schedules are not forced to be identical because they follow their standard training recipes. We therefore summarize the fairness boundary here and provide the full recipe table in Appendix B.4.

4.2 Overall Few-Shot Performance

4.2.1 Results on UCR

Table 4 reports the main results on 128 UCR datasets. TimeMorph achieves the best mean accuracy in all four shot settings, obtaining 48.31, 55.55, 64.14, and 69.16 in the 1-, 2-, 5-, and 10-shot settings. It also obtains the best overall average accuracy and the best average rank. Compared with the strongest baseline, COSCO-RESNET, TimeMorph improves average accuracy from 58.39 to 59.29 and average rank from 4.70 to 4.24. The gains are larger over generic sequence models and LLM-style baselines: TimeMorph improves the average accuracy by 4.53 points over INFORMER and 18.36 points over GPT4TS. These results show that TimeMorph consistently benefits from combining temporal dynamics, morphological structure, and LLM-based class-token scoring.

4.2.2 Results on External Semantic-Label Datasets

We further evaluate TimeMorph on MIT-BIH and CinC2017 under 10-, 20-, and 30-shot settings. These datasets provide meaningful class names, allowing TimeMorph to use label-verbalizer priors during fine-tuning. Table 5 reports the complete results. TimeMorph obtains the best result in five of the six dataset-shot settings and remains the strongest method when results are averaged across shots. The result confirms that the semantic class-token interface transfers beyond UCR and can exploit informative label names when they are available.

4.3 Component and Interface Analysis

4.3.1 Effect of Staged Pretraining and LLM Interface

Table 6 reports the main component ablations on all 128 UCR datasets. The "$+/-$" columns indicate whether staged pretraining, the LLM pathway, and the morphology view are retained. Pretraining denotes the full Stage-I/Stage-II transfer route, the LLM column denotes the language-side interface with class-token scoring, and the morphology column denotes the visual waveform branch.

TABLE 3
Key training settings for TimeMorph.

Component	Setting
Base LLM	`meta-llama/Llama-3.2-1B`
Vision backbone	`facebook/dinov2-base`
Classification prompt suffix / answer	`Class:` / one class token, e.g., $\langle \mathrm{cls}_y \rangle$ followed by EOS
Morphology-map construction	Overlapping-window grid transform with per-sequence median/IQR normalization
Temporal patch length / stride	16 / 8
Temporal hidden size / heads	128 / 8
Branch projectors	LayerNorm–Linear–GELU–Dropout MLPs when dimensions differ
Language projector	MLP projector into the LLM hidden dimension followed by output LayerNorm
Target initialization checkpoint	Final Stage-II language-alignment checkpoint
Warm-up / joint fine-tuning epochs	5 / 55
Training / evaluation batch size	8 / 8
Gradient accumulation	4
LR (encoder / projector / LoRA)	2×10^{-4} / 1×10^{-4} / 1×10^{-4}
Language-side fine-tuning	LoRA on `q/k/v/o_proj`, `gate_proj`, `up_proj`, and `down_proj` ($r = 16$, $\alpha = 32$)
Class-token parameterization	Trainable class-token input/output rows
Anonymous class-token initialization	pretrained vocabulary-row mean plus small random perturbation
Prediction rule	anonymous class-token scoring; semantic-prior class-token scoring when label text is meaningful
Vision backbone	frozen
Downstream data augmentation	none

TABLE 4
Few-shot classification accuracy (%) on 128 UCR datasets, grouped by model family. Baseline method references are shown in the Model column. We report the mean over five runs. `Avg` and `AvgRank` average the four shot-specific mean accuracies and average ranks, respectively. Best is bold and second-best is underlined.

Family	Model	1-shot	2-shot	5-shot	10-shot	Avg	AvgRank
CNN	ResNet [48]	47.79	53.98	61.27	66.24	57.32	4.89
	TapNet [7]	45.40	50.79	59.47	64.93	55.15	5.45
	COSCO-ResNet [24]	<u>47.88</u>	<u>54.95</u>	<u>63.55</u>	<u>67.18</u>	<u>58.39</u>	<u>4.70</u>
Transformer/TS	DLinear [49]	44.21	48.01	52.30	55.43	49.99	7.24
	PatchTST [34]	42.39	48.43	56.94	62.72	52.62	6.54
	TimesNet [50]	45.16	49.95	57.86	62.79	53.94	5.90
	Autoformer [51]	36.78	39.45	43.20	46.04	41.37	9.47
	Crossformer [52]	44.31	49.45	57.20	61.94	53.23	6.04
	FEDformer [53]	37.71	41.71	47.30	51.50	44.56	8.68
	Informer [54]	46.42	51.14	58.35	63.13	54.76	5.59
LLM/foundation	GPT4TS [13]	35.61	38.30	42.94	46.88	40.93	9.26
Ours	TimeMorph	**48.31**	**55.55**	**64.14**	**69.16**	**59.29**	**4.24**

Full TimeMorph achieves the best average accuracy, 59.29%. Removing staged pretraining reduces the average to 56.94%, a 2.35-point drop, indicating that the full Stage-I/Stage-II transfer route provides a useful initialization for target fine-tuning. Further removing the LLM pathway lowers the average to 53.94%, and additionally removing the morphology branch gives the weakest result, 49.94%. The full cumulative removal path is 9.35 points below TimeMorph, showing that the final system benefits from retaining pretrained transfer, the language-side interface, and waveform morphology evidence. Because the rows remove components cumulatively, the intermediate differences indicate the effect of each ablation path rather than isolated single-factor effects.

4.3.2 Effect of Morphology-Map Construction

We next examine whether the morphology branch depends on the proposed sliding-window morphology map or on the broader act of converting a sequence into a two-dimensional view. Table 7 reports an additional full-model ablation on UCR under the 5-shot protocol with three independent runs. The comparison preserves the complete LLM-based TimeMorph pipeline, so it evaluates morphology construction inside the final model.

The sliding-window morphology map obtains the highest accuracy, best average rank, and largest number of best-performing datasets among the compared 1D-to-2D constructions. Recurrence plots and Gramian angular fields remain competitive but trail the proposed construction, while the rendered line plot and STFT spectrogram are weaker in this setting. These results indicate that the morphology branch benefits from a structured local-window representation of waveform shape, not simply from visual rendering of the raw signal.

TABLE 5
Few-shot classification accuracy (%) on two external semantic-label datasets. Each cell reports mean ± standard deviation over 10 runs under the same few-shot protocol. Best is bold and second-best is underlined within each dataset-shot column.

Model	MIT-BIH			CinC2017		
	10-shot	20-shot	30-shot	10-shot	20-shot	30-shot
TimeMorph	57.05 ± 10.96	**66.62 ± 6.54**	**68.87 ± 11.23**	**34.72 ± 3.15**	**35.24 ± 1.24**	**38.86 ± 7.00**
ResNet	37.97 ± 14.13	55.66 ± 8.58	57.07 ± 9.05	30.95 ± 12.41	33.73 ± 7.43	36.20 ± 4.84
TapNet	31.94 ± 7.18	40.15 ± 10.28	44.11 ± 11.14	30.05 ± 6.73	32.80 ± 8.81	31.01 ± 11.54
COSCO-ResNet	32.33 ± 14.25	35.08 ± 13.75	41.17 ± 8.94	31.41 ± 4.48	31.04 ± 6.97	31.98 ± 8.37
PatchTST	38.94 ± 14.35	55.36 ± 8.84	57.08 ± 9.40	26.07 ± 6.24	30.35 ± 7.51	31.21 ± 3.36
OneFitsAll	32.92 ± 16.63	42.48 ± 10.82	54.87 ± 10.04	30.80 ± 6.07	27.56 ± 4.33	29.23 ± 3.03
TimesNet	51.01 ± 5.84	54.21 ± 7.81	56.15 ± 8.77	27.25 ± 2.81	28.41 ± 4.24	29.43 ± 2.44
DLinear	55.57 ± 9.85	57.45 ± 7.95	58.84 ± 8.69	24.92 ± 1.16	25.17 ± 1.22	24.53 ± 1.70
Autoformer	38.05 ± 5.65	42.49 ± 8.58	44.39 ± 7.25	28.37 ± 5.44	27.67 ± 2.76	26.30 ± 4.36
Crossformer	53.48 ± 9.36	58.55 ± 10.43	58.24 ± 9.71	24.02 ± 6.10	25.50 ± 2.86	27.35 ± 2.35
FEDformer	**58.88 ± 5.77**	56.80 ± 9.98	55.12 ± 10.09	20.35 ± 8.32	26.90 ± 3.54	27.91 ± 4.87
Informer	53.19 ± 6.03	54.20 ± 7.61	52.35 ± 10.80	26.93 ± 9.30	28.14 ± 4.01	30.12 ± 5.95

TABLE 6
Cumulative component ablations on the 128 UCR datasets. "+" means the component is retained, and "−" means it is removed along the cumulative path. Each value is mean accuracy (%) over the datasets. Δ denotes the average accuracy decrease relative to TimeMorph.

Setting	Pretrain	LLM	Morph.	1-shot	2-shot	5-shot	10-shot	Avg	Δ
TimeMorph	+	+	+	**48.31**	**55.55**	**64.14**	**69.16**	**59.29**	–
TimeMorph-nopretrain	−	+	+	46.70	51.81	62.65	66.60	56.94	2.35
TimeMorph-nopretrain-nollm	−	−	+	44.90	49.77	58.36	62.73	53.94	5.35
TimeMorph-nopretrain-nollm-novision	−	−	−	40.26	45.54	54.17	59.79	49.94	9.35

TABLE 7
Ablation of 1D-to-2D morphology construction on UCR using the full TimeMorph model under the 5-shot protocol with three independent runs. Acc. reports mean accuracy (%), Std. reports the corresponding standard deviation, AvgRank is the average dataset-level rank, Best counts the number of datasets on which each construction performs best, and W/T/L is measured against the reference morphology-map construction.

Method	Construction	Acc.	Std.	AvgRank	Best	W/T/L vs. Ref.
Morphology map	Sliding-window morphology map	**65.18**	3.02	**2.21**	**53**	–
Recurrence plot	Pairwise recurrence map	62.88	18.93	2.68	35	43/6/79
GAF	Gramian angular field	63.01	19.71	2.74	28	46/9/73
STFT spectrogram	Short-time Fourier spectrogram	58.05	2.85	3.59	15	29/5/94
Line plot	Rendered waveform line plot	57.20	3.65	3.79	14	26/2/100

4.3.3 Effect of Semantic Label Priors

For external datasets with meaningful label names, Table 8 compares anonymous class-token rows with semantic-prior rows. Both variants start from the same final Stage-II checkpoint; the semantic-prior variant differs only in label-verbalizer initialization and semantic regularization of the class-token rows. Semantic priors improve five of the six dataset-shot settings and increase the overall average from 47.84 to 50.23. The gains are dataset- and shot-dependent, with a decrease on CinC2017 20-shot, but the averaged result supports the semantic-prior class-token space as a useful extension of supervised closed-set decoding.

Overall, the ablations support the central design: staged transfer improves initialization, the LLM pathway strengthens support-set fine-tuning, the morphology view supplies complementary waveform evidence, and semantic-prior row initialization further improves transfer on datasets with meaningful label names.

TABLE 8
Semantic-prior class-token ablation on two external semantic-label datasets (accuracy, %). Each cell reports mean ± standard deviation over 10 runs under the same few-shot protocol. The better variant within each dataset-shot column is bold. Δ denotes Semantic prior minus Anonymous labels.

Dataset	Variant	10-shot	20-shot	30-shot
MIT-BIH	Anonymous	55.03 ± 9.40	63.95 ± 13.80	64.15 ± 5.32
	Semantic prior	**57.05 ± 10.96**	**66.62 ± 6.54**	**68.87 ± 11.23**
	Δ	+2.02	+2.67	+4.72
CinC2017	Anonymous	32.80 ± 9.08	**36.98 ± 4.27**	34.13 ± 9.73
	Semantic prior	**34.72 ± 3.15**	35.24 ± 1.24	**38.86 ± 7.00**
	Δ	+1.92	-1.74	+4.73

4.4 Qualitative and Representation Analysis

The preceding results quantify the contribution of each component. We further examine post-hoc qualitative and representation-level evidence after few-shot fine-tuning on

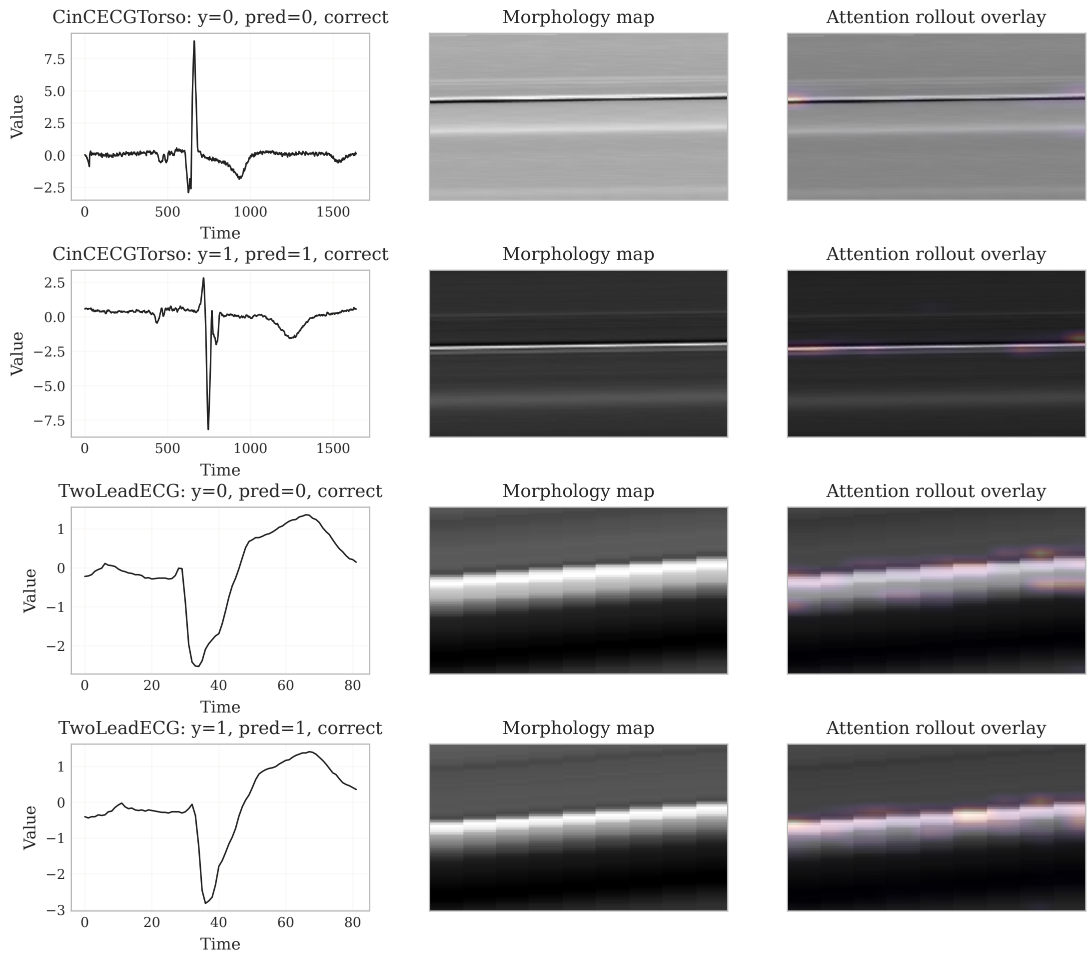

Fig. 4. Attention-rollout analysis of the morphology branch. Each row shows a normalized raw waveform, the corresponding morphology map, and the attention-rollout overlay from the frozen DINOv2 encoder. Highlighted regions indicate morphology-map areas receiving higher aggregated attention.

held-out query examples. These analyses complement the quantitative ablations and are not used for training, hyperparameter selection, or model selection.

4.4.1 Morphology-Branch Attention

Figure 4 visualizes attention rollout [55] on the frozen DINOv2 morphology encoder. For each query sample, the raw waveform is shown together with the morphology map constructed from overlapping local windows and the corresponding attention overlay. High-response regions usually appear near high-contrast structures in the morphology map, which are associated with salient local peaks, valleys, or cross-window shape transitions in the original signal. This pattern indicates that the frozen visual encoder attends to structured waveform morphology exposed by the map representation. Since attention rollout aggregates internal transformer attention, we interpret it as qualitative evidence of encoder focus rather than causal importance of individual timestamps or image pixels.

4.4.2 Representation Structure and Branch Complementarity

Figure 5 compares the temporal pooled representation, morphology pooled representation, and final LLM decision state using t-SNE [56]. As a low-dimensional projection, t-SNE is used qualitatively to compare how different representations organize the same query samples. The temporal and morphology views form visibly different neighborhood structures, indicating that they emphasize different aspects of the signal, namely chronological dynamics and local waveform shape. After fusion and language-side processing, the final decision states show clearer class organization in the representative visualization. The branch-correctness bars provide a complementary sample-level view: temporal-only and morphology-only inference do not fail on exactly the same query examples, and fused dual-view inference recovers part of the errors made by either single branch. Together with the quantitative ablation in Table 6, where removing the morphology pathway lowers the average UCR accuracy, these analyses support the interpretation that the two views provide complementary evidence for few-shot classification.

Figure 6 further reports mutual k-nearest-neighbor alignment among temporal, morphology, and fused decision representations. For two representation spaces, the score is the fraction of shared k nearest neighbors for the same query sample, averaged over query samples and representative datasets. The lower temporal–morphology alignment

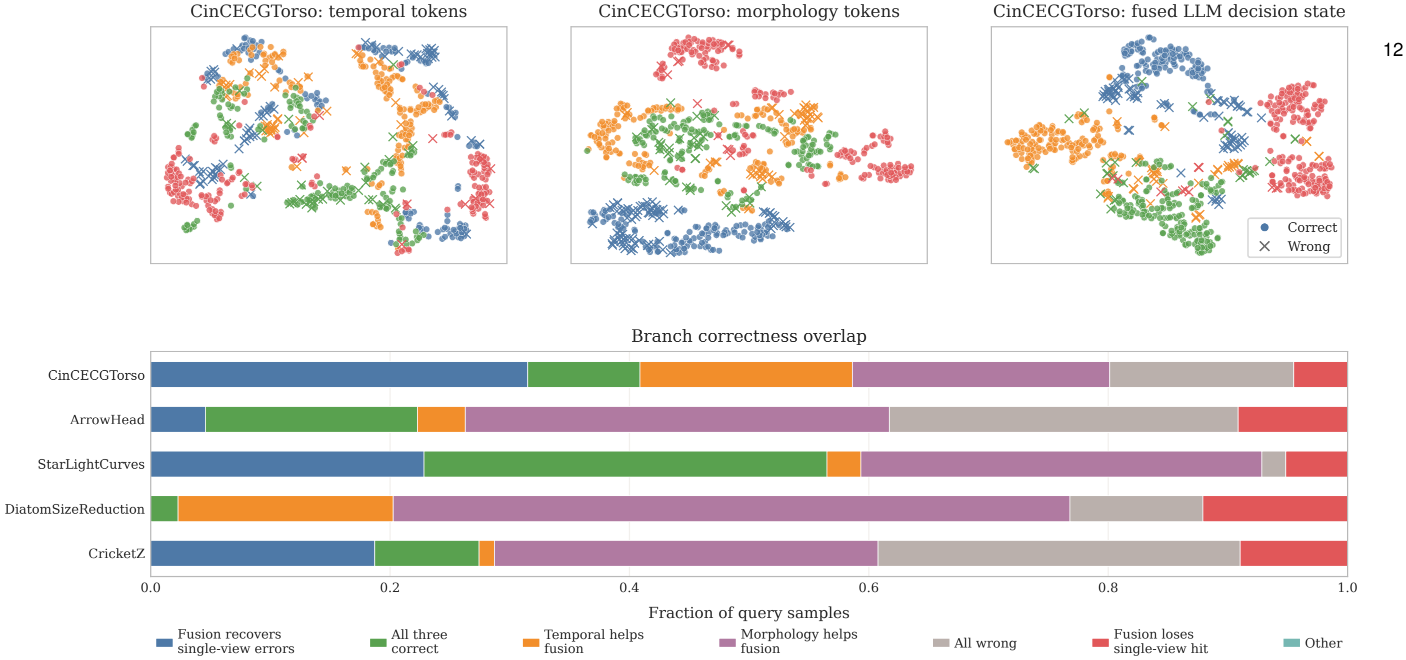

Fig. 5. Representation analysis. The top row gives t-SNE visualizations [56] of temporal, morphology, and fused LLM decision representations. Points are colored by ground-truth class, and misclassified samples are marked separately. The stacked bars compare correctness overlap among temporal-only, morphology-only, and fused dual-view inference on the same fine-tuned checkpoint.

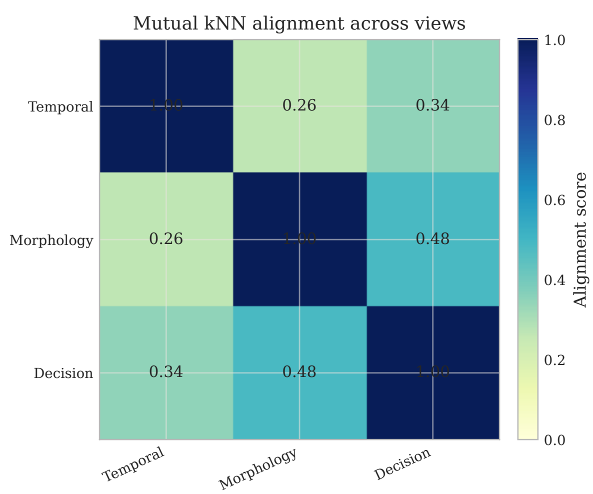

Fig. 6. Mutual k-nearest-neighbor alignment among temporal, morphology, and fused decision representations. Lower temporal–morphology alignment indicates complementary neighborhood structures, while the fused decision state remains aligned with both branches.

indicates different local neighborhood structures, while the higher branch–fused alignment shows that the final decision state retains information from both views. This diagnostic measures representation overlap, not classification accuracy.

Additional t-SNE plots are provided in Appendix C.1.

4.5 Dataset-Level Analysis

4.5.1 Win–Tie–Loss Analysis

We next examine whether the aggregate gains are distributed across datasets. Figure 7 shows win–tie–loss statistics against representative baselines. TimeMorph wins on substantially more datasets than it loses against PATCHTST, TIMESNET, and GPT4TS. The comparison with specialized CNN and metric-learning methods is closer, but TimeMorph still achieves the strongest overall average and rank.

4.5.2 Per-Dataset Accuracy Distribution

Figure 8 reports the per-dataset accuracy distributions of representative baselines and TimeMorph. TimeMorph has the highest mean accuracy in all four shot settings, with a distribution that is especially separated from the Transformer-style and LLM-style baselines. The comparison with specialized CNN and metric-learning baselines is closer, matching the aggregate results in Table 4: TimeMorph provides broad gains over generic sequence models and remains competitive with specialized few-shot classifiers.

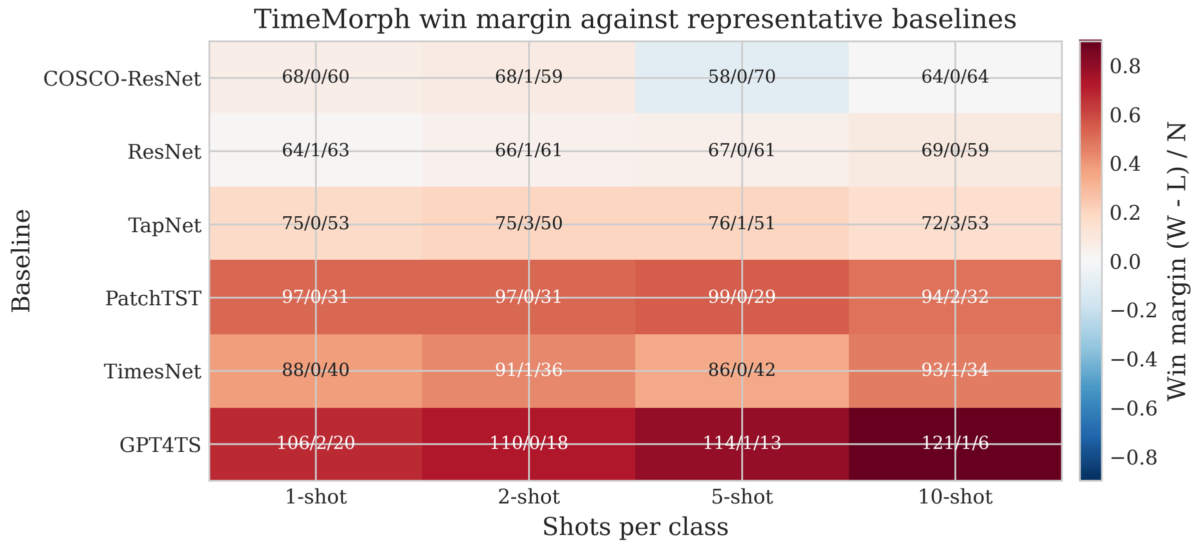

Fig. 7. Win–tie–loss heatmap against representative UCR baselines. Each cell reports wins/ties/losses over 128 datasets, with color indicating the normalized win margin $(W - L)/N$.

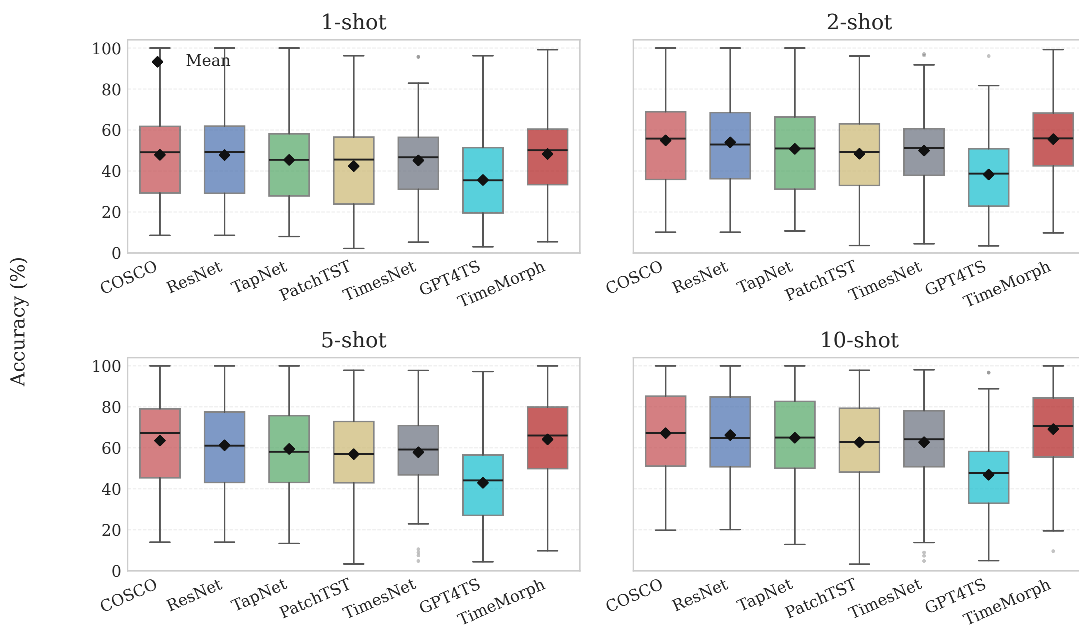

Fig. 8. Per-dataset accuracy distribution on the 128 UCR datasets. Each box summarizes one method's accuracy across datasets for a fixed shot setting; diamonds mark mean accuracies.

4.6 Efficiency and Practicality

Table 9 reports efficiency on one UCR task and two external semantic-label tasks under the 10-shot protocol. Although TimeMorph uses a large frozen LLM backbone, fine-tuning updates only about 12.2M parameters through LoRA and task-specific class-token components. The LLM interface increases memory and latency relative to compact CNN baselines, but it remains parameter-efficient and gives the best accuracy on all three representative tasks. Additional UCR efficiency statistics, including throughput and memory usage, are reported in Appendix B.3.

TABLE 9
Efficiency and accuracy comparison on representative 10-shot tasks. Accuracy is averaged over five runs for GunPoint and ten runs for MIT-BIH and CinC2017. Peak memory and inference latency are measured on a single RTX 3090 under the same evaluation setting. Updated parameters count only parameters fine-tuned on the support set. †TimeMorph uses gradient checkpointing on CinC2017; PatchTST exceeds the 24GB memory budget under the same setting.

Model	Updated params	GunPoint			MIT-BIH			CinC2017		
		Acc.	Peak GB	ms/ex.	Acc.	Peak GB	ms/ex.	Acc.	Peak GB	ms/ex.
TimeMorph	12.2M	**97.47**	13.24	31.07	**57.05**	15.05	38.02	**34.72**	12.21†	54.44
COSCO-ResNet	661K	91.73	0.06	0.19	32.33	0.16	0.21	31.41	1.34	0.45
ResNet	628K	85.73	0.06	0.14	37.97	0.16	0.20	30.95	1.29	0.43
TapNet	911K	82.13	0.09	0.16	31.94	0.13	0.23	30.05	1.17	1.94
PatchTST	597K	77.60	0.13	0.31	38.94	0.27	0.29	26.07	OOM	–
GPT4TS	1.1M	62.00	0.38	0.58	32.92	0.45	0.59	30.80	1.19	0.59

5 Conclusion

This paper presented **TimeMorph**, a dual-view LLM-based framework for few-shot time-series classification. TimeMorph addresses three key issues in adapting pretrained language models to scarce-label time-series data: preserving complementary temporal and morphological evidence, aligning continuous time-series representations with the LLM embedding space, and enforcing valid closed-set predictions. To this end, TimeMorph combines a temporal branch for chronological dynamics with a morphology branch that exposes waveform structures through visual pseudo-images, projects the fused tokens into a frozen causal LLM through staged cross-modal transfer, and replaces open-ended label generation with supervised class-token scoring over the valid label set. When informative class names are available, semantic label priors further regularize the class-token space without relying on unstable multi-token generation.

Extensive experiments on the full UCR few-shot benchmark and external semantic-label datasets show that TimeMorph consistently improves over task-specific, Transformer-based, and LLM/foundation-model baselines. The ablation studies further confirm that staged transfer, the LLM pathway, dual-view modeling, and semantic label priors contribute to the final performance. These results suggest that LLMs can be effectively fine-tuned for few-shot TSC when the modality interface and prediction space are carefully designed. Future work will extend TimeMorph to broader multivariate, streaming, and domain-specific time-series scenarios while further improving fine-tuning efficiency.

APPENDIX A
FULL UCR BENCHMARK RESULTS

This appendix reports the complete per-dataset few-shot benchmark results for all 128 UCR datasets. Each cell is shown as mean ± standard deviation over five runs, and the summary rows at the end of each table report average accuracy, average rank, and the number of datasets attaining the best mean accuracy (counting ties).

TABLE 10: Per-dataset 1-shot classification results on 128 UCR datasets. Each value is reported as mean ± standard deviation over 5 runs.

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	TimeMorph
ACSF1	**53.80 ± 4.97**	42.00 ± 4.95	48.40 ± 7.30	24.00 ± 4.24	35.60 ± 3.21	27.67 ± 7.02	24.33 ± 9.71	28.33 ± 2.52	27.80 ± 7.09	31.67 ± 9.81	18.60 ± 4.51	44.00 ± 5.20
Adiac	33.81 ± 1.23	12.84 ± 3.09	33.81 ± 1.23	8.85 ± 2.36	11.97 ± 1.69	21.40 ± 4.24	34.53 ± 1.77	8.10 ± 1.92	40.82 ± 3.61	24.21 ± 3.24	3.02 ± 1.05	**45.17 ± 5.28**
AllGestureWiimoteX	25.71 ± 4.01	**26.86 ± 2.52**	24.43 ± 2.55	15.57 ± 2.95	21.86 ± 4.59	20.71 ± 2.87	11.33 ± 1.19	16.76 ± 2.52	13.43 ± 1.74	18.90 ± 1.08	21.86 ± 4.59	25.11 ± 4.65
AllGestureWiimoteY	**26.37 ± 2.89**	26.17 ± 2.88	22.71 ± 4.62	17.49 ± 2.02	25.17 ± 5.50	21.81 ± 2.78	12.29 ± 1.27	20.29 ± 1.98	13.06 ± 1.34	21.33 ± 2.38	25.17 ± 5.50	25.54 ± 4.42
AllGestureWiimoteZ	**27.11 ± 1.95**	22.66 ± 4.13	25.49 ± 2.57	18.26 ± 2.01	19.71 ± 2.87	19.24 ± 1.57	13.29 ± 0.43	18.76 ± 2.79	14.14 ± 1.72	19.71 ± 1.79	19.71 ± 2.87	22.26 ± 3.96
ArrowHead	47.66 ± 8.94	46.63 ± 9.45	48.00 ± 7.74	52.00 ± 6.38	49.71 ± 1.40	47.62 ± 9.46	38.29 ± 4.00	50.86 ± 1.71	35.89 ± 2.60	50.86 ± 6.05	41.49 ± 12.34	**52.57 ± 3.43**
BME	49.60 ± 2.85	50.80 ± 2.13	49.73 ± 3.42	56.67 ± 3.43	62.93 ± 5.35	55.33 ± 7.86	51.56 ± 3.15	61.33 ± 4.06	55.87 ± 7.71	58.89 ± 6.19	35.73 ± 3.22	**66.40 ± 3.88**
Beef	36.00 ± 6.41	31.33 ± 5.06	27.33 ± 7.96	38.67 ± 9.89	32.00 ± 7.67	40.00 ± 8.82	34.44 ± 6.94	**41.11 ± 1.92**	30.67 ± 6.83	40.00 ± 6.67	31.33 ± 9.31	34.00 ± 2.79
BeetleFly	55.00 ± 24.24	60.00 ± 25.00	61.00 ± 24.85	67.00 ± 4.47	61.00 ± 9.62	60.00 ± 13.23	55.00 ± 5.00	61.67 ± 11.55	48.00 ± 2.74	60.00 ± 13.23	**70.00 ± 11.18**	**70.00 ± 11.73**
BirdChicken	62.00 ± 8.37	**71.00 ± 9.62**	70.00 ± 17.68	56.00 ± 4.18	56.00 ± 4.18	51.67 ± 2.89	51.67 ± 12.58	53.33 ± 5.77	55.00 ± 5.00	51.67 ± 7.64	59.00 ± 9.62	68.00 ± 14.83
CBF	81.67 ± 6.47	**87.62 ± 2.73**	71.71 ± 7.91	54.02 ± 3.44	66.38 ± 7.40	55.85 ± 2.16	33.63 ± 1.20	58.00 ± 5.61	36.89 ± 2.37	54.78 ± 1.68	35.20 ± 3.75	72.48 ± 4.85
Car	37.00 ± 4.62	39.00 ± 4.80	41.67 ± 9.28	38.67 ± 11.63	27.33 ± 9.90	42.22 ± 5.09	24.44 ± 3.47	43.89 ± 5.85	27.00 ± 4.15	40.00 ± 7.26	30.00 ± 9.65	**55.67 ± 8.94**
Chinatown	66.53 ± 19.14	63.73 ± 26.98	62.51 ± 27.20	67.41 ± 23.86	67.35 ± 11.32	77.07 ± 16.93	81.83 ± 2.19	77.94 ± 9.20	77.61 ± 8.88	75.41 ± 7.92	56.68 ± 20.52	**93.59 ± 3.92**
ChlorineConcentration	34.61 ± 2.26	35.41 ± 4.36	34.04 ± 2.01	33.26 ± 2.96	35.54 ± 3.20	33.64 ± 3.96	32.92 ± 4.26	33.42 ± 5.09	33.78 ± 2.74	34.11 ± 3.50	**39.93 ± 15.64**	32.61 ± 3.30
CinCECGTorso	37.20 ± 9.73	36.03 ± 6.56	36.51 ± 8.82	40.06 ± 3.86	35.72 ± 7.09	44.86 ± 6.03	35.36 ± 5.77	**50.53 ± 4.66**	34.97 ± 4.68	45.68 ± 5.50	42.49 ± 6.76	48.74 ± 6.47
Coffee	78.57 ± 9.78	**92.86 ± 8.38**	84.29 ± 13.02	72.86 ± 14.42	74.29 ± 18.80	63.10 ± 10.91	65.48 ± 7.43	71.43 ± 16.37	76.43 ± 4.07	64.29 ± 7.14	51.43 ± 4.07	75.71 ± 13.69
Computers	57.52 ± 7.64	**60.40 ± 6.13**	59.12 ± 4.47	47.12 ± 5.87	52.64 ± 10.26	50.93 ± 1.29	48.80 ± 0.69	51.87 ± 2.57	48.96 ± 2.79	52.27 ± 3.23	45.20 ± 5.79	56.40 ± 5.77
CricketX	26.15 ± 3.29	26.10 ± 3.67	20.56 ± 4.67	17.28 ± 1.96	18.15 ± 4.50	20.34 ± 0.39	8.89 ± 0.82	17.52 ± 1.26	11.38 ± 2.35	20.68 ± 0.74	7.28 ± 1.05	**26.26 ± 5.29**
CricketY	**26.97 ± 3.03**	24.46 ± 4.25	22.72 ± 1.74	18.56 ± 4.13	17.90 ± 3.61	22.82 ± 1.56	10.34 ± 1.89	20.94 ± 3.72	11.23 ± 0.80	21.20 ± 3.98	9.54 ± 2.33	23.33 ± 5.19
CricketZ	**32.51 ± 2.32**	28.72 ± 2.15	26.56 ± 4.19	19.64 ± 2.73	22.26 ± 3.91	20.34 ± 4.11	11.03 ± 1.54	20.00 ± 2.31	12.26 ± 1.25	20.34 ± 2.57	7.64 ± 1.05	29.28 ± 4.82
Crop	25.10 ± 4.94	28.14 ± 2.90	25.10 ± 4.94	10.23 ± 2.66	17.27 ± 1.43	27.73 ± 2.89	24.10 ± 4.30	26.19 ± 2.96	27.27 ± 3.58	30.31 ± 5.00	3.80 ± 0.58	**36.48 ± 3.38**
DiatomSizeReduction	79.67 ± 5.51	55.36 ± 19.14	84.12 ± 4.82	**89.22 ± 8.91**	49.61 ± 24.86	75.71 ± 7.74	55.01 ± 11.77	56.43 ± 20.40	50.07 ± 12.66	76.47 ± 11.70	41.44 ± 13.25	85.82 ± 8.55
DistalPhalanxOutlineAgeGroup	43.02 ± 17.16	43.02 ± 18.34	43.60 ± 14.92	42.73 ± 19.01	46.76 ± 16.49	45.80 ± 19.81	46.28 ± 21.27	**56.59 ± 9.36**	50.94 ± 19.66	46.28 ± 20.78	35.68 ± 14.61	42.01 ± 16.44
DistalPhalanxOutlineCorrect	62.10 ± 1.32	57.32 ± 9.22	**63.04 ± 2.44**	57.54 ± 11.13	53.55 ± 11.63	53.99 ± 14.78	53.26 ± 12.27	53.62 ± 11.97	55.14 ± 10.52	53.26 ± 12.97	56.74 ± 7.30	60.58 ± 2.98
DistalPhalanxTW	51.22 ± 5.15	44.03 ± 11.70	53.38 ± 4.02	43.74 ± 5.17	51.22 ± 6.58	45.32 ± 9.89	42.69 ± 4.40	38.13 ± 7.51	48.20 ± 5.84	47.48 ± 5.62	18.56 ± 14.51	**54.53 ± 1.72**
DodgerLoopDay	32.50 ± 7.34	25.00 ± 4.05	29.75 ± 7.68	**41.50 ± 2.05**	26.00 ± 6.58	40.42 ± 4.73	20.42 ± 1.91	32.92 ± 3.15	21.50 ± 4.45	40.42 ± 7.94	26.00 ± 6.58	34.50 ± 4.38
DodgerLoopGame	52.17 ± 6.30	55.07 ± 17.38	49.13 ± 8.90	**60.87 ± 12.35**	59.86 ± 22.05	51.45 ± 9.96	51.21 ± 5.14	56.52 ± 10.82	50.43 ± 2.44	51.45 ± 9.59	59.86 ± 22.05	56.96 ± 11.69
DodgerLoopWeekend	84.49 ± 8.37	74.78 ± 12.27	83.19 ± 10.24	93.62 ± 3.81	**96.23 ± 1.19**	95.65 ± 1.26	63.77 ± 4.41	94.44 ± 0.42	68.41 ± 4.57	93.72 ± 5.86	**96.23 ± 1.19**	85.36 ± 4.30
ECG200	63.00 ± 12.61	**72.00 ± 6.75**	66.20 ± 8.32	71.80 ± 4.44	61.60 ± 12.82	61.33 ± 10.97	65.00 ± 11.79	63.00 ± 20.66	64.20 ± 14.29	68.67 ± 7.09	51.40 ± 12.93	60.60 ± 6.91
ECG5000	54.22 ± 21.00	50.05 ± 20.59	61.53 ± 13.66	62.80 ± 19.19	53.35 ± 20.14	**69.07 ± 12.52**	54.44 ± 9.79	57.20 ± 16.09	62.35 ± 12.99	68.84 ± 13.99	19.70 ± 25.30	67.18 ± 10.57
ECGFiveDays	58.65 ± 7.69	57.24 ± 6.51	54.80 ± 13.00	57.63 ± 2.63	**62.21 ± 8.35**	57.99 ± 4.84	54.86 ± 3.17	55.87 ± 1.56	54.08 ± 6.73	58.46 ± 5.08	51.54 ± 6.38	61.77 ± 6.61
EOGHorizontalSignal	20.11 ± 2.01	22.38 ± 1.86	22.60 ± 2.77	22.43 ± 4.01	21.22 ± 3.15	20.63 ± 4.63	15.29 ± 2.63	23.57 ± 5.45	10.39 ± 2.26	22.01 ± 4.37	27.24 ± 2.99	**29.06 ± 4.16**
EOGVerticalSignal	11.27 ± 3.67	14.09 ± 3.23	13.26 ± 3.09	24.20 ± 3.27	23.76 ± 3.73	23.39 ± 2.79	11.23 ± 0.70	24.86 ± 4.17	11.27 ± 3.83	23.11 ± 4.07	19.06 ± 6.08	**27.35 ± 1.90**
Earthquakes	50.07 ± 8.55	51.08 ± 6.33	50.94 ± 6.46	49.35 ± 5.05	45.61 ± 20.94	48.68 ± 19.16	49.40 ± 1.10	49.40 ± 14.03	47.05 ± 5.90	47.00 ± 24.65	46.04 ± 13.71	**55.54 ± 11.57**
ElectricDevices	28.83 ± 9.01	32.82 ± 12.31	28.35 ± 12.24	17.59 ± 1.67	27.21 ± 7.38	21.93 ± 4.54	17.11 ± 1.87	15.12 ± 2.81	17.04 ± 2.05	15.61 ± 7.10	17.06 ± 3.10	**33.14 ± 10.96**
EthanolLevel	24.44 ± 1.30	25.40 ± 0.85	25.28 ± 1.57	24.92 ± 1.50	23.80 ± 1.65	26.13 ± 0.58	26.53 ± 1.55	26.67 ± 1.29	25.32 ± 1.50	25.80 ± 1.39	24.76 ± 1.40	**26.68 ± 1.86**
FaceAll	**42.17 ± 4.79**	32.36 ± 9.47	**42.17 ± 4.79**	28.66 ± 9.09	19.36 ± 3.07	29.11 ± 4.19	33.12 ± 5.46	25.94 ± 1.75	34.05 ± 3.40	32.92 ± 2.10	8.64 ± 3.46	32.84 ± 6.86
FaceFour	68.64 ± 10.56	47.50 ± 11.12	56.36 ± 4.30	64.32 ± 4.73	54.09 ± 8.34	70.08 ± 4.30	52.65 ± 3.99	67.05 ± 7.10	51.36 ± 4.28	**73.11 ± 3.99**	42.95 ± 6.60	54.77 ± 8.25
FacesUCR	**54.83 ± 5.21**	37.63 ± 5.89	43.65 ± 3.88	35.69 ± 7.47	20.81 ± 1.95	36.24 ± 3.26	41.98 ± 2.95	40.05 ± 2.80	41.87 ± 3.09	42.63 ± 2.58	16.92 ± 7.54	50.37 ± 2.98
FiftyWords	13.27 ± 1.89	12.79 ± 3.45	13.27 ± 1.89	32.70 ± 4.33	19.60 ± 4.90	35.90 ± 3.31	3.44 ± 1.59	**38.97 ± 6.44**	11.30 ± 2.86	36.70 ± 4.41	3.74 ± 2.40	36.48 ± 2.66
Fish	**49.03 ± 5.96**	21.26 ± 3.60	45.71 ± 1.71	36.00 ± 3.98	23.20 ± 3.80	32.57 ± 4.57	16.95 ± 2.93	30.29 ± 2.62	21.71 ± 2.77	31.81 ± 4.76	20.80 ± 5.10	34.74 ± 4.91
FordA	59.23 ± 9.68	57.68 ± 10.00	**59.83 ± 8.76**	50.73 ± 1.01	56.70 ± 8.04	50.30 ± 0.33	49.57 ± 1.42	49.72 ± 0.63	49.68 ± 1.11	49.92 ± 0.80	50.61 ± 1.68	50.71 ± 0.98
FordB	52.54 ± 6.44	52.64 ± 6.88	52.40 ± 4.53	50.12 ± 1.83	**53.60 ± 4.09**	49.96 ± 0.94	50.53 ± 0.72	50.21 ± 0.80	50.81 ± 0.65	49.88 ± 0.98	50.05 ± 1.16	49.75 ± 2.56
FreezerRegularTrain	61.33 ± 21.55	58.22 ± 20.21	56.76 ± 19.80	61.31 ± 21.84	57.07 ± 18.69	59.91 ± 27.91	48.78 ± 1.68	53.81 ± 23.62	49.55 ± 8.37	60.27 ± 28.68	61.45 ± 8.09	**61.90 ± 22.25**
FreezerSmallTrain	67.85 ± 12.31	66.48 ± 12.45	67.42 ± 12.38	67.47 ± 12.29	63.19 ± 10.91	**73.32 ± 3.65**	55.33 ± 0.97	69.60 ± 5.96	57.00 ± 2.79	73.03 ± 4.44	65.92 ± 7.33	66.53 ± 10.64
Fungi	93.23 ± 4.00	35.81 ± 4.97	64.62 ± 2.56	68.39 ± 5.78	67.20 ± 19.03	**95.70 ± 0.93**	73.66 ± 2.69	86.74 ± 1.89	75.16 ± 1.03	**95.70 ± 0.93**	14.09 ± 7.48	79.35 ± 5.30
GestureMidAirD1	17.08 ± 4.26	19.38 ± 4.43	17.08 ± 4.26	34.31 ± 3.87	19.08 ± 6.45	31.54 ± 2.04	17.44 ± 0.89	34.62 ± 4.07	21.85 ± 4.63	**34.87 ± 2.47**	19.08 ± 6.45	32.00 ± 3.94
GestureMidAirD2	13.23 ± 5.14	17.08 ± 3.90	13.23 ± 5.14	29.08 ± 4.19	20.46 ± 5.77	29.74 ± 3.20	11.79 ± 2.22	29.23 ± 5.38	17.38 ± 5.95	**36.41 ± 0.89**	20.46 ± 5.77	33.85 ± 3.03
GestureMidAirD3	8.62 ± 1.84	9.08 ± 2.52	8.62 ± 1.84	20.00 ± 3.17	10.92 ± 1.26	**24.87 ± 4.24**	9.49 ± 1.60	23.08 ± 0.77	14.77 ± 4.47	22.56 ± 2.91	10.92 ± 1.26	18.46 ± 2.61
GesturePebbleZ1	49.07 ± 5.03	38.37 ± 6.03	**53.26 ± 6.97**	46.16 ± 13.08	23.60 ± 6.95	40.50 ± 15.60	20.93 ± 3.54	42.44 ± 7.05	18.95 ± 2.08	43.99 ± 8.75	23.60 ± 6.95	35.81 ± 4.48
GesturePebbleZ2	55.19 ± 11.38	42.91 ± 15.92	**57.47 ± 8.89**	48.35 ± 13.33	21.01 ± 3.48	47.68 ± 13.93	18.35 ± 4.94	43.88 ± 16.17	16.20 ± 5.17	55.06 ± 14.60	21.01 ± 3.48	33.29 ± 5.85
GunPoint	55.47 ± 4.79	58.13 ± 9.13	55.87 ± 5.32	53.20 ± 9.11	56.40 ± 7.58	**64.00 ± 8.08**	58.89 ± 10.59	63.33 ± 4.81	58.93 ± 8.45	62.89 ± 3.01	52.40 ± 11.44	58.00 ± 7.79
GunPointAgeSpan	49.94 ± 8.02	48.67 ± 3.53	49.11 ± 3.99	60.82 ± 22.09	58.73 ± 19.72	59.70 ± 21.56	52.11 ± 7.60	59.07 ± 20.81	52.41 ± 6.32	**61.92 ± 19.97**	48.10 ± 2.61	60.51 ± 10.32
GunPointMaleVersusFemale	63.99 ± 11.97	63.29 ± 11.59	64.30 ± 12.20	54.56 ± 4.47	56.65 ± 3.27	55.91 ± 6.97	50.95 ± 7.17	55.59 ± 9.92	50.19 ± 6.98	56.75 ± 8.76	58.04 ± 11.38	**81.01 ± 10.12**
GunPointOldVersusYoung	**92.38 ± 10.44**	89.97 ± 13.74	90.35 ± 13.23	53.90 ± 25.42	53.33 ± 25.91	58.31 ± 30.15	54.81 ± 4.33	57.46 ± 29.14	54.79 ± 8.98	58.10 ± 28.87	78.41 ± 20.62	57.97 ± 10.78
Ham	50.10 ± 9.32	51.24 ± 7.01	47.81 ± 11.28	55.24 ± 11.13	49.33 ± 2.81	48.89 ± 5.25	53.65 ± 3.06	51.43 ± 7.44	**55.81 ± 6.79**	50.16 ± 4.89	51.81 ± 10.55	52.76 ± 9.30
HandOutlines	46.81 ± 15.76	58.86 ± 12.82	55.84 ± 9.47	70.59 ± 7.20	75.68 ± 8.02	71.17 ± 7.55	51.08 ± 2.34	**76.76 ± 10.41**	52.81 ± 15.40	74.86 ± 11.22	61.19 ± 14.73	61.73 ± 6.31
Haptics	23.96 ± 5.22	22.66 ± 2.12	22.86 ± 4.94	**26.49 ± 6.85**	22.66 ± 4.38	25.11 ± 9.19	19.16 ± 1.81	23.16 ± 7.32	23.44 ± 3.46	25.43 ± 9.51	25.65 ± 5.26	25.65 ± 4.99
Herring	49.69 ± 6.39	49.69 ± 10.50	50.94 ± 11.51	48.12 ± 8.44	49.69 ± 7.44	47.40 ± 10.40	**51.04 ± 2.39**	49.48 ± 9.55	48.75 ± 8.44	48.44 ± 9.50	48.12 ± 10.39	50.00 ± 10.00
HouseTwenty	**80.84 ± 7.67**	74.62 ± 5.52	68.40 ± 16.13	50.25 ± 3.68	57.65 ± 8.25	68.63 ± 3.50	49.30 ± 7.15	61.62 ± 2.57	46.05 ± 6.81	66.67 ± 1.28	53.61 ± 8.52	62.18 ± 8.04
InlineSkate	16.40 ± 1.60	15.64 ± 0.76	18.00 ± 0.73	15.89 ± 3.32	14.69 ± 2.92	18.06 ± 2.11	15.15 ± 0.73	18.00 ± 2.36	15.02 ± 3.34	**18.67 ± 3.56**	14.87 ± 3.22	17.13 ± 2.24
InsectEPGRegularTrain	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	43.53 ± 11.83	50.44 ± 13.06	48.19 ± 5.92	42.03 ± 6.13	45.25 ± 3.22	46.27 ± 8.59	49.26 ± 6.69	87.15 ± 20.64	55.34 ± 14.47
InsectEPGSmallTrain	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	47.95 ± 6.69	55.10 ± 5.49	51.14 ± 8.16	39.36 ± 1.75	52.88 ± 6.50	37.67 ± 14.87	52.48 ± 11.47	85.78 ± 19.79	50.76 ± 7.86
InsectWingbeatSound	16.18 ± 3.00	14.23 ± 3.87	14.35 ± 2.82	39.88 ± 4.03	22.79 ± 2.65	43.42 ± 0.26	17.56 ± 4.92	43.16 ± 3.28	21.14 ± 2.99	**45.61 ± 5.13**	12.52 ± 3.37	25.27 ± 2.20
ItalyPowerDemand	61.83 ± 17.15	62.29 ± 17.84	64.74 ± 16.58	55.32 ± 7.10	60.35 ± 16.10	69.81 ± 17.24	68.35 ± 9.65	**78.33 ± 24.61**	68.98 ± 15.67	71.62 ± 21.13	47.31 ± 2.37	58.52 ± 15.27
LargeKitchenAppliances	48.48 ± 12.88	52.85 ± 11.74	46.72 ± 15.63	33.39 ± 3.38	**54.88 ± 8.28**	40.09 ± 11.55	34.93 ± 5.70	32.53 ± 3.23	31.47 ± 3.69	41.69 ± 12.56	33.81 ± 2.87	45.44 ± 5.60

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TABLE 10: Per-dataset 1-shot classification results on 128 UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	TimeMorph
Lightning2	50.49 ± 6.80	55.08 ± 10.15	51.80 ± 4.27	56.72 ± 11.39	51.80 ± 12.52	58.47 ± 11.16	48.63 ± 3.41	57.38 ± 11.36	51.48 ± 5.26	59.02 ± 8.67	55.08 ± 13.25	**59.67 ± 8.87**
Lightning7	44.93 ± 4.48	44.38 ± 4.60	40.82 ± 7.47	43.01 ± 4.18	42.19 ± 7.34	53.42 ± 10.70	15.07 ± 6.28	45.66 ± 6.76	18.63 ± 4.40	**56.16 ± 8.33**	24.11 ± 7.72	41.64 ± 7.60
Mallat	64.03 ± 14.52	50.42 ± 9.84	64.49 ± 9.43	80.37 ± 4.76	56.96 ± 10.09	**82.32 ± 3.73**	54.67 ± 2.15	75.96 ± 4.82	64.55 ± 4.53	75.72 ± 2.60	37.30 ± 5.46	68.24 ± 7.54
Meat	42.00 ± 9.96	44.00 ± 10.18	58.33 ± 10.99	**77.67 ± 17.26**	52.67 ± 11.94	46.11 ± 9.18	38.33 ± 8.66	33.33 ± 0.00	36.00 ± 2.53	66.11 ± 20.97	33.67 ± 0.75	55.67 ± 9.62
MedicalImages	27.92 ± 5.84	28.45 ± 4.75	24.95 ± 5.95	26.34 ± 8.67	24.16 ± 4.26	22.32 ± 5.35	26.84 ± 12.59	23.86 ± 6.12	**28.53 ± 8.23**	23.16 ± 6.10	4.21 ± 2.27	27.66 ± 4.41
MelbournePedestrian	**51.29 ± 5.58**	44.27 ± 4.81	47.47 ± 6.50	9.85 ± 0.23	31.99 ± 4.95	21.74 ± 7.37	46.44 ± 8.26	40.62 ± 4.65	43.40 ± 7.11	43.73 ± 5.33	31.99 ± 4.95	39.16 ± 5.22
MiddlePhalanxOutlineAgeGroup	38.83 ± 7.74	37.53 ± 10.57	38.05 ± 7.83	34.03 ± 9.03	35.97 ± 8.21	37.45 ± 11.78	29.87 ± 4.55	38.53 ± 15.64	33.90 ± 7.76	32.90 ± 7.86	**41.69 ± 18.94**	39.48 ± 7.40
MiddlePhalanxOutlineCorrect	53.68 ± 13.63	52.10 ± 14.21	54.43 ± 13.33	53.75 ± 17.89	56.49 ± 9.58	**58.42 ± 17.26**	55.56 ± 15.16	54.07 ± 23.81	54.50 ± 10.22	57.50 ± 21.09	52.85 ± 12.08	51.68 ± 5.00
MiddlePhalanxTW	34.03 ± 16.27	44.68 ± 7.76	32.47 ± 17.97	30.52 ± 10.75	32.34 ± 10.78	29.00 ± 15.96	38.96 ± 3.37	**46.10 ± 3.62**	35.45 ± 13.57	42.21 ± 5.07	15.19 ± 11.64	37.14 ± 16.84
MixedShapesRegularTrain	45.36 ± 5.29	40.59 ± 8.80	43.58 ± 10.39	47.60 ± 10.74	39.59 ± 5.62	43.05 ± 15.67	17.31 ± 1.15	45.53 ± 11.76	25.71 ± 3.77	42.76 ± 16.81	45.03 ± 12.47	**56.06 ± 5.34**
MixedShapesSmallTrain	52.78 ± 8.02	39.55 ± 6.53	48.84 ± 9.40	53.90 ± 7.66	45.57 ± 7.91	52.81 ± 6.78	17.32 ± 0.61	53.84 ± 4.25	24.17 ± 3.24	53.31 ± 5.69	48.73 ± 8.44	**63.30 ± 3.34**
MoteStrain	62.99 ± 13.96	60.67 ± 12.58	64.35 ± 11.70	63.40 ± 15.50	63.27 ± 13.76	68.45 ± 11.94	56.26 ± 12.55	63.82 ± 16.98	61.65 ± 9.57	**68.66 ± 11.89**	57.99 ± 9.98	54.36 ± 12.77
NonInvasiveFetalECGThorax1	19.05 ± 4.42	8.04 ± 2.52	19.05 ± 4.42	38.23 ± 1.81	12.07 ± 7.41	36.76 ± 3.30	12.57 ± 2.38	28.62 ± 3.97	14.44 ± 2.07	**42.58 ± 4.62**	5.58 ± 2.48	39.16 ± 2.28
NonInvasiveFetalECGThorax2	20.69 ± 5.30	9.71 ± 2.65	20.69 ± 5.30	47.48 ± 2.72	14.81 ± 5.33	46.67 ± 0.18	20.97 ± 3.22	34.59 ± 7.78	23.69 ± 1.73	**49.89 ± 4.96**	8.30 ± 1.72	47.48 ± 2.06
OSULeaf	**45.62 ± 7.45**	30.41 ± 8.35	38.35 ± 12.00	24.05 ± 2.14	22.15 ± 5.63	27.82 ± 3.13	17.77 ± 1.80	26.03 ± 1.89	20.00 ± 5.09	26.17 ± 1.56	17.60 ± 2.44	29.83 ± 5.82
OliveOil	28.00 ± 23.64	26.67 ± 11.06	47.33 ± 13.21	**65.33 ± 15.20**	36.00 ± 22.66	32.22 ± 16.44	44.44 ± 10.72	28.89 ± 11.71	32.00 ± 9.60	61.11 ± 15.40	27.33 ± 11.88	28.00 ± 14.26
PLAID	15.46 ± 6.16	13.97 ± 4.45	16.95 ± 4.17	17.77 ± 3.34	21.79 ± 0.97	21.54 ± 5.77	13.78 ± 3.08	19.93 ± 5.41	14.41 ± 2.63	20.30 ± 4.75	21.79 ± 0.97	**32.10 ± 5.72**
PhalangesOutlinesCorrect	**54.90 ± 6.98**	51.28 ± 10.93	54.41 ± 7.38	53.40 ± 8.33	52.54 ± 10.38	48.87 ± 4.48	51.01 ± 10.34	53.38 ± 8.94	52.73 ± 9.70	52.49 ± 9.30	54.83 ± 6.32	51.93 ± 9.54
Phoneme	**10.43 ± 1.42**	8.46 ± 1.00	**10.43 ± 1.42**	4.21 ± 0.54	5.61 ± 2.38	6.14 ± 1.81	2.39 ± 1.01	5.87 ± 1.54	3.20 ± 2.06	4.98 ± 3.20	3.42 ± 0.41	6.87 ± 1.15
PickupGestureWiimoteZ	35.60 ± 7.80	44.80 ± 4.82	42.00 ± 5.10	40.80 ± 5.40	36.00 ± 3.16	46.67 ± 9.87	17.33 ± 5.03	39.33 ± 6.43	20.40 ± 4.98	**48.67 ± 8.33**	36.00 ± 3.16	47.20 ± 7.29
PigAirwayPressure	23.94 ± 3.76	9.13 ± 2.66	**38.65 ± 2.92**	3.65 ± 1.47	3.75 ± 1.33	5.29 ± 0.83	4.01 ± 0.73	4.01 ± 1.11	3.27 ± 0.92	4.81 ± 0.96	6.15 ± 1.64	5.48 ± 2.19
PigArtPressure	40.58 ± 5.14	10.10 ± 2.36	**60.10 ± 2.28**	9.90 ± 2.42	2.21 ± 0.80	10.58 ± 2.10	8.01 ± 2.00	10.58 ± 1.73	4.81 ± 0.76	9.94 ± 2.27	15.58 ± 3.05	15.38 ± 4.07
PigCVP	22.50 ± 3.47	14.33 ± 2.79	**50.29 ± 2.90**	5.00 ± 1.25	3.27 ± 0.71	5.93 ± 0.73	3.53 ± 1.39	6.89 ± 0.56	2.02 ± 0.22	4.33 ± 0.83	8.94 ± 1.94	11.44 ± 2.08
Plane	**98.86 ± 2.56**	91.62 ± 6.26	95.43 ± 3.65	87.24 ± 8.67	85.14 ± 9.49	82.86 ± 4.15	78.73 ± 5.25	88.57 ± 9.09	86.48 ± 4.39	88.57 ± 6.67	26.10 ± 13.02	93.90 ± 6.92
PowerCons	76.89 ± 9.68	**80.22 ± 9.41**	78.11 ± 13.88	60.22 ± 11.82	61.00 ± 7.74	62.96 ± 3.70	49.44 ± 7.47	58.33 ± 2.78	51.33 ± 4.56	62.78 ± 6.41	55.22 ± 21.33	60.89 ± 11.09
ProximalPhalanxOutlineAgeGroup	48.49 ± 34.51	48.20 ± 31.13	48.68 ± 30.82	39.61 ± 19.82	39.12 ± 18.60	43.74 ± 32.89	49.59 ± 25.02	**60.81 ± 40.85**	38.15 ± 22.87	46.50 ± 27.14	32.68 ± 16.57	48.98 ± 23.27
ProximalPhalanxOutlineCorrect	54.43 ± 13.05	53.40 ± 15.85	51.20 ± 16.57	56.22 ± 5.47	53.26 ± 10.68	44.67 ± 13.09	46.96 ± 12.63	**58.76 ± 6.63**	52.78 ± 12.27	43.99 ± 6.30	57.59 ± 13.02	52.23 ± 12.03
ProximalPhalanxTW	54.63 ± 10.31	57.56 ± 9.33	58.15 ± 6.64	54.34 ± 11.16	49.76 ± 11.67	**60.81 ± 11.71**	55.93 ± 9.23	38.37 ± 0.75	55.80 ± 7.24	57.07 ± 10.70	17.37 ± 16.23	55.22 ± 6.63
RefrigerationDevices	36.05 ± 6.49	35.68 ± 5.16	37.44 ± 5.37	34.99 ± 2.77	33.97 ± 4.76	**41.16 ± 5.10**	34.58 ± 1.47	34.93 ± 0.92	34.72 ± 0.99	38.84 ± 5.04	32.75 ± 2.24	29.76 ± 11.02
Rock	29.20 ± 8.07	34.40 ± 7.92	28.40 ± 6.54	41.60 ± 16.52	26.40 ± 8.53	38.67 ± 25.32	**57.33 ± 5.03**	44.67 ± 18.90	36.80 ± 8.07	40.00 ± 24.25	36.40 ± 14.93	39.20 ± 13.75
ScreenType	33.92 ± 4.73	34.19 ± 7.39	**34.29 ± 6.61**	33.87 ± 2.72	33.87 ± 4.61	32.44 ± 5.27	32.18 ± 3.99	28.00 ± 3.40	30.99 ± 3.01	33.60 ± 4.19	30.61 ± 2.66	32.53 ± 4.84
SemgHandGenderCh2	50.87 ± 2.35	55.37 ± 5.38	53.17 ± 4.35	50.17 ± 11.45	**61.00 ± 15.57**	42.78 ± 13.47	52.78 ± 1.84	39.89 ± 8.47	44.67 ± 12.14	35.61 ± 1.06	55.73 ± 4.74	51.90 ± 14.92
SemgHandMovementCh2	28.89 ± 4.22	**29.87 ± 4.86**	29.60 ± 5.24	25.29 ± 1.13	24.89 ± 3.57	20.37 ± 0.13	21.70 ± 2.31	22.52 ± 2.76	18.71 ± 1.91	20.07 ± 3.46	27.02 ± 5.10	28.13 ± 4.95
SemgHandSubjectCh2	28.89 ± 5.77	29.07 ± 4.60	28.00 ± 2.65	35.60 ± 4.19	29.91 ± 10.77	26.37 ± 1.89	23.11 ± 2.89	26.67 ± 4.87	21.56 ± 1.94	26.96 ± 3.68	27.91 ± 3.11	**37.42 ± 8.82**
ShakeGestureWiimoteZ	63.20 ± 7.69	58.00 ± 11.83	**67.60 ± 11.61**	40.40 ± 11.26	43.20 ± 9.55	45.33 ± 3.06	16.00 ± 7.21	40.00 ± 3.46	24.40 ± 4.98	46.00 ± 6.00	43.20 ± 9.55	60.40 ± 7.92
ShapeletSim	66.33 ± 12.25	**87.00 ± 16.91**	73.89 ± 23.85	50.11 ± 2.27	75.22 ± 8.30	53.89 ± 1.67	52.96 ± 1.40	48.89 ± 5.64	50.44 ± 1.82	51.11 ± 0.96	47.78 ± 3.07	62.44 ± 7.44
ShapesAll	23.63 ± 4.60	15.60 ± 1.98	23.63 ± 4.60	45.70 ± 2.59	19.27 ± 4.58	44.72 ± 2.58	6.00 ± 0.58	43.94 ± 2.14	8.07 ± 3.90	44.06 ± 1.11	8.13 ± 2.75	**50.20 ± 3.03**
SmallKitchenAppliances	40.37 ± 15.51	**46.24 ± 15.42**	44.32 ± 15.50	42.19 ± 3.16	38.13 ± 22.42	32.98 ± 5.23	32.80 ± 3.67	35.82 ± 4.27	35.41 ± 3.13	36.89 ± 7.74	38.88 ± 7.80	42.88 ± 15.27
SmoothSubspace	**74.93 ± 5.59**	74.40 ± 5.13	74.80 ± 8.48	31.73 ± 3.96	50.00 ± 10.52	46.67 ± 1.76	53.33 ± 9.45	50.22 ± 10.96	62.27 ± 6.56	60.67 ± 10.35	56.67 ± 5.19	68.53 ± 8.95
SonyAIBORobotSurface1	77.57 ± 10.30	**80.80 ± 8.99**	80.37 ± 5.89	59.80 ± 6.73	64.53 ± 18.83	52.86 ± 10.38	56.96 ± 5.82	55.35 ± 9.37	56.07 ± 4.43	57.68 ± 9.32	55.61 ± 9.09	57.20 ± 8.62
SonyAIBORobotSurface2	80.29 ± 6.43	**84.09 ± 5.18**	80.90 ± 6.99	67.39 ± 19.08	73.03 ± 13.78	66.18 ± 11.61	60.37 ± 27.05	61.35 ± 27.10	69.49 ± 15.13	68.07 ± 16.06	57.31 ± 15.03	66.09 ± 10.78
StarLightCurves	63.75 ± 3.45	62.30 ± 9.88	57.59 ± 9.08	57.39 ± 13.40	57.55 ± 15.28	**67.54 ± 17.11**	37.78 ± 3.53	57.92 ± 15.55	27.99 ± 0.00	67.36 ± 17.40	64.81 ± 14.61	63.03 ± 14.23
Strawberry	52.27 ± 4.26	51.84 ± 3.86	54.43 ± 4.33	54.86 ± 4.34	53.62 ± 3.45	58.29 ± 0.95	54.86 ± 13.21	**60.18 ± 0.68**	56.11 ± 9.26	57.93 ± 0.83	45.89 ± 11.15	51.46 ± 5.21
SwedishLeaf	**65.95 ± 6.01**	46.08 ± 6.36	**65.95 ± 6.01**	27.42 ± 3.44	40.83 ± 4.40	36.69 ± 1.99	27.20 ± 2.42	24.53 ± 4.08	31.65 ± 2.48	38.19 ± 0.81	8.10 ± 1.23	57.98 ± 2.72
Symbols	**85.77 ± 4.83**	70.17 ± 4.91	76.68 ± 7.53	79.60 ± 7.65	72.50 ± 6.12	81.54 ± 3.92	37.15 ± 0.97	82.18 ± 0.42	46.13 ± 9.06	80.57 ± 0.95	51.54 ± 6.72	80.94 ± 7.54
SyntheticControl	72.53 ± 6.61	76.40 ± 5.24	68.67 ± 10.59	38.47 ± 6.28	80.33 ± 9.12	47.78 ± 5.87	44.89 ± 3.20	48.22 ± 0.38	42.13 ± 2.87	50.00 ± 4.48	24.53 ± 7.68	**84.20 ± 6.54**
ToeSegmentation1	**69.82 ± 7.39**	69.65 ± 4.27	65.09 ± 8.05	51.67 ± 1.47	55.79 ± 6.85	49.27 ± 1.10	49.71 ± 3.29	46.64 ± 4.73	52.46 ± 2.82	48.39 ± 3.23	49.47 ± 2.47	58.16 ± 3.70
ToeSegmentation2	68.92 ± 8.63	70.31 ± 3.38	70.92 ± 10.76	50.31 ± 5.37	65.54 ± 9.84	50.26 ± 2.91	35.90 ± 10.81	50.51 ± 4.24	33.23 ± 16.47	52.05 ± 6.54	41.54 ± 14.95	**75.23 ± 12.13**
Trace	61.00 ± 4.64	65.20 ± 9.09	65.80 ± 6.61	49.60 ± 3.65	74.40 ± 6.15	51.33 ± 5.51	42.33 ± 4.51	47.67 ± 0.58	44.80 ± 7.69	51.33 ± 4.93	37.80 ± 6.42	**99.20 ± 1.79**
TwoLeadECG	**69.48 ± 12.74**	64.97 ± 10.26	66.51 ± 13.23	54.03 ± 2.33	57.82 ± 5.04	59.32 ± 3.05	53.32 ± 7.30	52.62 ± 3.32	54.47 ± 5.41	60.32 ± 1.06	50.73 ± 1.02	64.88 ± 11.28
TwoPatterns	48.77 ± 2.41	**61.51 ± 6.85**	46.17 ± 11.65	31.86 ± 3.24	36.86 ± 6.10	29.76 ± 2.56	25.52 ± 1.93	28.23 ± 0.74	26.06 ± 1.27	29.87 ± 1.68	26.32 ± 2.53	45.74 ± 2.64
UMD	43.06 ± 4.02	**54.58 ± 9.53**	45.56 ± 6.14	46.81 ± 7.70	50.14 ± 6.28	46.30 ± 5.02	45.60 ± 3.21	47.69 ± 4.19	48.61 ± 2.55	47.45 ± 6.56	31.39 ± 4.35	45.47 ± 8.73
UWaveGestureLibraryAll	29.36 ± 4.20	24.46 ± 5.19	28.34 ± 3.82	66.77 ± 5.89	41.63 ± 5.40	69.19 ± 5.01	25.78 ± 1.68	67.23 ± 2.91	26.67 ± 3.53	**69.21 ± 6.24**	50.70 ± 5.42	33.37 ± 0.26
UWaveGestureLibraryX	26.45 ± 4.19	26.60 ± 3.44	24.71 ± 1.83	47.27 ± 6.01	35.36 ± 5.04	**49.30 ± 7.17**	22.37 ± 0.14	47.22 ± 8.02	24.34 ± 5.30	49.17 ± 7.61	17.28 ± 8.20	30.33 ± 0.23
UWaveGestureLibraryY	25.37 ± 4.63	26.13 ± 4.45	24.62 ± 4.93	43.95 ± 4.51	38.35 ± 7.26	47.05 ± 4.02	24.58 ± 1.89	46.37 ± 4.93	26.81 ± 2.91	**47.18 ± 5.20**	20.37 ± 4.71	40.36 ± 0.25
UWaveGestureLibraryZ	24.77 ± 4.74	27.03 ± 3.18	24.45 ± 3.15	42.90 ± 1.96	36.37 ± 3.72	44.61 ± 1.78	22.47 ± 2.82	44.70 ± 2.64	25.10 ± 2.80	44.75 ± 1.63	22.77 ± 3.47	**47.44 ± 2.61**
Wafer	50.39 ± 4.56	44.94 ± 10.31	49.91 ± 13.57	**67.87 ± 4.81**	51.56 ± 17.59	63.90 ± 3.73	64.15 ± 3.76	63.99 ± 9.60	63.82 ± 9.82	65.48 ± 1.03	43.02 ± 26.21	46.27 ± 7.11
Wine	**57.41 ± 14.70**	54.07 ± 4.61	54.81 ± 11.31	52.96 ± 8.34	50.74 ± 6.23	48.77 ± 3.85	50.00 ± 0.00	50.00 ± 0.00	52.22 ± 4.97	37.65 ± 19.80	51.48 ± 3.31	51.11 ± 7.48
WordSynonyms	14.92 ± 3.03	13.70 ± 5.79	14.92 ± 3.03	19.78 ± 2.70	16.96 ± 4.46	19.75 ± 1.90	6.79 ± 1.46	**21.16 ± 1.50**	10.06 ± 3.54	20.53 ± 2.41	4.98 ± 1.34	20.91 ± 2.73
Worms	**36.10 ± 9.87**	33.25 ± 6.66	35.06 ± 11.43	17.14 ± 4.04	23.90 ± 9.39	19.91 ± 6.54	23.38 ± 4.50	18.61 ± 5.86	17.66 ± 7.03	19.05 ± 5.25	17.66 ± 2.36	24.68 ± 8.95
WormsTwoClass	44.79 ± 3.48	49.09 ± 6.96	52.99 ± 6.71	49.09 ± 3.72	50.39 ± 3.23	49.35 ± 3.44	48.92 ± 10.50	52.38 ± 1.98	53.51 ± 4.81	**53.68 ± 3.75**	52.99 ± 2.82	47.53 ± 10.69
Yoga	**55.60 ± 4.57**	50.68 ± 5.18	53.54 ± 4.28	49.99 ± 1.44	49.75 ± 4.01	49.48 ± 0.91	51.11 ± 3.62	49.62 ± 0.72	51.12 ± 3.30	48.62 ± 0.52	49.35 ± 2.85	50.42 ± 4.51
Avg. Acc.	47.79	45.40	47.88	44.21	42.39	45.16	36.78	44.31	37.71	46.42	35.61	**48.31**
Avg. Rank	5.11	5.80	5.19	6.45	6.92	6.04	9.11	6.17	8.51	5.30	8.82	**4.58**
#Best	25	16	14	7	5	11	2	11	2	15	4	**26**

TABLE 11: Per-dataset 2-shot classification results on 128 UCR datasets. Each value is reported as mean ± standard deviation over 5 runs.

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	TimeMorph
ACSF1	**61.60 ± 7.86**	47.20 ± 6.30	59.20 ± 5.26	25.80 ± 4.38	42.00 ± 3.16	40.33 ± 2.31	30.67 ± 2.52	44.00 ± 6.24	35.80 ± 3.19	40.67 ± 5.69	24.80 ± 4.66	54.33 ± 4.16
Adiac	35.45 ± 2.60	14.27 ± 3.74	35.45 ± 2.60	18.87 ± 2.86	17.80 ± 5.08	27.79 ± 2.14	44.67 ± 0.39	16.62 ± 1.77	55.24 ± 4.67	32.91 ± 3.49	3.48 ± 1.27	**56.01 ± 1.25**
AllGestureWiimoteX	**32.43 ± 2.94**	27.69 ± 3.48	32.09 ± 3.58	19.46 ± 2.47	23.77 ± 3.63	24.43 ± 4.71	11.86 ± 1.00	22.67 ± 2.29	14.09 ± 1.28	21.10 ± 5.46	23.77 ± 3.63	26.51 ± 4.36
AllGestureWiimoteY	**35.11 ± 5.43**	28.57 ± 4.31	28.91 ± 5.81	20.54 ± 2.69	33.31 ± 2.65	24.33 ± 2.50	12.00 ± 1.38	21.90 ± 1.33	16.14 ± 1.67	22.57 ± 1.00	33.31 ± 2.65	29.43 ± 1.69
AllGestureWiimoteZ	**29.80 ± 2.53**	26.94 ± 2.84	29.31 ± 4.56	16.77 ± 2.02	22.97 ± 5.35	19.86 ± 4.12	12.76 ± 0.36	19.57 ± 2.48	14.97 ± 0.98	20.29 ± 3.48	22.97 ± 5.35	27.91 ± 5.21
ArrowHead	50.17 ± 5.13	44.80 ± 3.82	45.14 ± 5.25	**57.03 ± 4.92**	49.14 ± 13.02	53.14 ± 6.02	34.10 ± 6.27	50.67 ± 2.01	42.29 ± 4.92	55.43 ± 6.44	43.77 ± 8.76	**57.03 ± 2.81**
BME	48.40 ± 4.91	55.33 ± 7.02	54.40 ± 8.12	50.80 ± 5.11	67.33 ± 7.90	48.89 ± 2.14	64.00 ± 5.21	58.89 ± 3.91	67.33 ± 7.44	54.89 ± 9.20	50.80 ± 8.05	**79.47 ± 11.40**
Beef	42.00 ± 2.98	36.67 ± 8.50	36.70 ± 9.60	49.33 ± 7.23	39.33 ± 6.41	48.89 ± 5.09	40.00 ± 3.33	34.44 ± 5.09	42.67 ± 4.94	**51.11 ± 7.70**	32.67 ± 7.23	43.33 ± 5.27
BeetleFly	50.00 ± 9.35	67.00 ± 18.23	55.00 ± 12.75	52.00 ± 11.51	57.00 ± 16.43	45.00 ± 10.00	48.33 ± 12.58	38.33 ± 7.64	45.00 ± 9.35	56.67 ± 7.64	57.00 ± 4.47	**75.00 ± 18.03**
BirdChicken	71.00 ± 6.52	72.00 ± 4.47	**76.00 ± 7.42**	52.00 ± 14.40	65.00 ± 10.61	45.00 ± 8.66	51.67 ± 7.64	46.67 ± 5.77	53.00 ± 2.74	50.00 ± 13.23	49.00 ± 2.24	69.00 ± 22.19
CBF	85.76 ± 5.83	**90.62 ± 7.16**	82.18 ± 10.67	64.78 ± 6.67	82.69 ± 10.89	68.07 ± 6.00	34.89 ± 2.41	73.78 ± 6.88	36.78 ± 1.75	70.00 ± 6.65	42.84 ± 7.09	85.91 ± 4.23
Car	42.00 ± 15.43	41.67 ± 13.49	52.67 ± 7.87	55.33 ± 10.63	43.00 ± 11.45	54.44 ± 6.74	32.78 ± 0.96	52.78 ± 10.84	37.00 ± 7.76	55.56 ± 4.19	31.00 ± 10.71	**63.33 ± 7.45**
Chinatown	95.74 ± 1.26	**96.73 ± 1.19**	**96.73 ± 0.81**	54.52 ± 24.75	91.08 ± 5.39	85.33 ± 8.50	86.10 ± 2.71	84.55 ± 3.36	80.70 ± 5.41	82.80 ± 4.37	45.01 ± 24.65	91.84 ± 4.85
ChlorineConcentration	34.61 ± 4.95	35.24 ± 5.79	35.19 ± 5.79	34.54 ± 2.93	34.28 ± 2.22	31.22 ± 6.21	33.24 ± 2.95	33.84 ± 2.69	33.91 ± 2.07	33.69 ± 2.59	**39.08 ± 9.47**	32.47 ± 3.03
CinCECGTorso	46.52 ± 12.75	48.35 ± 9.58	51.06 ± 10.06	39.77 ± 4.46	54.91 ± 12.73	48.14 ± 2.63	36.18 ± 3.15	53.94 ± 2.51	50.30 ± 12.26	48.89 ± 2.81	49.00 ± 5.40	**62.95 ± 6.66**
Coffee	**94.29 ± 7.41**	88.57 ± 12.73	**94.29 ± 3.19**	92.14 ± 3.91	90.71 ± 10.89	73.81 ± 8.99	63.10 ± 7.43	85.71 ± 3.57	84.29 ± 10.29	77.38 ± 4.12	65.71 ± 15.07	88.57 ± 4.66
Computers	52.24 ± 10.15	54.40 ± 10.81	52.72 ± 12.14	48.08 ± 7.70	53.28 ± 11.25	48.00 ± 10.51	48.67 ± 4.60	48.00 ± 11.45	50.40 ± 1.72	46.67 ± 11.22	50.16 ± 7.53	**57.73 ± 17.44**
CricketX	**36.51 ± 3.27**	34.10 ± 3.98	33.03 ± 3.86	20.62 ± 2.07	30.56 ± 3.87	27.35 ± 2.06	8.12 ± 1.41	26.15 ± 1.85	12.15 ± 1.24	27.44 ± 2.96	11.54 ± 1.86	35.98 ± 3.13
CricketY	**37.54 ± 5.48**	30.62 ± 2.69	33.28 ± 3.86	22.56 ± 4.44	24.21 ± 2.69	26.50 ± 7.27	9.06 ± 0.39	25.90 ± 6.29	12.67 ± 0.94	27.52 ± 6.80	10.00 ± 3.24	30.43 ± 5.23
CricketZ	**40.41 ± 4.15**	35.49 ± 2.72	33.69 ± 5.21	20.36 ± 1.13	24.92 ± 3.09	26.75 ± 2.72	8.12 ± 1.50	24.44 ± 0.78	12.97 ± 2.47	26.50 ± 2.74	11.18 ± 2.42	33.33 ± 3.02
Crop	31.31 ± 3.87	37.82 ± 2.09	31.31 ± 3.87	14.67 ± 0.98	24.66 ± 2.54	36.33 ± 2.63	35.06 ± 3.15	37.89 ± 1.28	35.83 ± 2.39	38.66 ± 2.99	4.63 ± 0.75	**43.94 ± 2.11**
DiatomSizeReduction	78.43 ± 9.98	84.44 ± 4.38	89.74 ± 8.40	83.73 ± 5.41	63.01 ± 13.25	84.42 ± 6.89	59.91 ± 2.78	84.97 ± 5.38	59.28 ± 18.44	82.35 ± 7.47	35.10 ± 12.79	**90.39 ± 5.62**
DistalPhalanxOutlineAgeGroup	66.91 ± 6.08	**67.77 ± 3.31**	65.76 ± 6.53	55.54 ± 12.17	58.42 ± 11.15	57.07 ± 19.57	50.84 ± 14.56	56.12 ± 13.32	56.26 ± 12.92	57.07 ± 14.47	38.99 ± 8.80	**67.77 ± 3.39**
DistalPhalanxOutlineCorrect	52.32 ± 12.15	51.59 ± 11.49	52.54 ± 12.44	49.57 ± 11.83	51.45 ± 15.36	**57.49 ± 13.09**	56.76 ± 11.56	55.80 ± 11.93	51.38 ± 12.34	56.64 ± 12.97	50.65 ± 9.31	55.80 ± 14.95
DistalPhalanxTW	49.50 ± 8.83	**55.54 ± 4.21**	54.10 ± 4.92	39.28 ± 8.61	53.96 ± 7.56	41.97 ± 11.56	42.45 ± 10.74	39.81 ± 18.49	47.05 ± 8.79	45.80 ± 16.17	22.45 ± 8.85	51.37 ± 8.97
DodgerLoopDay	28.75 ± 2.34	23.75 ± 2.50	28.00 ± 5.70	**44.00 ± 4.09**	38.50 ± 7.42	39.58 ± 4.02	23.75 ± 2.50	42.50 ± 7.60	19.75 ± 4.09	42.50 ± 3.31	38.50 ± 7.42	35.50 ± 8.78
DodgerLoopGame	54.93 ± 6.45	63.04 ± 8.02	58.41 ± 11.09	53.62 ± 8.51	65.65 ± 18.08	67.87 ± 7.26	48.55 ± 4.53	**71.50 ± 0.84**	50.14 ± 3.81	68.36 ± 3.42	65.65 ± 18.08	61.01 ± 13.03
DodgerLoopWeekend	92.75 ± 2.40	87.68 ± 5.59	92.03 ± 3.69	94.64 ± 2.74	96.09 ± 1.41	96.38 ± 1.26	64.98 ± 4.71	**97.34 ± 0.42**	65.22 ± 8.45	96.14 ± 0.84	96.09 ± 1.41	84.64 ± 2.73
ECG200	69.40 ± 7.77	75.20 ± 7.05	71.80 ± 5.54	76.80 ± 3.56	71.80 ± 4.38	**78.33 ± 1.53**	74.33 ± 5.69	75.67 ± 3.21	73.60 ± 3.65	75.33 ± 0.58	52.20 ± 18.20	73.00 ± 3.46
ECG5000	61.90 ± 13.09	69.05 ± 11.38	61.86 ± 10.65	**74.23 ± 10.96**	58.61 ± 14.82	72.42 ± 4.03	65.61 ± 11.76	69.40 ± 4.84	67.93 ± 11.62	66.36 ± 9.81	30.14 ± 19.40	67.68 ± 9.76
ECGFiveDays	68.78 ± 9.46	56.79 ± 4.91	71.73 ± 8.76	57.12 ± 2.23	67.29 ± 3.58	66.55 ± 6.20	74.64 ± 13.99	**75.18 ± 2.37**	68.76 ± 10.24	68.52 ± 6.54	57.79 ± 7.24	69.12 ± 9.82
EOGHorizontalSignal	20.66 ± 1.70	23.43 ± 5.48	21.38 ± 4.60	25.86 ± 4.54	28.07 ± 2.90	30.66 ± 2.91	12.06 ± 1.96	31.68 ± 2.61	9.56 ± 1.50	29.01 ± 2.49	29.83 ± 3.23	**37.35 ± 4.07**
EOGVerticalSignal	13.37 ± 1.32	16.57 ± 3.25	14.86 ± 2.77	26.80 ± 1.98	29.56 ± 4.11	26.98 ± 4.28	11.88 ± 1.54	26.34 ± 4.16	10.83 ± 1.45	26.61 ± 6.09	20.88 ± 2.94	**31.82 ± 3.47**
Earthquakes	52.95 ± 7.17	57.84 ± 11.11	49.35 ± 10.38	53.24 ± 4.93	**63.88 ± 6.42**	63.31 ± 10.45	49.16 ± 1.10	54.92 ± 17.17	56.55 ± 8.34	60.43 ± 21.88	57.12 ± 15.34	55.25 ± 14.33
ElectricDevices	37.63 ± 9.44	41.80 ± 2.08	34.70 ± 5.95	21.27 ± 5.73	37.69 ± 10.00	28.63 ± 3.21	22.56 ± 2.12	20.63 ± 2.87	22.76 ± 4.51	29.88 ± 3.14	14.36 ± 4.97	**47.14 ± 3.99**
EthanolLevel	25.32 ± 0.64	25.52 ± 1.56	25.12 ± 2.09	25.76 ± 1.71	26.96 ± 0.82	26.87 ± 2.61	**29.20 ± 1.91**	25.67 ± 3.18	25.24 ± 0.98	26.47 ± 2.08	25.12 ± 0.81	28.00 ± 3.11
FaceAll	**55.98 ± 6.19**	45.22 ± 8.18	**55.98 ± 6.19**	40.27 ± 5.76	24.12 ± 4.85	38.56 ± 3.11	38.84 ± 3.81	36.71 ± 2.55	38.89 ± 3.64	41.40 ± 3.79	11.35 ± 5.01	39.09 ± 1.28
FaceFour	77.73 ± 4.99	52.05 ± 10.37	81.82 ± 7.14	74.09 ± 5.98	59.32 ± 5.76	79.55 ± 9.02	57.58 ± 7.31	80.68 ± 3.41	62.50 ± 7.14	**84.85 ± 2.37**	40.45 ± 14.16	68.55 ± 4.91
FacesUCR	**67.20 ± 6.42**	51.90 ± 6.85	62.04 ± 8.47	41.67 ± 3.36	25.42 ± 6.93	44.55 ± 2.52	45.95 ± 4.27	49.07 ± 5.45	48.56 ± 3.79	49.27 ± 6.47	10.96 ± 5.26	55.08 ± 2.32
FiftyWords	19.03 ± 1.51	13.49 ± 1.95	19.03 ± 1.51	40.09 ± 3.22	28.35 ± 3.05	44.69 ± 2.97	5.05 ± 0.44	46.74 ± 2.43	16.53 ± 1.75	44.18 ± 3.70	15.16 ± 3.75	**47.30 ± 2.89**
Fish	**50.97 ± 10.77**	26.97 ± 9.88	50.51 ± 16.65	50.17 ± 6.36	31.54 ± 5.18	45.90 ± 8.88	20.00 ± 4.12	39.05 ± 5.64	22.97 ± 3.62	45.71 ± 10.47	22.86 ± 2.83	49.91 ± 10.25
FordA	**64.98 ± 8.59**	62.27 ± 9.82	62.82 ± 9.33	49.92 ± 1.84	60.48 ± 7.29	50.30 ± 1.31	50.66 ± 0.90	50.15 ± 0.79	50.27 ± 1.01	50.58 ± 1.43	50.76 ± 2.11	53.35 ± 2.84
FordB	**61.19 ± 3.67**	58.10 ± 1.40	58.67 ± 2.63	49.09 ± 1.35	56.89 ± 2.99	49.84 ± 0.63	50.99 ± 2.05	50.25 ± 1.29	49.95 ± 1.77	50.58 ± 0.68	50.77 ± 0.87	50.42 ± 2.72
FreezerRegularTrain	68.01 ± 9.57	67.73 ± 22.81	61.99 ± 17.00	67.56 ± 20.68	71.73 ± 5.42	59.79 ± 26.51	53.13 ± 0.84	60.39 ± 25.70	58.85 ± 2.14	60.74 ± 25.80	72.39 ± 6.35	**79.14 ± 7.94**
FreezerSmallTrain	66.47 ± 12.08	65.70 ± 14.41	66.00 ± 12.69	66.60 ± 13.71	64.67 ± 11.73	**71.47 ± 6.03**	56.34 ± 1.10	68.49 ± 7.01	56.88 ± 4.22	71.17 ± 6.80	65.57 ± 16.06	64.30 ± 13.28
Fungi	97.10 ± 2.45	34.84 ± 7.53	66.24 ± 2.06	68.17 ± 5.25	68.17 ± 13.75	**97.13 ± 0.82**	75.99 ± 1.64	82.62 ± 2.42	74.73 ± 0.85	96.95 ± 0.31	8.06 ± 3.93	81.97 ± 3.24
GestureMidAirD1	24.62 ± 1.44	29.08 ± 3.94	20.62 ± 1.44	38.31 ± 3.66	29.38 ± 2.74	37.69 ± 3.35	17.95 ± 2.70	36.67 ± 1.78	35.69 ± 2.53	36.92 ± 2.77	29.38 ± 2.74	**44.92 ± 6.31**
GestureMidAirD2	16.46 ± 2.22	21.85 ± 4.16	16.46 ± 2.22	35.69 ± 3.47	33.69 ± 3.66	42.05 ± 0.44	12.05 ± 1.60	44.36 ± 2.47	30.92 ± 4.26	**45.64 ± 3.20**	33.69 ± 3.66	43.85 ± 4.03
GestureMidAirD3	10.15 ± 1.99	11.85 ± 3.91	10.15 ± 1.99	25.08 ± 3.59	18.00 ± 2.64	26.41 ± 1.18	14.62 ± 4.07	**26.92 ± 1.54**	22.00 ± 3.63	**26.92 ± 2.77**	18.00 ± 2.64	21.08 ± 1.85
GesturePebbleZ1	62.44 ± 5.24	41.40 ± 5.40	**64.88 ± 2.42**	53.26 ± 4.18	29.53 ± 10.46	58.91 ± 3.41	16.86 ± 3.54	57.17 ± 0.89	24.19 ± 4.00	61.82 ± 3.41	29.53 ± 10.46	37.91 ± 6.35
GesturePebbleZ2	**66.58 ± 5.17**	44.05 ± 14.70	61.70 ± 6.13	51.90 ± 9.19	35.32 ± 9.94	55.06 ± 7.46	15.82 ± 0.63	46.84 ± 8.79	20.76 ± 3.15	56.75 ± 4.12	35.32 ± 9.94	41.39 ± 4.24
GunPoint	67.07 ± 8.15	65.20 ± 5.95	66.00 ± 5.89	60.13 ± 7.74	65.47 ± 7.68	55.33 ± 6.57	64.44 ± 5.18	58.44 ± 9.46	68.13 ± 5.80	57.11 ± 7.95	54.00 ± 6.53	**79.33 ± 16.25**
GunPointAgeSpan	65.38 ± 2.37	63.73 ± 4.38	**68.61 ± 1.37**	67.41 ± 13.19	61.08 ± 13.41	61.08 ± 8.21	52.22 ± 8.74	55.49 ± 10.60	52.41 ± 8.13	58.76 ± 12.22	47.41 ± 14.13	64.49 ± 7.90
GunPointMaleVersusFemale	79.30 ± 6.81	84.62 ± 8.74	81.65 ± 8.32	57.15 ± 7.06	65.95 ± 9.57	60.65 ± 4.86	50.11 ± 5.57	60.76 ± 10.27	52.59 ± 5.35	65.82 ± 3.12	65.00 ± 5.79	**87.66 ± 5.66**
GunPointOldVersusYoung	**91.94 ± 11.58**	89.97 ± 13.74	91.37 ± 12.08	72.89 ± 4.20	73.14 ± 5.24	73.54 ± 7.07	56.51 ± 2.29	72.49 ± 6.74	59.30 ± 3.02	75.87 ± 9.07	78.03 ± 17.26	70.92 ± 5.05
Ham	45.71 ± 9.69	50.86 ± 8.18	48.00 ± 10.27	54.86 ± 11.40	53.33 ± 7.28	54.92 ± 1.98	**58.73 ± 5.25**	50.48 ± 13.83	55.24 ± 4.62	57.78 ± 9.77	52.57 ± 10.93	50.86 ± 8.07
HandOutlines	57.19 ± 12.29	63.30 ± 2.05	60.65 ± 3.52	69.19 ± 14.14	63.57 ± 17.16	67.21 ± 16.82	52.16 ± 12.39	68.11 ± 20.52	41.57 ± 12.57	67.48 ± 15.69	**69.89 ± 13.52**	64.59 ± 10.14
Haptics	29.16 ± 3.65	24.35 ± 4.42	25.32 ± 2.75	28.31 ± 7.72	28.25 ± 7.08	**29.33 ± 7.26**	23.81 ± 0.68	29.22 ± 2.66	20.97 ± 3.07	29.00 ± 7.31	26.95 ± 8.77	26.62 ± 4.64
Herring	52.81 ± 11.18	**56.88 ± 7.13**	52.81 ± 11.23	51.25 ± 5.89	52.19 ± 12.13	47.92 ± 6.51	46.35 ± 3.61	45.83 ± 3.25	50.00 ± 4.69	48.96 ± 4.51	45.00 ± 9.84	55.00 ± 4.34
HouseTwenty	**76.30 ± 9.04**	69.24 ± 11.83	67.39 ± 13.31	60.00 ± 7.10	63.03 ± 13.56	68.91 ± 7.47	47.34 ± 6.31	59.94 ± 1.75	52.10 ± 8.21	71.15 ± 7.81	60.34 ± 11.20	71.93 ± 10.30
InlineSkate	18.58 ± 2.99	19.56 ± 1.81	**19.93 ± 2.98**	18.29 ± 2.68	17.05 ± 3.64	16.85 ± 1.00	15.76 ± 0.93	15.82 ± 1.62	16.29 ± 1.31	17.03 ± 0.46	16.36 ± 2.46	18.98 ± 2.54
InsectEPGRegularTrain	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	51.97 ± 7.58	58.39 ± 6.76	58.37 ± 4.44	37.08 ± 10.70	54.89 ± 2.35	37.99 ± 13.89	59.71 ± 4.91	81.69 ± 17.66	68.76 ± 9.60
InsectEPGSmallTrain	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	51.33 ± 4.40	58.47 ± 6.25	57.56 ± 4.71	41.37 ± 7.24	58.23 ± 4.08	42.65 ± 10.05	58.10 ± 4.04	80.40 ± 17.17	67.79 ± 2.45
InsectWingbeatSound	20.38 ± 3.08	16.62 ± 4.40	18.70 ± 3.47	47.87 ± 2.61	33.18 ± 5.98	50.44 ± 4.17	20.69 ± 3.83	49.41 ± 1.90	27.56 ± 2.55	**51.03 ± 4.83**	16.68 ± 7.51	31.49 ± 3.72
ItalyPowerDemand	68.88 ± 12.96	75.63 ± 13.47	73.61 ± 12.06	50.03 ± 0.16	79.18 ± 15.47	54.49 ± 7.77	75.87 ± 16.00	76.35 ± 22.01	**83.15 ± 15.39**	73.57 ± 21.06	50.07 ± 0.34	65.03 ± 13.70
LargeKitchenAppliances	45.65 ± 8.51	**54.19 ± 10.41**	50.93 ± 13.42	33.60 ± 7.89	44.85 ± 7.66	37.96 ± 12.60	32.00 ± 3.23	34.40 ± 7.77	32.96 ± 7.01	36.62 ± 11.05	32.00 ± 6.67	42.56 ± 5.00
Lightning2	57.05 ± 8.63	53.77 ± 7.90	59.34 ± 7.64	54.10 ± 10.50	51.80 ± 7.11	57.38 ± 4.34	49.73 ± 9.88	51.91 ± 4.13	52.13 ± 11.90	55.19 ± 6.21	50.16 ± 8.80	**63.28 ± 8.25**
Lightning7	45.48 ± 6.31	43.01 ± 5.94	42.74 ± 6.38	38.90 ± 8.91	47.95 ± 8.16	**59.36 ± 11.07**	19.63 ± 3.45	50.23 ± 12.43	27.40 ± 4.94	56.16 ± 10.34	20.82 ± 11.60	46.03 ± 6.17
Mallat	73.35 ± 15.31	47.53 ± 14.01	80.84 ± 4.13	86.79 ± 1.54	65.16 ± 19.15	**91.78 ± 0.99**	63.31 ± 2.66	78.79 ± 4.89	70.90 ± 5.59	87.65 ± 2.43	49.21 ± 6.11	80.67 ± 4.06
Meat	54.33 ± 7.42	42.00 ± 7.76	73.00 ± 16.39	**89.67 ± 7.85**	52.00 ± 12.16	53.33 ± 6.01	43.33 ± 8.33	41.11 ± 7.52	51.33 ± 18.61	57.22 ± 7.88	36.33 ± 5.82	57.67 ± 17.30
MedicalImages	33.45 ± 8.28	25.74 ± 5.92	33.61 ± 5.62	24.45 ± 6.71	27.45 ± 4.51	28.46 ± 3.05	27.85 ± 6.61	28.55 ± 2.80	**34.53 ± 1.25**	30.00 ± 1.71	29.82 ± 16.29	31.63 ± 5.59
MelbournePedestrian	**63.62 ± 4.54**	57.50 ± 8.49	63.00 ± 2.07	14.05 ± 5.17	42.42 ± 4.22	40.29 ± 3.90	51.77 ± 5.29	50.24 ± 2.74	54.32 ± 5.40	46.37 ± 2.70	42.42 ± 4.22	49.78 ± 4.68
MiddlePhalanxOutlineAgeGroup	47.01 ± 9.52	44.55 ± 12.64	45.45 ± 9.45	**50.39 ± 7.58**	44.29 ± 10.12	37.45 ± 10.11	46.75 ± 13.26	45.24 ± 16.76	48.57 ± 12.10	48.92 ± 14.26	22.86 ± 4.12	42.99 ± 12.95
MiddlePhalanxOutlineCorrect	47.01 ± 7.12	51.13 ± 7.83	49.42 ± 8.07	53.75 ± 12.47	44.54 ± 7.47	52.92 ± 7.99	47.54 ± 10.52	**54.07 ± 11.30**	48.59 ± 8.66	53.26 ± 15.07	50.72 ± 11.13	48.87 ± 3.56
MiddlePhalanxTW	37.92 ± 11.48	44.68 ± 3.99	42.73 ± 9.27	38.96 ± 14.59	39.22 ± 13.51	45.45 ± 5.95	43.07 ± 3.27	36.36 ± 13.61	**46.36 ± 2.32**	45.24 ± 3.58	16.36 ± 7.24	44.81 ± 7.78
MixedShapesRegularTrain	55.18 ± 4.21	33.76 ± 13.30	55.82 ± 8.68	58.38 ± 6.27	54.48 ± 12.34	62.60 ± 3.23	19.42 ± 0.39	63.00 ± 3.88	22.36 ± 6.85	63.64 ± 2.51	54.05 ± 4.44	**67.98 ± 4.06**

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TABLE 11: Per-dataset 2-shot classification results on 128 UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	TimeMorph
MixedShapesSmallTrain	45.73 ± 9.51	37.76 ± 12.10	51.77 ± 7.21	61.69 ± 3.51	54.28 ± 10.55	63.00 ± 2.46	19.05 ± 0.77	64.04 ± 1.32	24.01 ± 6.26	62.64 ± 3.39	57.53 ± 4.77	**68.35 ± 3.26**
MoteStrain	81.95 ± 4.07	75.34 ± 9.74	78.95 ± 3.25	70.97 ± 10.57	75.59 ± 8.13	78.41 ± 9.87	68.53 ± 3.31	82.48 ± 1.96	68.63 ± 10.07	**83.65 ± 4.68**	55.75 ± 11.79	58.56 ± 15.12
NonInvasiveFetalECGThorax1	25.55 ± 2.02	14.44 ± 3.90	25.55 ± 2.02	**55.96 ± 3.30**	18.44 ± 9.38	50.77 ± 4.84	18.37 ± 2.16	51.65 ± 2.21	21.42 ± 2.01	53.89 ± 1.99	13.55 ± 0.51	50.58 ± 2.13
NonInvasiveFetalECGThorax2	27.07 ± 5.56	22.30 ± 6.32	27.07 ± 5.56	**62.09 ± 2.37**	28.27 ± 8.71	55.69 ± 4.28	27.23 ± 2.03	56.05 ± 0.94	30.73 ± 3.23	58.63 ± 2.42	18.29 ± 1.75	59.72 ± 2.76
OSULeaf	55.70 ± 9.28	38.35 ± 6.31	**56.03 ± 5.89**	25.79 ± 2.77	25.29 ± 5.19	29.89 ± 3.34	18.18 ± 1.43	28.10 ± 2.07	21.90 ± 3.33	28.51 ± 4.37	23.31 ± 3.85	36.28 ± 6.97
OliveOil	64.67 ± 10.70	32.67 ± 5.48	71.33 ± 10.70	**80.67 ± 3.65**	17.33 ± 10.90	32.22 ± 11.71	38.89 ± 7.70	25.56 ± 12.62	24.00 ± 10.90	58.89 ± 22.69	22.00 ± 13.04	23.33 ± 9.72
PLAID	19.29 ± 3.99	16.65 ± 2.54	20.89 ± 4.30	22.98 ± 3.80	35.53 ± 7.89	27.56 ± 4.82	17.44 ± 2.97	25.88 ± 4.40	19.81 ± 3.24	28.68 ± 3.91	35.53 ± 7.89	**40.19 ± 5.06**
PhalangesOutlinesCorrect	52.91 ± 9.24	53.03 ± 10.79	52.61 ± 9.16	56.64 ± 11.15	54.13 ± 11.04	**60.64 ± 3.81**	58.70 ± 2.16	60.37 ± 4.20	52.26 ± 8.88	60.06 ± 3.62	49.04 ± 14.25	50.84 ± 2.93
Phoneme	**12.22 ± 1.88**	10.75 ± 2.12	11.22 ± 1.88	5.41 ± 0.40	7.31 ± 0.69	7.77 ± 0.84	4.13 ± 1.74	7.00 ± 0.45	6.19 ± 2.34	5.29 ± 0.61	5.20 ± 0.75	10.02 ± 0.59
PickupGestureWiimoteZ	43.20 ± 11.37	53.20 ± 9.55	**62.40 ± 1.67**	43.20 ± 6.72	40.40 ± 8.29	52.67 ± 10.07	21.33 ± 1.15	53.33 ± 4.16	27.60 ± 3.58	51.33 ± 6.43	40.40 ± 8.29	58.80 ± 3.03
PigAirwayPressure	28.37 ± 3.10	13.08 ± 1.90	**49.52 ± 0.76**	5.00 ± 0.26	3.65 ± 0.87	4.49 ± 0.28	3.21 ± 1.21	7.05 ± 0.73	4.62 ± 1.05	6.09 ± 0.73	5.10 ± 0.80	9.81 ± 0.43
PigArtPressure	50.58 ± 2.62	23.65 ± 3.56	**69.42 ± 3.86**	7.88 ± 0.26	5.29 ± 1.63	10.10 ± 0.48	8.01 ± 1.39	9.78 ± 1.11	7.12 ± 1.24	8.33 ± 0.56	19.13 ± 1.04	30.67 ± 1.38
PigCVP	31.54 ± 0.94	18.08 ± 2.27	**61.92 ± 1.78**	7.02 ± 0.43	3.94 ± 1.10	6.73 ± 0.00	4.01 ± 0.56	7.85 ± 1.00	4.13 ± 0.80	5.29 ± 0.83	10.10 ± 1.40	19.33 ± 0.99
Plane	**100.00 ± 0.00**	94.29 ± 3.75	99.05 ± 1.17	86.10 ± 6.69	91.62 ± 3.39	86.98 ± 7.02	84.44 ± 7.02	81.90 ± 1.90	87.24 ± 4.98	86.98 ± 6.34	21.71 ± 8.89	99.24 ± 0.80
PowerCons	82.11 ± 6.61	**83.78 ± 3.63**	82.33 ± 7.91	72.11 ± 2.98	70.78 ± 14.65	69.81 ± 4.10	52.22 ± 1.47	70.93 ± 7.46	50.67 ± 4.38	70.00 ± 8.73	65.89 ± 16.73	66.78 ± 15.52
ProximalPhalanxOutlineAgeGroup	**68.10 ± 13.07**	64.29 ± 28.01	64.78 ± 22.07	58.44 ± 28.63	68.00 ± 16.32	53.17 ± 31.21	56.26 ± 31.37	56.91 ± 37.90	59.22 ± 23.54	51.38 ± 35.67	33.95 ± 22.63	66.24 ± 17.89
ProximalPhalanxOutlineCorrect	**65.43 ± 5.55**	60.07 ± 16.64	63.85 ± 6.55	63.37 ± 8.02	62.96 ± 6.50	54.75 ± 14.29	54.30 ± 16.96	60.94 ± 6.65	59.86 ± 17.09	55.21 ± 16.86	44.88 ± 18.17	59.93 ± 12.82
ProximalPhalanxTW	50.83 ± 4.70	56.00 ± 9.92	**67.12 ± 8.12**	49.76 ± 11.43	53.37 ± 7.43	58.70 ± 5.35	53.98 ± 16.76	42.28 ± 24.98	56.00 ± 8.06	48.29 ± 19.01	16.98 ± 12.37	57.76 ± 7.59
RefrigerationDevices	**41.39 ± 9.49**	40.37 ± 10.57	41.23 ± 9.97	33.92 ± 0.58	35.31 ± 3.72	36.71 ± 1.61	33.87 ± 4.03	35.64 ± 1.78	33.81 ± 1.50	33.60 ± 3.03	33.07 ± 2.39	33.39 ± 7.13
Rock	32.00 ± 5.48	38.40 ± 8.65	37.80 ± 13.31	67.60 ± 10.53	44.80 ± 18.25	58.67 ± 13.61	**70.00 ± 11.14**	66.67 ± 8.08	47.60 ± 12.28	60.00 ± 14.00	44.40 ± 14.24	64.00 ± 7.07
ScreenType	36.91 ± 4.95	**40.21 ± 6.72**	38.40 ± 6.31	35.25 ± 3.07	36.75 ± 2.52	39.56 ± 4.47	33.78 ± 3.74	38.67 ± 2.63	33.71 ± 2.21	39.11 ± 4.89	33.81 ± 3.79	35.63 ± 4.40
SemgHandGenderCh2	55.13 ± 5.29	58.23 ± 11.97	54.27 ± 6.20	53.27 ± 13.91	**62.87 ± 8.65**	49.22 ± 10.20	55.56 ± 1.55	62.44 ± 3.53	43.87 ± 10.45	56.22 ± 13.04	53.70 ± 4.90	59.73 ± 10.87
SemgHandMovementCh2	33.51 ± 7.12	31.29 ± 5.18	**36.00 ± 6.56**	29.78 ± 3.16	23.56 ± 5.26	24.44 ± 2.52	18.89 ± 0.59	24.44 ± 1.39	18.71 ± 1.13	24.07 ± 5.35	27.91 ± 3.59	31.16 ± 4.70
SemgHandSubjectCh2	32.00 ± 3.88	30.71 ± 4.13	33.56 ± 5.04	**45.91 ± 5.66**	40.67 ± 4.39	29.56 ± 4.02	23.56 ± 2.47	32.52 ± 7.62	24.09 ± 3.07	30.81 ± 0.84	29.47 ± 4.28	43.20 ± 6.72
ShakeGestureWiimoteZ	76.00 ± 6.63	76.80 ± 7.43	**81.60 ± 0.89**	47.20 ± 7.95	58.40 ± 5.18	54.00 ± 6.00	26.00 ± 4.00	54.67 ± 4.62	32.40 ± 2.61	56.00 ± 5.29	58.40 ± 5.18	**81.60 ± 2.61**
ShapeletSim	78.89 ± 14.16	**98.33 ± 2.08**	92.11 ± 5.26	50.11 ± 2.87	80.22 ± 16.23	47.59 ± 3.16	50.74 ± 3.06	48.15 ± 1.79	49.22 ± 2.93	46.48 ± 3.06	49.67 ± 1.50	67.44 ± 10.92
ShapesAll	36.80 ± 3.60	28.17 ± 3.75	36.80 ± 3.60	50.73 ± 1.71	32.23 ± 5.06	55.78 ± 1.50	6.11 ± 0.63	53.17 ± 1.00	18.33 ± 5.95	53.33 ± 2.62	27.63 ± 3.10	**63.03 ± 2.22**
SmallKitchenAppliances	48.85 ± 7.91	50.40 ± 8.87	50.19 ± 7.71	41.49 ± 3.33	47.15 ± 6.04	49.60 ± 6.54	35.20 ± 2.93	40.09 ± 2.14	42.03 ± 4.49	44.09 ± 8.36	34.61 ± 4.49	**55.68 ± 8.74**
SmoothSubspace	78.93 ± 10.17	77.87 ± 5.55	80.13 ± 5.80	33.33 ± 0.00	51.20 ± 5.65	58.22 ± 4.73	68.89 ± 7.31	59.11 ± 4.23	70.67 ± 4.88	81.33 ± 6.36	48.27 ± 9.05	**85.07 ± 5.20**
SonyAIBORobotSurface1	90.42 ± 4.34	**94.08 ± 1.77**	89.38 ± 5.35	64.83 ± 12.29	70.05 ± 3.77	62.90 ± 13.91	61.90 ± 7.89	66.61 ± 10.89	59.10 ± 5.18	62.73 ± 9.45	52.65 ± 8.42	71.85 ± 8.28
SonyAIBORobotSurface2	**88.00 ± 2.82**	84.87 ± 6.58	83.67 ± 9.41	74.59 ± 3.28	78.70 ± 2.68	69.60 ± 7.04	77.26 ± 1.79	78.49 ± 0.86	78.30 ± 1.49	79.29 ± 1.17	56.81 ± 17.39	68.21 ± 5.28
StarLightCurves	**76.49 ± 11.97**	66.10 ± 9.91	66.22 ± 10.70	67.92 ± 13.52	71.57 ± 12.97	63.90 ± 14.70	35.37 ± 11.93	65.31 ± 14.87	34.69 ± 14.98	63.78 ± 14.15	68.45 ± 12.08	74.47 ± 15.08
Strawberry	62.81 ± 6.83	57.14 ± 6.89	60.22 ± 9.39	51.84 ± 8.28	56.59 ± 2.41	53.33 ± 7.10	**66.58 ± 2.89**	52.97 ± 8.18	59.62 ± 11.05	54.59 ± 4.47	51.03 ± 11.13	59.41 ± 6.38
SwedishLeaf	**72.45 ± 3.37**	54.30 ± 6.89	**72.45 ± 3.37**	34.02 ± 3.00	43.26 ± 7.26	48.91 ± 2.97	33.28 ± 3.94	46.24 ± 1.25	38.85 ± 4.16	47.52 ± 2.78	7.94 ± 0.74	64.99 ± 3.83
Symbols	**90.49 ± 1.42**	75.88 ± 7.72	88.48 ± 4.41	81.23 ± 0.99	76.18 ± 6.87	81.37 ± 1.84	37.42 ± 3.12	82.04 ± 0.58	58.95 ± 6.86	80.00 ± 1.74	51.50 ± 7.93	82.69 ± 3.43
SyntheticControl	79.67 ± 4.05	85.07 ± 4.63	82.60 ± 6.01	41.40 ± 6.37	83.80 ± 5.24	55.00 ± 3.38	44.11 ± 1.39	59.89 ± 3.15	47.93 ± 4.58	62.89 ± 3.15	20.80 ± 4.95	**87.93 ± 3.95**
ToeSegmentation1	**68.42 ± 7.02**	60.79 ± 8.14	61.58 ± 7.54	53.95 ± 2.19	58.60 ± 10.97	51.90 ± 4.30	53.80 ± 2.21	53.95 ± 0.44	52.37 ± 4.44	51.61 ± 3.08	50.70 ± 2.53	60.26 ± 8.54
ToeSegmentation2	76.15 ± 9.44	**79.69 ± 9.24**	72.15 ± 7.68	48.92 ± 5.64	68.62 ± 12.68	66.41 ± 15.71	51.54 ± 5.38	63.08 ± 15.78	47.08 ± 8.33	63.59 ± 15.79	54.31 ± 14.96	76.15 ± 12.44
Trace	76.00 ± 3.00	76.40 ± 3.05	73.80 ± 2.95	52.40 ± 4.04	76.60 ± 4.39	47.33 ± 3.21	51.00 ± 4.00	49.00 ± 3.00	55.60 ± 4.62	49.00 ± 5.00	46.00 ± 6.78	**98.40 ± 3.05**
TwoLeadECG	**82.53 ± 6.47**	79.19 ± 7.45	70.63 ± 4.92	55.68 ± 3.84	59.95 ± 3.74	62.36 ± 5.86	56.60 ± 1.27	55.90 ± 1.72	55.49 ± 2.80	60.14 ± 7.03	48.48 ± 5.21	74.63 ± 8.09
TwoPatterns	52.71 ± 4.44	71.75 ± 5.17	65.30 ± 11.45	32.62 ± 1.84	44.61 ± 9.45	31.66 ± 1.59	24.88 ± 1.30	28.57 ± 1.14	25.14 ± 0.61	29.88 ± 2.34	27.30 ± 2.15	**78.23 ± 0.41**
UMD	54.86 ± 7.95	71.67 ± 9.36	68.75 ± 16.74	54.17 ± 4.55	62.50 ± 12.00	52.31 ± 6.22	47.69 ± 9.93	60.19 ± 1.45	53.06 ± 4.83	56.25 ± 6.05	47.50 ± 4.16	**78.86 ± 14.11**
UWaveGestureLibraryAll	32.50 ± 3.38	28.30 ± 5.01	31.51 ± 4.14	**69.44 ± 6.55**	43.65 ± 6.39	66.54 ± 6.96	26.72 ± 0.50	66.58 ± 7.48	30.82 ± 2.10	66.61 ± 7.59	54.22 ± 7.86	42.54 ± 0.28
UWaveGestureLibraryX	31.94 ± 2.72	28.97 ± 3.44	30.99 ± 3.77	50.42 ± 3.39	42.01 ± 2.19	**51.69 ± 1.72**	21.35 ± 1.20	51.05 ± 1.76	28.72 ± 2.91	50.93 ± 1.76	22.53 ± 5.80	42.54 ± 0.28
UWaveGestureLibraryY	28.00 ± 4.17	26.42 ± 2.97	27.96 ± 5.26	43.36 ± 3.42	39.43 ± 5.20	42.54 ± 7.86	19.88 ± 2.66	**43.61 ± 6.37**	28.02 ± 2.78	41.75 ± 7.40	28.22 ± 7.42	42.54 ± 0.28
UWaveGestureLibraryZ	31.19 ± 2.60	30.19 ± 2.10	28.42 ± 4.12	44.60 ± 3.60	41.59 ± 3.78	42.17 ± 5.87	23.93 ± 2.40	43.23 ± 7.30	29.59 ± 1.71	40.53 ± 6.44	23.38 ± 7.02	**59.88 ± 2.03**
Wafer	47.08 ± 18.17	57.95 ± 22.06	52.16 ± 16.40	56.77 ± 13.38	49.59 ± 17.01	67.22 ± 1.00	55.02 ± 6.93	66.32 ± 1.28	56.21 ± 9.73	**67.64 ± 1.29**	54.99 ± 20.43	55.01 ± 15.62
Wine	**60.74 ± 8.43**	50.74 ± 7.92	60.48 ± 12.52	53.70 ± 8.88	47.41 ± 11.54	53.70 ± 3.21	53.09 ± 5.35	50.00 ± 0.00	50.74 ± 1.66	54.94 ± 26.28	50.74 ± 1.66	48.89 ± 5.34
WordSynonyms	20.63 ± 4.03	16.05 ± 1.67	20.63 ± 4.03	21.35 ± 4.96	20.06 ± 3.25	29.05 ± 0.77	6.11 ± 1.90	29.73 ± 1.73	13.92 ± 2.48	29.57 ± 1.41	6.43 ± 4.20	**34.92 ± 2.64**
Worms	**54.03 ± 11.78**	44.94 ± 11.42	52.73 ± 7.32	22.60 ± 2.99	28.57 ± 6.43	19.48 ± 1.30	17.32 ± 5.41	17.75 ± 5.25	15.58 ± 4.00	20.35 ± 3.75	18.70 ± 5.00	38.70 ± 4.63
WormsTwoClass	55.32 ± 9.03	52.47 ± 8.50	**55.84 ± 3.31**	48.57 ± 2.69	49.09 ± 4.63	49.78 ± 2.70	51.52 ± 8.84	47.19 ± 3.97	47.27 ± 4.37	46.32 ± 1.50	48.83 ± 7.82	54.03 ± 7.72
Yoga	**52.99 ± 4.56**	50.68 ± 2.33	50.48 ± 3.16	50.86 ± 0.91	51.49 ± 3.75	50.12 ± 2.91	51.48 ± 2.93	50.70 ± 2.90	51.46 ± 3.58	50.58 ± 2.12	51.69 ± 4.15	50.64 ± 4.90
Avg. Acc.	53.98	50.79	54.95	48.01	48.43	49.95	39.45	49.45	41.71	51.14	38.30	**55.55**
Avg. Rank	4.84	5.63	4.83	6.78	6.42	6.13	9.36	6.29	8.63	5.64	9.17	**4.28**
#Best	**33**	13	19	10	2	9	4	6	3	7	2	32

TABLE 12: Per-dataset 5-shot classification results on 128 UCR datasets. Each value is reported as mean ± standard deviation over 5 runs.

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	TimeMorph
ACSF1	**76.00 ± 2.92**	64.60 ± 5.18	72.80 ± 1.64	36.20 ± 3.49	50.40 ± 2.61	60.00 ± 2.00	40.00 ± 1.00	62.67 ± 2.52	46.80 ± 3.70	58.67 ± 1.53	30.60 ± 4.83	68.33 ± 2.89
Adiac	56.21 ± 4.18	42.81 ± 4.46	56.21 ± 4.18	35.29 ± 1.32	34.58 ± 2.33	43.82 ± 1.54	46.97 ± 3.07	38.62 ± 1.68	64.50 ± 5.47	43.99 ± 3.86	7.11 ± 1.15	**69.51 ± 2.42**
AllGestureWiimoteX	**40.46 ± 4.50**	34.14 ± 3.58	38.71 ± 5.08	23.63 ± 2.19	35.40 ± 4.31	32.24 ± 3.43	13.57 ± 1.57	29.95 ± 4.86	18.17 ± 3.28	29.38 ± 3.89	35.40 ± 4.31	39.40 ± 2.86
AllGestureWiimoteY	43.20 ± 1.70	40.60 ± 1.51	36.20 ± 3.42	27.89 ± 2.07	40.97 ± 3.06	36.76 ± 2.07	15.38 ± 0.81	34.43 ± 2.02	19.49 ± 2.37	35.43 ± 1.55	40.97 ± 3.06	**48.11 ± 2.89**
AllGestureWiimoteZ	**41.46 ± 4.33**	35.71 ± 5.12	36.46 ± 5.30	20.20 ± 1.22	27.11 ± 2.51	26.57 ± 0.25	13.86 ± 1.71	26.67 ± 2.65	17.37 ± 2.23	26.86 ± 0.99	27.11 ± 2.51	36.14 ± 0.97
ArrowHead	66.40 ± 5.44	61.03 ± 5.46	64.23 ± 7.62	59.43 ± 5.27	58.86 ± 12.66	60.19 ± 7.46	42.67 ± 0.87	60.76 ± 1.19	48.11 ± 2.87	63.43 ± 8.42	54.06 ± 9.79	**73.60 ± 3.71**
BME	55.60 ± 7.80	56.93 ± 3.90	68.00 ± 9.90	58.53 ± 7.20	89.60 ± 10.65	58.89 ± 5.55	73.33 ± 8.11	76.89 ± 7.34	80.00 ± 6.70	73.33 ± 5.93	53.07 ± 5.63	**97.73 ± 1.53**
Beef	54.67 ± 4.47	47.33 ± 2.79	55.33 ± 2.98	52.67 ± 7.60	43.33 ± 4.71	64.44 ± 3.85	50.00 ± 3.33	52.22 ± 8.39	62.00 ± 8.03	**84.44 ± 1.92**	35.33 ± 4.47	51.33 ± 6.91
BeetleFly	72.00 ± 7.58	60.00 ± 11.73	79.00 ± 10.84	71.00 ± 14.32	57.00 ± 8.37	70.00 ± 8.66	51.67 ± 2.89	66.67 ± 16.07	49.00 ± 6.52	70.00 ± 10.00	59.00 ± 18.51	**91.67 ± 2.89**
BirdChicken	69.00 ± 12.94	77.00 ± 9.75	**90.00 ± 3.54**	57.00 ± 5.70	61.00 ± 16.73	63.33 ± 7.64	36.67 ± 2.89	65.00 ± 10.00	54.00 ± 8.94	68.33 ± 15.28	54.00 ± 4.18	85.00 ± 11.73
CBF	95.87 ± 2.59	95.58 ± 3.67	**95.89 ± 3.27**	61.47 ± 7.78	95.67 ± 4.48	74.52 ± 5.66	39.04 ± 0.80	80.11 ± 4.97	42.29 ± 1.09	80.59 ± 4.09	41.20 ± 4.62	93.76 ± 3.90
Car	59.33 ± 9.02	51.00 ± 7.96	65.67 ± 5.73	52.33 ± 6.41	45.67 ± 2.79	68.89 ± 6.31	31.67 ± 5.00	48.89 ± 4.19	41.00 ± 12.56	57.78 ± 5.85	36.00 ± 4.50	**69.67 ± 3.98**
Chinatown	96.15 ± 2.16	96.97 ± 0.49	**97.43 ± 0.38**	50.96 ± 33.10	96.03 ± 2.24	84.94 ± 15.12	91.06 ± 1.94	93.00 ± 5.85	89.56 ± 2.00	87.95 ± 8.75	73.00 ± 0.76	93.46 ± 10.56
ChlorineConcentration	35.97 ± 2.09	35.63 ± 3.63	36.15 ± 2.53	35.57 ± 3.02	35.80 ± 3.16	35.02 ± 2.94	36.15 ± 1.24	35.49 ± 4.56	37.02 ± 1.05	35.96 ± 1.50	38.60 ± 5.65	**39.58 ± 3.20**
CinCECGTorso	49.41 ± 5.43	57.77 ± 5.88	56.17 ± 3.91	43.81 ± 1.84	69.26 ± 4.67	63.41 ± 2.73	37.80 ± 1.42	64.95 ± 2.40	64.83 ± 11.83	68.41 ± 5.18	54.58 ± 3.46	**82.56 ± 7.22**
Coffee	86.43 ± 14.81	97.86 ± 1.96	**100.00 ± 0.00**	92.86 ± 3.57	97.86 ± 1.96	86.90 ± 2.06	76.19 ± 2.06	82.14 ± 9.45	92.86 ± 10.41	85.71 ± 0.00	50.00 ± 6.19	94.29 ± 1.96
Computers	58.80 ± 7.81	61.60 ± 5.76	60.96 ± 6.58	46.64 ± 4.37	54.48 ± 7.54	49.87 ± 5.33	50.53 ± 1.29	48.00 ± 8.43	49.04 ± 4.09	48.67 ± 4.31	46.56 ± 5.23	**62.67 ± 4.74**
CricketX	45.90 ± 4.41	43.08 ± 3.97	43.08 ± 2.15	25.23 ± 1.80	36.10 ± 5.19	34.87 ± 4.89	11.79 ± 1.43	30.77 ± 0.51	16.92 ± 1.32	33.68 ± 3.44	11.33 ± 1.41	**49.06 ± 3.21**
CricketY	42.67 ± 4.49	43.08 ± 3.40	43.28 ± 6.01	30.41 ± 3.78	36.92 ± 6.92	37.86 ± 2.42	9.06 ± 1.21	36.58 ± 2.06	15.23 ± 1.85	37.26 ± 2.44	12.05 ± 2.18	**49.66 ± 1.41**
CricketZ	51.59 ± 3.02	49.64 ± 3.11	47.79 ± 6.26	26.31 ± 2.67	36.31 ± 4.35	37.18 ± 1.18	10.77 ± 0.51	34.87 ± 1.43	15.44 ± 1.54	37.18 ± 0.77	10.92 ± 1.59	**53.68 ± 2.06**
Crop	49.93 ± 1.45	46.93 ± 2.27	49.93 ± 1.45	31.91 ± 2.21	38.13 ± 2.76	46.92 ± 2.69	44.47 ± 0.97	47.82 ± 2.55	47.76 ± 1.51	48.69 ± 2.69	8.15 ± 3.22	**52.18 ± 2.24**
DiatomSizeReduction	85.75 ± 6.98	70.26 ± 16.84	**95.95 ± 1.26**	61.57 ± 2.74	65.49 ± 10.17	87.69 ± 0.50	68.63 ± 2.55	66.34 ± 8.52	73.53 ± 9.37	87.25 ± 0.00	41.83 ± 13.83	93.66 ± 1.97
DistalPhalanxOutlineAgeGroup	65.76 ± 4.54	68.63 ± 3.96	67.19 ± 5.75	58.99 ± 14.81	63.31 ± 4.46	64.75 ± 3.74	67.63 ± 2.16	69.30 ± 3.24	68.20 ± 2.35	68.59 ± 2.31	53.67 ± 12.57	**69.35 ± 1.20**
DistalPhalanxOutlineCorrect	66.81 ± 3.92	61.96 ± 7.85	**68.62 ± 6.57**	56.23 ± 10.81	60.07 ± 6.07	62.44 ± 3.65	62.68 ± 3.32	58.94 ± 3.37	63.41 ± 7.40	59.30 ± 3.08	55.87 ± 8.58	63.77 ± 11.63
DistalPhalanxTW	58.99 ± 2.69	57.12 ± 1.40	58.56 ± 3.40	49.64 ± 7.01	**60.58 ± 2.61**	57.31 ± 0.42	51.32 ± 3.62	59.47 ± 4.40	54.10 ± 4.24	56.59 ± 5.05	31.94 ± 15.91	56.40 ± 4.87
DodgerLoopDay	37.75 ± 3.35	40.50 ± 4.11	37.00 ± 3.26	46.75 ± 4.01	44.25 ± 2.09	**52.50 ± 5.45**	22.08 ± 0.72	45.42 ± 2.60	24.25 ± 3.49	50.83 ± 4.73	44.25 ± 2.09	45.75 ± 3.26
DodgerLoopGame	66.52 ± 4.02	82.32 ± 3.38	72.17 ± 3.64	76.09 ± 5.15	**84.49 ± 1.89**	75.36 ± 5.02	50.00 ± 2.61	78.99 ± 7.63	53.33 ± 2.09	77.29 ± 4.82	**84.49 ± 1.89**	72.61 ± 10.25
DodgerLoopWeekend	95.51 ± 1.65	92.46 ± 3.46	94.78 ± 1.88	97.39 ± 0.40	97.25 ± 0.61	96.62 ± 0.84	66.43 ± 4.25	**97.83 ± 0.00**	72.90 ± 1.82	96.86 ± 0.42	97.25 ± 0.61	93.62 ± 2.37
ECG200	76.40 ± 3.58	77.80 ± 6.26	76.20 ± 4.60	75.00 ± 1.58	62.80 ± 16.36	77.67 ± 3.06	76.67 ± 3.79	78.00 ± 2.00	77.60 ± 3.21	**79.33 ± 4.93**	69.60 ± 4.51	67.40 ± 8.91
ECG5000	78.85 ± 9.03	75.51 ± 9.29	75.56 ± 6.88	**84.88 ± 3.51**	71.69 ± 7.99	82.01 ± 5.12	76.33 ± 7.90	83.48 ± 5.73	74.58 ± 8.80	82.23 ± 5.84	18.64 ± 16.84	73.89 ± 8.21
ECGFiveDays	71.54 ± 13.30	70.27 ± 8.41	70.22 ± 9.52	69.38 ± 6.83	77.54 ± 3.65	72.94 ± 9.66	**87.65 ± 8.42**	75.22 ± 2.60	83.30 ± 11.51	75.61 ± 4.26	60.49 ± 6.59	70.96 ± 3.91
EOGHorizontalSignal	24.97 ± 1.88	32.71 ± 1.62	27.35 ± 3.43	29.56 ± 3.95	37.73 ± 6.23	36.92 ± 1.62	10.96 ± 1.15	38.40 ± 2.19	9.94 ± 1.90	35.17 ± 4.76	31.99 ± 1.89	**46.35 ± 2.27**
EOGVerticalSignal	18.18 ± 3.62	22.10 ± 3.22	18.78 ± 4.22	29.01 ± 2.45	**38.23 ± 4.43**	31.77 ± 5.00	8.47 ± 1.80	31.95 ± 3.43	10.39 ± 1.59	31.22 ± 4.29	22.82 ± 2.72	36.74 ± 2.24
Earthquakes	50.94 ± 12.41	50.22 ± 8.85	52.23 ± 11.61	**56.83 ± 2.92**	52.09 ± 5.05	56.35 ± 11.87	51.32 ± 9.11	56.12 ± 3.14	53.38 ± 4.56	54.68 ± 7.61	52.81 ± 9.99	54.68 ± 9.31
ElectricDevices	36.69 ± 5.52	36.63 ± 6.97	36.08 ± 5.74	23.65 ± 1.86	38.77 ± 8.33	38.73 ± 7.16	22.89 ± 3.05	25.09 ± 5.42	31.29 ± 6.08	38.61 ± 5.70	11.47 ± 4.02	**47.06 ± 4.18**
EthanolLevel	27.52 ± 1.92	26.36 ± 0.86	26.60 ± 1.66	27.88 ± 1.25	24.88 ± 0.18	28.40 ± 1.73	28.60 ± 2.78	26.53 ± 0.12	25.40 ± 0.75	**29.53 ± 0.64**	26.32 ± 1.07	25.92 ± 1.40
FaceAll	67.63 ± 4.66	**67.75 ± 7.28**	67.63 ± 4.66	53.96 ± 2.14	41.22 ± 3.29	55.76 ± 2.36	53.79 ± 5.19	55.44 ± 2.51	51.51 ± 5.08	54.30 ± 2.41	17.64 ± 1.73	54.54 ± 1.26
FaceFour	88.18 ± 2.85	61.82 ± 13.74	**89.77 ± 4.62**	78.86 ± 3.17	57.50 ± 8.82	86.36 ± 1.14	64.77 ± 2.27	87.88 ± 1.74	71.82 ± 1.24	85.61 ± 1.74	41.14 ± 6.88	75.68 ± 4.37
FacesUCR	**78.67 ± 2.69**	73.00 ± 4.70	76.25 ± 3.16	50.94 ± 2.11	43.00 ± 7.53	68.33 ± 3.31	58.46 ± 2.45	68.99 ± 1.59	60.62 ± 1.49	67.61 ± 2.78	27.20 ± 3.34	68.58 ± 2.57
FiftyWords	30.07 ± 3.14	27.91 ± 3.50	30.07 ± 3.14	47.03 ± 2.26	37.71 ± 3.50	52.23 ± 1.46	9.45 ± 3.91	56.19 ± 3.63	25.10 ± 2.19	49.38 ± 1.41	40.84 ± 2.35	**61.58 ± 2.55**
Fish	61.26 ± 15.63	48.69 ± 6.91	67.66 ± 6.55	56.80 ± 6.22	42.97 ± 5.32	**68.76 ± 3.49**	25.52 ± 2.93	56.57 ± 5.51	40.80 ± 3.78	67.05 ± 1.44	26.40 ± 8.24	64.57 ± 5.92
FordA	71.41 ± 10.22	**71.94 ± 9.09**	70.55 ± 10.29	50.33 ± 1.09	63.62 ± 6.82	50.28 ± 2.11	51.67 ± 1.61	49.67 ± 1.72	51.11 ± 2.04	49.82 ± 1.42	50.91 ± 1.25	54.00 ± 2.22
FordB	**61.26 ± 2.75**	58.96 ± 1.47	59.04 ± 3.16	50.00 ± 1.41	55.88 ± 4.39	50.86 ± 0.77	49.51 ± 0.81	48.81 ± 1.26	49.56 ± 0.80	50.21 ± 1.36	50.27 ± 2.02	52.32 ± 2.78
FreezerRegularTrain	71.93 ± 7.16	**86.24 ± 2.60**	79.17 ± 4.44	76.42 ± 0.69	77.88 ± 9.36	76.43 ± 0.19	57.27 ± 4.31	76.76 ± 0.69	62.60 ± 6.21	71.04 ± 6.76	59.41 ± 11.73	85.11 ± 3.83
FreezerSmallTrain	**77.10 ± 3.40**	75.64 ± 1.31	75.24 ± 0.61	75.24 ± 1.24	71.21 ± 6.86	75.59 ± 1.06	60.63 ± 1.62	76.02 ± 0.68	66.32 ± 5.59	73.06 ± 2.85	63.24 ± 10.73	75.05 ± 3.10
Fungi	96.34 ± 4.07	34.19 ± 4.54	66.45 ± 5.80	74.09 ± 4.22	74.73 ± 9.52	**97.13 ± 0.31**	75.63 ± 1.35	90.14 ± 2.24	75.27 ± 0.54	96.06 ± 0.82	10.75 ± 6.51	81.18 ± 3.16
GestureMidAirD1	37.54 ± 4.63	52.92 ± 3.41	37.54 ± 4.63	46.00 ± 0.64	43.38 ± 3.51	53.85 ± 0.77	21.03 ± 3.11	52.05 ± 3.11	40.46 ± 4.85	52.82 ± 1.94	43.38 ± 3.51	**60.31 ± 2.35**
GestureMidAirD2	28.15 ± 3.51	42.15 ± 7.00	28.15 ± 3.51	48.31 ± 1.58	46.46 ± 3.63	50.00 ± 1.54	12.82 ± 3.55	49.49 ± 1.60	33.08 ± 9.04	50.00 ± 0.77	46.46 ± 3.63	**54.62 ± 1.96**
GestureMidAirD3	14.00 ± 2.69	19.85 ± 2.33	14.00 ± 2.69	30.92 ± 2.20	26.92 ± 3.31	25.13 ± 1.18	14.10 ± 3.11	31.79 ± 1.60	26.31 ± 4.82	27.18 ± 1.78	26.92 ± 3.31	**32.15 ± 3.33**
GesturePebbleZ1	71.74 ± 3.41	64.53 ± 3.05	73.60 ± 2.12	65.35 ± 3.15	63.95 ± 8.94	65.89 ± 3.50	23.45 ± 1.21	73.64 ± 2.20	28.95 ± 4.06	**76.16 ± 6.07**	63.95 ± 8.94	63.84 ± 8.99
GesturePebbleZ2	67.34 ± 3.97	52.53 ± 17.40	**73.04 ± 4.07**	58.35 ± 6.18	48.10 ± 7.12	62.03 ± 7.30	19.41 ± 4.45	64.98 ± 9.05	24.81 ± 3.02	60.34 ± 8.06	48.10 ± 7.12	60.89 ± 8.06
GunPoint	82.53 ± 12.46	77.33 ± 12.19	79.87 ± 13.86	59.60 ± 4.87	73.07 ± 2.03	66.00 ± 6.36	55.11 ± 6.19	70.22 ± 2.52	65.87 ± 7.58	66.00 ± 2.91	62.40 ± 5.79	**90.13 ± 10.49**
GunPointAgeSpan	75.70 ± 7.16	69.81 ± 6.17	76.46 ± 8.05	75.51 ± 2.81	73.67 ± 12.74	76.79 ± 5.43	49.26 ± 6.34	75.74 ± 4.37	60.38 ± 3.16	69.20 ± 7.84	65.70 ± 10.14	**81.52 ± 9.45**
GunPointMaleVersusFemale	91.20 ± 8.64	**96.65 ± 4.17**	90.32 ± 9.07	61.14 ± 7.95	87.53 ± 12.91	74.37 ± 13.69	55.17 ± 0.48	75.00 ± 13.22	61.20 ± 4.73	77.95 ± 14.78	63.92 ± 13.57	94.62 ± 6.24
GunPointOldVersusYoung	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	75.87 ± 2.49	70.29 ± 8.33	70.79 ± 4.48	56.93 ± 3.97	73.23 ± 4.31	58.98 ± 5.43	70.48 ± 8.59	92.38 ± 8.67	80.44 ± 8.63
Ham	60.95 ± 6.21	60.00 ± 6.77	62.10 ± 4.59	62.86 ± 11.45	58.86 ± 7.48	60.00 ± 11.70	63.49 ± 0.55	56.51 ± 8.59	63.62 ± 3.26	**66.35 ± 16.28**	56.76 ± 9.56	60.57 ± 9.91
HandOutlines	46.86 ± 11.04	52.86 ± 13.07	67.24 ± 3.85	80.76 ± 4.40	74.49 ± 7.28	**83.06 ± 3.92**	47.84 ± 8.25	81.62 ± 4.95	35.95 ± 0.00	82.34 ± 4.91	75.57 ± 5.75	68.54 ± 6.21
Haptics	32.14 ± 2.79	28.25 ± 2.44	31.49 ± 1.92	35.39 ± 2.49	28.18 ± 3.44	36.58 ± 0.19	21.32 ± 3.65	**37.12 ± 1.14**	24.68 ± 3.46	**37.12 ± 0.94**	31.82 ± 7.39	26.95 ± 4.88
Herring	50.94 ± 5.70	55.00 ± 8.93	52.50 ± 7.46	51.88 ± 6.76	50.94 ± 6.68	48.96 ± 5.02	**57.29 ± 4.51**	51.56 ± 7.16	46.88 ± 13.58	49.48 ± 2.39	49.06 ± 7.13	45.00 ± 4.87
HouseTwenty	83.70 ± 6.31	81.18 ± 9.08	81.85 ± 9.95	58.99 ± 5.13	76.81 ± 7.55	73.67 ± 7.34	54.90 ± 2.57	66.95 ± 4.78	59.66 ± 6.85	72.83 ± 4.23	63.03 ± 4.28	**84.37 ± 7.51**
InlineSkate	20.15 ± 4.70	22.95 ± 2.93	19.75 ± 2.66	21.05 ± 2.12	18.98 ± 2.35	24.85 ± 1.07	16.00 ± 1.31	24.06 ± 0.93	16.51 ± 0.15	24.06 ± 0.90	19.20 ± 1.41	**28.07 ± 1.19**
InsectEPGRegularTrain	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	56.47 ± 1.83	78.96 ± 9.81	59.57 ± 2.67	38.96 ± 2.13	57.30 ± 1.41	53.17 ± 6.32	61.31 ± 3.42	89.88 ± 9.24	80.24 ± 3.80
InsectEPGSmallTrain	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	53.98 ± 1.90	77.03 ± 8.38	59.44 ± 2.44	52.61 ± 2.78	60.51 ± 1.98	55.66 ± 1.65	62.12 ± 1.62	83.13 ± 0.00	83.05 ± 2.80
InsectWingbeatSound	25.39 ± 1.91	24.63 ± 1.27	26.10 ± 5.93	53.74 ± 0.54	42.18 ± 1.58	56.25 ± 3.65	25.84 ± 1.75	**58.96 ± 0.98**	38.75 ± 2.42	57.51 ± 1.69	20.02 ± 5.68	41.44 ± 2.25
ItalyPowerDemand	87.66 ± 4.88	89.29 ± 4.25	88.36 ± 4.24	62.66 ± 11.58	93.39 ± 3.05	58.50 ± 7.42	89.57 ± 5.31	93.78 ± 2.95	**95.72 ± 1.14**	85.46 ± 2.44	56.64 ± 14.68	84.26 ± 5.94
LargeKitchenAppliances	60.00 ± 4.27	58.51 ± 9.39	**62.67 ± 10.64**	37.55 ± 2.23	60.48 ± 4.88	44.09 ± 3.07	36.98 ± 2.46	40.44 ± 3.60	37.71 ± 3.74	44.09 ± 6.61	37.17 ± 1.99	50.13 ± 3.07
Lightning2	64.92 ± 6.82	62.30 ± 7.42	67.87 ± 5.51	60.33 ± 3.91	64.92 ± 4.86	**69.95 ± 2.50**	50.82 ± 5.68	68.85 ± 5.68	61.64 ± 7.91	68.31 ± 3.41	59.02 ± 6.86	69.51 ± 10.47
Lightning7	58.90 ± 7.12	54.25 ± 7.09	60.55 ± 3.80	51.23 ± 2.84	50.96 ± 8.07	67.58 ± 3.45	18.26 ± 3.95	61.64 ± 3.62	30.41 ± 0.61	**70.32 ± 2.85**	22.19 ± 2.63	66.30 ± 3.83
Mallat	71.03 ± 13.07	70.33 ± 5.20	93.72 ± 2.08	88.67 ± 2.40	77.56 ± 9.50	**94.66 ± 0.33**	65.67 ± 5.01	87.51 ± 2.78	79.72 ± 5.50	90.06 ± 1.99	56.21 ± 5.78	90.88 ± 1.15
Meat	64.33 ± 14.22	50.00 ± 9.93	81.00 ± 8.47	85.33 ± 10.70	49.33 ± 15.48	76.67 ± 10.41	47.78 ± 15.03	37.22 ± 5.36	50.00 ± 11.61	**88.89 ± 2.55**	34.00 ± 1.49	64.00 ± 11.82
MedicalImages	**45.61 ± 3.94**	43.95 ± 4.74	42.63 ± 4.22	27.26 ± 7.75	32.26 ± 2.93	37.41 ± 0.59	36.84 ± 3.23	38.03 ± 3.53	40.16 ± 6.60	38.20 ± 1.38	5.24 ± 3.29	42.03 ± 2.75
MelbournePedestrian	76.79 ± 2.31	74.96 ± 2.28	**77.84 ± 3.11**	11.65 ± 1.29	58.25 ± 3.82	62.66 ± 1.83	70.73 ± 4.35	71.76 ± 3.72	70.23 ± 2.95	67.90 ± 5.27	58.25 ± 3.82	65.81 ± 3.77
MiddlePhalanxOutlineAgeGroup	47.01 ± 8.99	54.03 ± 8.97	50.78 ± 7.01	52.08 ± 8.63	49.09 ± 6.85	50.22 ± 4.61	51.73 ± 7.75	53.90 ± 7.87	**55.06 ± 4.76**	54.11 ± 9.46	31.82 ± 16.75	49.74 ± 4.72
MiddlePhalanxOutlineCorrect	53.20 ± 7.83	52.71 ± 15.62	58.14 ± 12.45	46.94 ± 12.76	48.93 ± 10.44	49.60 ± 3.67	**58.65 ± 6.95**	55.44 ± 15.07	51.68 ± 10.94	49.60 ± 9.23	50.93 ± 8.48	48.45 ± 7.93
MiddlePhalanxTW	**45.84 ± 9.64**	**45.84 ± 3.45**	45.71 ± 4.74	**45.84 ± 4.10**	43.12 ± 4.67	44.37 ± 0.37	40.04 ± 2.62	43.72 ± 4.56	44.16 ± 4.95	44.59 ± 2.28	22.86 ± 6.13	43.64 ± 5.71
MixedShapesRegularTrain	63.93 ± 9.77	61.94 ± 4.99	68.21 ± 8.35	68.04 ± 1.34	68.63 ± 3.58	67.15 ± 2.55	20.78 ± 0.21	70.05 ± 0.70	30.67 ± 6.49	67.15 ± 1.60	64.11 ± 1.37	**76.68 ± 5.11**

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TABLE 12: Per-dataset 5-shot classification results on 128 UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	TimeMorph
MixedShapesSmallTrain	66.24 ± 12.23	57.21 ± 4.26	72.37 ± 7.91	66.37 ± 2.62	66.13 ± 3.66	68.30 ± 1.42	22.61 ± 0.88	67.60 ± 3.67	25.23 ± 4.04	66.78 ± 1.26	60.00 ± 5.54	**78.71 ± 1.70**
MoteStrain	84.74 ± 4.04	83.85 ± 4.63	85.10 ± 2.03	80.19 ± 8.46	77.96 ± 3.58	**85.73 ± 4.20**	72.68 ± 2.25	80.09 ± 3.68	77.11 ± 6.04	85.36 ± 1.87	60.30 ± 15.64	80.37 ± 4.50
NonInvasiveFetalECGThorax1	40.80 ± 1.72	31.63 ± 3.55	40.80 ± 1.72	**72.10 ± 1.96**	51.79 ± 9.05	70.09 ± 3.02	22.04 ± 0.40	70.72 ± 1.13	32.89 ± 2.08	69.19 ± 0.16	28.56 ± 4.30	67.42 ± 0.56
NonInvasiveFetalECGThorax2	41.15 ± 5.46	43.13 ± 8.61	41.15 ± 5.46	**78.55 ± 0.96**	60.71 ± 3.55	75.84 ± 1.88	37.61 ± 1.37	75.40 ± 1.50	50.66 ± 2.46	74.93 ± 2.49	37.30 ± 2.25	75.31 ± 0.82
OSULeaf	57.85 ± 5.97	50.74 ± 10.99	**72.23 ± 4.84**	32.40 ± 1.86	35.37 ± 3.13	38.84 ± 1.09	16.80 ± 1.04	32.64 ± 2.71	24.05 ± 2.60	36.91 ± 1.04	23.06 ± 4.07	49.92 ± 6.46
OliveOil	59.33 ± 13.21	28.67 ± 13.66	**74.00 ± 6.83**	31.33 ± 9.60	26.67 ± 10.00	42.22 ± 25.02	40.00 ± 10.00	32.22 ± 13.47	40.00 ± 21.73	32.22 ± 22.19	26.67 ± 10.00	28.67 ± 11.69
PLAID	24.02 ± 4.96	24.51 ± 3.44	22.87 ± 2.95	30.06 ± 4.34	47.93 ± 3.93	36.56 ± 2.69	26.88 ± 2.01	40.16 ± 2.24	27.26 ± 3.48	37.93 ± 4.09	47.93 ± 3.93	**52.96 ± 5.39**
PhalangesOutlinesCorrect	56.22 ± 7.89	55.22 ± 9.82	**56.85 ± 6.96**	52.75 ± 11.57	53.15 ± 7.53	51.83 ± 11.55	50.35 ± 9.47	51.20 ± 13.21	53.05 ± 9.33	50.85 ± 11.66	53.61 ± 9.97	53.61 ± 8.69
Phoneme	**19.05 ± 2.97**	18.15 ± 2.47	**19.05 ± 2.97**	6.61 ± 0.25	9.00 ± 1.08	10.58 ± 0.95	6.87 ± 0.83	9.00 ± 0.29	9.61 ± 1.88	9.30 ± 1.34	8.21 ± 0.75	15.63 ± 1.56
PickupGestureWiimoteZ	56.00 ± 10.20	66.80 ± 5.93	**73.60 ± 4.56**	52.40 ± 1.67	56.80 ± 9.55	57.33 ± 3.06	27.33 ± 1.15	58.00 ± 3.46	42.00 ± 2.00	58.67 ± 1.15	56.80 ± 9.55	68.00 ± 2.45
PigAirwayPressure	25.00 ± 2.61	13.37 ± 2.05	**48.17 ± 0.40**	5.10 ± 0.26	4.62 ± 0.55	4.81 ± 0.00	4.17 ± 2.22	6.73 ± 0.48	4.04 ± 0.87	5.93 ± 0.56	4.42 ± 0.86	9.81 ± 0.73
PigArtPressure	50.96 ± 5.81	20.77 ± 3.29	**68.17 ± 4.63**	8.46 ± 0.65	5.10 ± 1.82	8.97 ± 1.21	6.57 ± 2.00	8.97 ± 0.73	6.92 ± 1.65	9.29 ± 1.11	18.17 ± 2.24	33.08 ± 1.33
PigCVP	29.33 ± 2.90	18.75 ± 2.40	**62.98 ± 1.73**	6.83 ± 0.22	3.37 ± 1.32	7.53 ± 0.56	4.65 ± 1.11	7.53 ± 0.28	3.75 ± 1.24	4.81 ± 0.48	10.38 ± 1.21	19.52 ± 0.55
Plane	**100.00 ± 0.00**	98.10 ± 2.43	99.62 ± 0.52	88.95 ± 7.93	95.24 ± 3.16	97.78 ± 0.55	96.51 ± 1.98	95.87 ± 1.45	97.14 ± 1.51	95.87 ± 0.55	23.05 ± 11.32	99.24 ± 1.24
PowerCons	80.56 ± 3.36	**84.44 ± 5.02**	84.00 ± 6.30	70.11 ± 5.26	81.11 ± 10.74	75.37 ± 5.45	54.26 ± 3.90	80.93 ± 10.72	52.22 ± 4.39	78.52 ± 10.32	68.89 ± 15.97	76.11 ± 6.55
ProximalPhalanxOutlineAgeGroup	73.37 ± 8.19	81.76 ± 3.12	79.90 ± 3.95	65.56 ± 9.78	75.41 ± 5.61	79.02 ± 7.19	79.02 ± 8.46	**84.07 ± 1.71**	74.34 ± 7.73	80.00 ± 3.41	43.61 ± 12.69	76.88 ± 5.79
ProximalPhalanxOutlineCorrect	**69.35 ± 11.27**	69.00 ± 5.91	69.28 ± 9.35	64.47 ± 0.52	63.71 ± 8.48	65.18 ± 3.44	64.60 ± 1.57	63.69 ± 1.73	63.16 ± 5.61	66.67 ± 4.47	55.26 ± 17.89	65.84 ± 4.70
ProximalPhalanxTW	62.34 ± 7.16	66.44 ± 4.62	**69.37 ± 2.52**	53.37 ± 11.45	65.27 ± 5.74	58.05 ± 7.67	58.21 ± 7.39	55.45 ± 4.15	60.00 ± 4.28	55.28 ± 3.66	27.32 ± 18.90	66.24 ± 3.19
RefrigerationDevices	42.83 ± 1.65	43.20 ± 6.13	**44.00 ± 6.18**	33.33 ± 2.73	37.92 ± 6.42	38.93 ± 3.24	31.20 ± 3.23	32.98 ± 2.22	32.32 ± 2.29	38.04 ± 3.74	34.61 ± 1.83	40.96 ± 5.59
Rock	42.00 ± 5.29	36.80 ± 7.16	38.40 ± 4.34	80.40 ± 2.97	60.80 ± 5.93	80.67 ± 1.15	81.33 ± 3.06	**88.67 ± 2.31**	59.20 ± 3.63	85.33 ± 3.06	33.60 ± 5.90	76.00 ± 3.16
ScreenType	**42.56 ± 7.61**	41.33 ± 8.02	40.64 ± 8.17	35.41 ± 2.78	36.16 ± 4.77	35.56 ± 2.39	32.18 ± 2.24	36.09 ± 2.30	32.53 ± 1.41	35.02 ± 0.41	34.08 ± 2.12	34.51 ± 4.14
SemgHandGenderCh2	58.13 ± 2.49	66.47 ± 7.67	56.07 ± 2.82	65.13 ± 3.61	**67.13 ± 2.65**	50.28 ± 5.31	55.83 ± 13.67	55.06 ± 2.04	54.63 ± 4.30	47.94 ± 16.07	59.97 ± 7.08	66.80 ± 7.25
SemgHandMovementCh2	30.89 ± 3.99	33.91 ± 3.59	34.40 ± 6.06	32.84 ± 5.27	26.49 ± 1.89	27.70 ± 2.36	20.59 ± 1.85	29.48 ± 2.44	21.64 ± 2.79	23.41 ± 3.03	33.16 ± 2.38	**36.31 ± 5.92**
SemgHandSubjectCh2	37.11 ± 4.58	44.09 ± 1.20	35.60 ± 3.13	50.76 ± 1.90	49.02 ± 0.83	46.67 ± 2.12	27.78 ± 0.59	51.19 ± 2.74	30.62 ± 1.93	41.48 ± 6.03	36.62 ± 4.30	**54.31 ± 3.37**
ShakeGestureWiimoteZ	80.40 ± 4.34	91.60 ± 2.61	89.20 ± 1.10	56.40 ± 1.67	72.80 ± 5.22	62.00 ± 0.00	35.33 ± 1.15	60.67 ± 3.06	44.00 ± 2.45	55.33 ± 2.31	72.80 ± 5.22	**92.00 ± 0.00**
ShapeletSim	82.78 ± 10.96	**99.67 ± 0.75**	92.56 ± 6.55	49.44 ± 2.69	88.22 ± 10.06	55.93 ± 2.31	49.81 ± 2.80	49.63 ± 4.21	50.67 ± 1.82	49.44 ± 3.33	49.67 ± 3.98	92.00 ± 3.88
ShapesAll	57.57 ± 3.68	48.33 ± 5.58	57.57 ± 3.68	58.13 ± 1.68	51.50 ± 3.69	64.67 ± 1.64	9.22 ± 0.79	65.83 ± 3.88	13.83 ± 3.47	59.22 ± 2.28	56.47 ± 1.46	**76.20 ± 1.83**
SmallKitchenAppliances	59.47 ± 8.51	57.39 ± 6.38	**62.35 ± 6.85**	39.04 ± 4.11	57.55 ± 2.52	57.51 ± 6.62	38.67 ± 2.54	39.56 ± 5.26	46.67 ± 3.53	56.62 ± 3.96	44.00 ± 7.15	51.63 ± 12.25
SmoothSubspace	86.80 ± 2.13	82.00 ± 2.83	**88.67 ± 2.36**	33.87 ± 1.19	57.07 ± 4.58	65.78 ± 2.04	69.33 ± 3.71	61.11 ± 0.77	77.07 ± 5.51	84.00 ± 4.81	37.07 ± 6.74	88.00 ± 2.87
SonyAIBORobotSurface1	93.94 ± 2.22	**95.71 ± 1.41**	92.85 ± 2.46	61.83 ± 10.72	84.73 ± 2.02	71.21 ± 3.16	67.39 ± 1.20	71.60 ± 6.11	64.96 ± 2.40	64.89 ± 3.03	54.54 ± 12.34	84.86 ± 2.67
SonyAIBORobotSurface2	**92.51 ± 3.32**	91.14 ± 0.89	91.54 ± 4.19	78.15 ± 2.59	85.83 ± 4.10	79.96 ± 1.73	83.11 ± 2.73	81.78 ± 2.73	80.76 ± 4.57	83.21 ± 2.88	61.15 ± 14.84	77.21 ± 3.67
StarLightCurves	81.52 ± 8.85	60.67 ± 16.70	76.44 ± 1.22	74.70 ± 4.25	76.72 ± 3.78	75.63 ± 4.10	28.48 ± 9.45	75.33 ± 3.39	27.99 ± 0.00	75.69 ± 5.23	75.95 ± 2.54	**81.86 ± 7.93**
Strawberry	63.51 ± 2.99	64.22 ± 4.90	60.22 ± 9.83	58.49 ± 2.90	63.03 ± 7.09	61.89 ± 4.92	**67.21 ± 4.90**	59.82 ± 1.80	61.03 ± 3.64	65.77 ± 7.40	52.92 ± 6.34	58.05 ± 7.04
SwedishLeaf	**79.81 ± 1.70**	76.00 ± 4.97	**79.81 ± 1.70**	47.97 ± 1.68	65.66 ± 4.54	62.61 ± 4.53	49.23 ± 4.26	67.15 ± 4.00	56.93 ± 3.43	62.29 ± 1.68	15.90 ± 6.80	79.78 ± 2.56
Symbols	88.78 ± 7.99	88.74 ± 7.74	**91.24 ± 0.96**	83.44 ± 1.29	84.92 ± 3.31	84.29 ± 1.01	36.11 ± 0.23	86.40 ± 1.63	58.19 ± 7.40	83.75 ± 0.90	27.56 ± 4.43	88.04 ± 1.30
SyntheticControl	87.80 ± 2.38	**96.33 ± 0.33**	92.60 ± 1.01	48.53 ± 3.68	92.73 ± 2.94	73.22 ± 3.10	52.11 ± 3.20	66.22 ± 3.66	56.27 ± 3.45	72.78 ± 3.66	20.93 ± 4.87	95.07 ± 1.62
ToeSegmentation1	**85.96 ± 7.05**	77.37 ± 5.14	81.58 ± 5.14	54.74 ± 3.58	68.16 ± 3.79	55.56 ± 2.16	53.95 ± 2.74	51.61 ± 0.91	53.25 ± 3.38	54.82 ± 1.32	48.77 ± 4.31	70.53 ± 10.39
ToeSegmentation2	74.77 ± 12.53	77.85 ± 7.33	74.77 ± 8.79	50.77 ± 2.88	68.46 ± 6.25	58.97 ± 1.60	50.26 ± 0.44	59.49 ± 6.72	37.54 ± 3.14	64.10 ± 2.70	52.31 ± 9.12	**78.00 ± 6.36**
Trace	88.60 ± 7.37	99.00 ± 1.00	93.00 ± 8.00	53.00 ± 4.42	76.40 ± 5.77	51.67 ± 4.93	53.33 ± 2.08	49.67 ± 3.79	59.80 ± 2.59	55.67 ± 8.50	48.80 ± 10.71	**100.00 ± 0.00**
TwoLeadECG	**97.98 ± 4.17**	95.47 ± 5.89	75.84 ± 5.58	58.75 ± 2.38	82.55 ± 14.28	62.16 ± 1.52	60.81 ± 0.10	56.34 ± 2.59	61.88 ± 1.65	68.51 ± 9.61	52.10 ± 2.73	96.68 ± 5.58
TwoPatterns	64.62 ± 3.64	**74.98 ± 5.04**	74.94 ± 1.43	38.09 ± 3.07	51.91 ± 7.48	40.83 ± 3.89	26.00 ± 0.92	39.83 ± 3.80	26.50 ± 1.83	37.54 ± 1.63	27.50 ± 4.13	74.22 ± 2.81
UMD	87.08 ± 10.10	91.53 ± 5.28	**96.67 ± 2.05**	56.53 ± 3.69	74.44 ± 17.00	61.57 ± 1.45	62.96 ± 5.82	68.52 ± 10.79	71.81 ± 2.17	70.14 ± 10.69	48.47 ± 7.32	95.78 ± 2.33
UWaveGestureLibraryAll	42.53 ± 1.15	39.33 ± 5.11	44.53 ± 1.84	77.02 ± 5.21	57.17 ± 2.24	**78.18 ± 3.39**	26.42 ± 1.62	77.47 ± 3.61	35.67 ± 1.53	76.96 ± 4.32	65.94 ± 6.48	48.36 ± 0.30
UWaveGestureLibraryX	39.64 ± 3.12	46.31 ± 4.46	46.23 ± 4.40	56.88 ± 2.64	50.95 ± 2.22	58.42 ± 2.37	19.27 ± 3.18	**58.58 ± 2.34**	31.22 ± 3.55	57.86 ± 1.47	26.48 ± 5.59	47.36 ± 0.30
UWaveGestureLibraryY	37.00 ± 1.20	38.44 ± 2.57	40.23 ± 3.63	**52.69 ± 2.24**	48.41 ± 3.12	51.42 ± 3.10	21.22 ± 1.11	51.72 ± 2.98	30.92 ± 5.43	50.81 ± 4.53	19.79 ± 3.07	51.36 ± 0.31
UWaveGestureLibraryZ	38.74 ± 2.41	44.95 ± 4.51	40.85 ± 4.13	48.35 ± 3.41	47.76 ± 4.10	50.34 ± 2.37	20.23 ± 1.68	49.48 ± 1.66	31.23 ± 4.16	48.05 ± 3.36	25.20 ± 6.78	**55.54 ± 2.61**
Wafer	76.83 ± 9.15	69.99 ± 14.88	71.79 ± 15.46	55.60 ± 16.77	73.34 ± 19.66	80.42 ± 16.85	76.25 ± 9.34	**83.86 ± 16.37**	75.80 ± 12.54	83.26 ± 17.85	64.01 ± 22.29	71.65 ± 9.58
Wine	57.41 ± 13.03	48.15 ± 3.93	**61.11 ± 5.40**	55.93 ± 3.56	53.33 ± 5.62	47.53 ± 8.35	54.94 ± 8.55	50.00 ± 0.00	46.67 ± 6.47	37.65 ± 19.80	50.00 ± 0.00	51.48 ± 2.03
WordSynonyms	25.08 ± 1.54	21.22 ± 1.56	25.08 ± 1.54	29.56 ± 2.06	30.38 ± 2.74	45.87 ± 4.63	7.99 ± 1.24	44.67 ± 4.83	18.81 ± 2.07	41.33 ± 4.67	22.26 ± 2.64	**47.05 ± 3.87**
Worms	57.14 ± 9.76	55.58 ± 5.91	**62.86 ± 4.55**	24.68 ± 4.68	40.26 ± 5.11	22.94 ± 0.75	19.91 ± 3.27	25.97 ± 0.00	16.62 ± 2.32	22.08 ± 1.30	22.08 ± 5.11	44.42 ± 6.26
WormsTwoClass	**68.05 ± 7.21**	59.74 ± 6.69	62.08 ± 7.97	48.83 ± 3.13	51.17 ± 8.09	52.38 ± 2.70	42.86 ± 3.44	51.08 ± 5.25	48.83 ± 3.26	52.81 ± 3.27	48.83 ± 5.92	54.81 ± 7.59
Yoga	**55.71 ± 3.37**	53.57 ± 4.72	55.19 ± 2.57	50.13 ± 2.22	53.89 ± 3.78	49.79 ± 3.21	52.94 ± 1.57	47.50 ± 3.55	51.67 ± 1.99	49.68 ± 4.28	48.87 ± 2.05	51.31 ± 4.34
Avg. Acc.	61.27	59.47	63.55	52.30	56.94	57.86	43.20	57.20	47.30	58.35	42.94	**64.14**
Avg. Rank	4.77	5.34	4.23	7.73	6.60	5.69	9.55	5.94	8.70	5.79	9.48	**4.19**
#Best	21	13	30	6	4	8	4	7	2	8	1	**36**

TABLE 13: Per-dataset 10-shot classification results on 128 UCR datasets. Each value is reported as mean ± standard deviation over 5 runs.

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	TimeMorph
ACSF1	**85.40 ± 2.19**	76.80 ± 3.35	80.80 ± 0.84	38.60 ± 0.89	58.80 ± 2.59	64.67 ± 0.58	42.67 ± 1.53	67.00 ± 2.00	53.60 ± 2.51	67.00 ± 2.00	36.80 ± 3.27	81.33 ± 1.53
Adiac	62.46 ± 4.53	56.98 ± 3.84	62.46 ± 4.53	45.32 ± 1.06	47.72 ± 5.48	56.01 ± 2.77	48.68 ± 0.78	47.91 ± 6.68	71.71 ± 1.94	48.00 ± 3.58	12.84 ± 1.42	**76.42 ± 1.68**
AllGestureWiimoteX	49.37 ± 3.15	43.11 ± 2.57	39.26 ± 2.50	23.91 ± 1.60	43.54 ± 2.85	37.14 ± 0.52	15.00 ± 0.87	35.19 ± 1.33	25.80 ± 2.24	31.00 ± 1.57	43.54 ± 2.85	**52.31 ± 2.26**
AllGestureWiimoteY	54.89 ± 2.09	50.74 ± 4.62	40.17 ± 2.24	27.26 ± 1.11	48.03 ± 1.58	42.57 ± 3.19	16.95 ± 2.22	42.57 ± 3.81	30.37 ± 4.65	41.24 ± 2.84	48.03 ± 1.58	**56.34 ± 3.54**
AllGestureWiimoteZ	**49.34 ± 1.38**	46.03 ± 2.76	39.80 ± 2.97	22.43 ± 1.80	34.54 ± 4.20	31.29 ± 1.13	16.67 ± 1.15	29.67 ± 3.32	21.71 ± 1.69	32.19 ± 0.59	34.54 ± 4.20	44.43 ± 2.91
ArrowHead	72.23 ± 5.22	63.89 ± 7.06	65.49 ± 9.09	64.34 ± 0.65	62.86 ± 7.57	60.38 ± 1.19	48.19 ± 2.58	59.62 ± 2.88	57.83 ± 5.80	63.43 ± 3.48	45.83 ± 13.23	**81.49 ± 3.22**
BME	64.27 ± 9.29	69.33 ± 1.56	78.67 ± 4.64	64.53 ± 0.30	97.60 ± 3.29	71.11 ± 2.34	78.89 ± 1.02	88.44 ± 1.02	91.33 ± 2.31	86.67 ± 0.00	46.40 ± 7.46	**99.20 ± 0.30**
Beef	51.33 ± 6.91	44.00 ± 6.83	52.00 ± 2.98	61.33 ± 4.47	46.00 ± 6.41	70.00 ± 3.33	51.11 ± 5.09	54.44 ± 3.85	64.00 ± 7.23	**80.00 ± 0.00**	37.33 ± 4.35	68.00 ± 1.83
BeetleFly	71.00 ± 8.94	74.00 ± 6.52	75.00 ± 6.12	86.00 ± 4.18	86.00 ± 2.24	81.67 ± 2.89	48.33 ± 2.89	83.33 ± 2.89	49.00 ± 11.94	75.00 ± 5.00	70.00 ± 12.25	**96.67 ± 2.89**
BirdChicken	84.00 ± 12.45	92.00 ± 5.70	**94.00 ± 4.18**	68.00 ± 2.74	81.00 ± 9.62	71.67 ± 7.64	41.67 ± 7.64	68.33 ± 16.07	56.00 ± 5.48	56.67 ± 7.64	56.00 ± 4.18	89.00 ± 5.48
CBF	98.96 ± 0.23	98.27 ± 0.96	**99.24 ± 0.32**	67.49 ± 2.09	95.98 ± 4.64	90.07 ± 1.09	39.37 ± 1.51	91.44 ± 1.93	51.24 ± 7.49	90.11 ± 0.88	49.31 ± 9.05	97.71 ± 0.69
Car	57.67 ± 10.84	58.67 ± 3.21	65.33 ± 2.17	73.00 ± 2.98	62.67 ± 5.22	70.00 ± 2.89	27.22 ± 5.85	65.56 ± 0.96	37.33 ± 3.84	72.22 ± 1.92	50.33 ± 8.28	**76.33 ± 5.45**
Chinatown	98.19 ± 0.13	98.13 ± 0.16	**98.54 ± 0.21**	54.58 ± 24.80	97.73 ± 0.75	94.75 ± 2.10	95.63 ± 1.01	97.28 ± 0.61	96.62 ± 0.60	96.60 ± 1.10	63.56 ± 20.21	94.56 ± 3.09
ChlorineConcentration	**41.18 ± 3.66**	34.41 ± 8.63	37.86 ± 5.20	34.42 ± 4.40	35.51 ± 6.32	35.06 ± 1.39	38.85 ± 1.62	32.99 ± 4.01	39.05 ± 4.08	39.50 ± 4.07	33.01 ± 11.68	34.77 ± 6.93
CinCECGTorso	54.03 ± 3.03	65.94 ± 5.62	64.64 ± 5.11	45.36 ± 1.09	80.65 ± 7.24	81.59 ± 1.76	35.82 ± 2.41	84.42 ± 2.94	63.93 ± 1.27	85.19 ± 1.24	52.77 ± 4.47	**94.71 ± 4.65**
Coffee	97.14 ± 6.39	97.14 ± 1.60	**100.00 ± 0.00**	96.43 ± 4.37	97.86 ± 1.96	91.67 ± 2.06	86.90 ± 16.50	92.86 ± 0.00	98.57 ± 3.19	90.48 ± 2.06	56.43 ± 7.74	95.71 ± 2.99
Computers	66.48 ± 3.03	63.20 ± 6.81	**68.08 ± 4.62**	48.88 ± 2.52	56.40 ± 10.98	52.13 ± 9.60	52.13 ± 6.02	50.93 ± 12.73	49.68 ± 5.04	53.87 ± 12.94	51.52 ± 3.69	65.60 ± 6.66
CricketX	60.72 ± 2.35	55.18 ± 4.36	51.28 ± 3.40	30.15 ± 2.84	42.92 ± 1.99	46.50 ± 1.21	14.27 ± 1.71	45.38 ± 2.45	20.87 ± 2.94	45.38 ± 1.94	19.33 ± 2.95	**66.92 ± 2.71**
CricketY	56.92 ± 2.81	49.44 ± 1.94	47.69 ± 2.43	32.82 ± 4.21	42.15 ± 5.17	44.62 ± 0.44	9.32 ± 1.04	44.02 ± 2.09	17.90 ± 2.44	41.11 ± 0.53	23.95 ± 3.97	**57.95 ± 1.33**
CricketZ	63.18 ± 1.66	57.38 ± 2.58	48.82 ± 4.98	30.56 ± 3.52	43.18 ± 5.27	46.32 ± 1.71	12.91 ± 2.15	47.78 ± 1.41	18.26 ± 2.08	45.30 ± 1.50	21.95 ± 3.02	**65.30 ± 1.16**
Crop	57.92 ± 1.22	54.52 ± 1.67	57.92 ± 1.22	42.46 ± 2.10	44.96 ± 1.35	54.14 ± 2.07	52.84 ± 0.15	54.82 ± 1.79	54.74 ± 0.79	54.60 ± 1.45	25.02 ± 2.10	**57.97 ± 1.25**
DiatomSizeReduction	87.32 ± 3.50	71.05 ± 13.58	**96.80 ± 0.91**	44.77 ± 1.16	62.68 ± 4.64	87.04 ± 1.80	66.88 ± 2.86	51.96 ± 5.03	48.89 ± 18.63	87.47 ± 0.19	30.13 ± 0.15	92.61 ± 2.00
DistalPhalanxOutlineAgeGroup	68.20 ± 2.98	68.78 ± 4.24	70.50 ± 1.61	69.50 ± 1.66	69.50 ± 2.81	66.67 ± 1.10	65.71 ± 3.62	70.50 ± 1.25	68.78 ± 2.72	66.91 ± 1.25	48.06 ± 11.76	**70.65 ± 2.18**
DistalPhalanxOutlineCorrect	64.78 ± 12.49	65.07 ± 11.29	63.62 ± 12.87	50.00 ± 13.55	60.14 ± 9.92	58.94 ± 4.04	**67.15 ± 2.41**	55.43 ± 14.84	62.39 ± 9.46	61.71 ± 6.72	47.61 ± 8.80	60.51 ± 11.85
DistalPhalanxTW	55.83 ± 3.65	59.14 ± 2.12	**60.58 ± 3.58**	47.34 ± 6.42	57.55 ± 6.08	53.96 ± 3.14	53.00 ± 2.91	54.92 ± 0.83	52.52 ± 4.35	54.92 ± 2.91	32.37 ± 14.50	54.10 ± 3.11
DodgerLoopDay	49.25 ± 3.71	47.25 ± 4.28	39.75 ± 3.11	41.50 ± 2.24	47.00 ± 4.11	**49.58 ± 5.64**	28.75 ± 2.50	45.42 ± 1.91	29.00 ± 2.40	48.75 ± 1.25	47.00 ± 4.11	48.75 ± 1.77
DodgerLoopGame	68.70 ± 6.09	83.33 ± 5.28	75.07 ± 2.64	79.28 ± 0.97	**88.84 ± 1.67**	76.81 ± 1.26	48.79 ± 3.42	81.40 ± 1.51	53.48 ± 2.83	78.99 ± 1.45	**88.84 ± 1.67**	83.62 ± 0.65
DodgerLoopWeekend	94.06 ± 1.39	94.20 ± 2.11	96.81 ± 0.83	**97.83 ± 0.00**	96.81 ± 0.65	96.86 ± 0.42	72.71 ± 1.82	97.58 ± 0.42	76.38 ± 1.82	97.10 ± 0.00	96.81 ± 0.65	**97.83 ± 0.00**
ECG200	80.00 ± 4.00	**83.00 ± 2.00**	80.40 ± 4.28	77.60 ± 4.34	79.80 ± 6.53	79.33 ± 3.21	78.00 ± 3.61	81.00 ± 5.00	82.00 ± 2.92	82.00 ± 3.61	63.00 ± 15.26	77.40 ± 4.88
ECG5000	83.56 ± 5.09	78.23 ± 2.20	80.72 ± 4.37	**88.37 ± 1.22**	76.31 ± 5.83	79.59 ± 2.75	73.64 ± 2.21	84.75 ± 1.16	79.28 ± 5.11	82.19 ± 1.79	24.56 ± 15.15	73.78 ± 6.18
ECGFiveDays	95.40 ± 1.88	88.99 ± 4.44	**97.56 ± 1.20**	74.87 ± 4.61	94.61 ± 2.00	83.82 ± 7.05	96.44 ± 1.42	91.64 ± 1.90	97.51 ± 1.12	86.18 ± 8.29	57.47 ± 4.38	75.28 ± 1.71
EOGHorizontalSignal	30.61 ± 3.99	43.15 ± 5.84	30.94 ± 3.48	29.89 ± 1.60	38.78 ± 6.14	39.59 ± 1.67	8.01 ± 1.27	41.62 ± 1.42	10.61 ± 3.44	40.06 ± 3.72	39.56 ± 4.17	**49.17 ± 1.62**
EOGVerticalSignal	21.71 ± 2.53	28.23 ± 2.47	19.83 ± 3.08	31.55 ± 2.19	40.06 ± 1.98	36.00 ± 1.12	8.47 ± 0.84	37.75 ± 1.67	11.99 ± 1.33	37.02 ± 1.99	30.55 ± 3.17	**42.21 ± 2.26**
Earthquakes	53.81 ± 8.62	53.96 ± 9.50	51.37 ± 3.47	52.52 ± 6.49	54.82 ± 2.85	60.91 ± 4.63	49.16 ± 8.34	59.71 ± 6.86	54.68 ± 5.26	**61.87 ± 3.60**	47.91 ± 6.66	54.68 ± 7.63
ElectricDevices	40.12 ± 4.42	38.65 ± 3.96	36.00 ± 4.64	28.04 ± 1.63	39.08 ± 8.26	48.59 ± 6.21	30.07 ± 3.26	33.74 ± 3.43	37.44 ± 1.49	48.14 ± 2.51	16.08 ± 5.57	**49.42 ± 5.02**
EthanolLevel	26.56 ± 1.96	25.76 ± 1.05	26.16 ± 0.71	29.28 ± 1.78	25.92 ± 1.70	28.07 ± 2.55	25.20 ± 3.34	26.80 ± 0.69	25.60 ± 1.13	**29.87 ± 2.47**	25.56 ± 1.05	28.24 ± 0.50
FaceAll	77.73 ± 3.48	**80.30 ± 1.67**	77.73 ± 3.48	67.09 ± 2.40	54.49 ± 5.41	67.46 ± 1.82	59.68 ± 1.83	67.32 ± 2.82	60.69 ± 2.63	64.06 ± 2.32	32.89 ± 2.71	63.01 ± 1.34
FaceFour	87.95 ± 3.07	59.32 ± 7.51	**92.95 ± 2.03**	74.77 ± 0.95	66.36 ± 10.77	85.98 ± 1.31	65.91 ± 2.27	88.26 ± 0.66	69.55 ± 1.69	83.71 ± 1.31	34.55 ± 3.82	79.09 ± 1.90
FacesUCR	**87.80 ± 1.60**	84.26 ± 2.06	82.41 ± 2.48	63.80 ± 1.79	54.40 ± 3.05	80.54 ± 1.15	70.11 ± 1.05	83.69 ± 1.13	72.08 ± 1.16	79.51 ± 1.53	34.67 ± 3.28	79.08 ± 1.41
FiftyWords	45.36 ± 2.39	36.70 ± 1.89	45.36 ± 2.39	53.67 ± 1.47	50.46 ± 2.83	64.32 ± 1.87	16.85 ± 0.67	67.99 ± 1.84	37.05 ± 3.62	57.36 ± 1.54	56.40 ± 1.98	**71.34 ± 2.10**
Fish	**83.31 ± 3.78**	56.00 ± 4.28	71.54 ± 6.28	65.60 ± 3.49	62.74 ± 2.15	68.95 ± 4.29	24.38 ± 3.80	71.05 ± 1.44	57.14 ± 3.45	75.05 ± 2.31	38.17 ± 5.05	78.29 ± 3.92
FordA	78.26 ± 5.14	75.61 ± 0.71	**78.70 ± 3.26**	49.09 ± 0.93	67.88 ± 3.90	51.99 ± 2.52	52.22 ± 0.95	48.81 ± 1.31	53.05 ± 1.09	48.21 ± 1.67	50.06 ± 2.65	57.15 ± 1.85
FordB	**63.83 ± 5.74**	62.17 ± 3.33	61.70 ± 3.19	48.54 ± 0.72	59.06 ± 1.85	46.75 ± 0.61	49.79 ± 1.45	47.24 ± 1.24	50.47 ± 2.25	47.33 ± 0.14	49.33 ± 1.67	55.60 ± 1.91
FreezerRegularTrain	86.27 ± 5.59	**89.20 ± 7.73**	79.00 ± 5.98	76.33 ± 0.75	80.48 ± 6.98	76.36 ± 0.25	61.75 ± 7.11	76.22 ± 0.12	68.27 ± 6.23	74.90 ± 1.40	66.57 ± 12.21	**89.20 ± 5.65**
FreezerSmallTrain	78.20 ± 4.17	80.08 ± 7.30	74.46 ± 2.32	76.11 ± 0.45	68.74 ± 3.00	73.18 ± 1.45	65.15 ± 4.43	71.92 ± 2.00	68.25 ± 2.23	70.97 ± 2.25	65.53 ± 10.69	**82.86 ± 3.49**
Fungi	93.98 ± 3.37	38.06 ± 6.98	67.85 ± 1.59	67.74 ± 4.64	69.57 ± 7.88	**95.88 ± 1.64**	72.94 ± 1.89	91.22 ± 1.73	74.73 ± 0.76	**95.88 ± 0.31**	12.47 ± 9.15	79.57 ± 1.20
GestureMidAirD1	49.38 ± 3.66	60.92 ± 1.48	49.38 ± 3.66	49.69 ± 1.40	48.31 ± 2.52	54.36 ± 1.94	26.15 ± 2.31	58.21 ± 2.22	44.15 ± 6.68	57.44 ± 1.60	48.31 ± 2.52	**67.08 ± 1.00**
GestureMidAirD2	38.00 ± 5.29	47.23 ± 3.34	38.00 ± 5.29	50.46 ± 0.69	50.46 ± 2.75	53.08 ± 2.66	11.28 ± 2.47	55.13 ± 1.18	34.00 ± 4.05	51.79 ± 1.18	50.46 ± 2.75	**55.69 ± 1.85**
GestureMidAirD3	20.15 ± 2.40	25.23 ± 1.38	20.15 ± 2.40	32.92 ± 0.34	29.23 ± 3.17	28.46 ± 0.77	15.13 ± 2.70	33.08 ± 2.04	29.38 ± 4.79	31.28 ± 0.44	29.23 ± 3.17	**33.69 ± 2.63**
GesturePebbleZ1	76.86 ± 5.40	75.47 ± 8.13	77.67 ± 2.20	66.63 ± 2.78	68.84 ± 4.61	75.00 ± 1.16	22.29 ± 3.87	77.33 ± 2.10	38.95 ± 10.62	**81.40 ± 2.53**	68.84 ± 4.61	75.12 ± 5.19
GesturePebbleZ2	73.16 ± 4.51	63.42 ± 5.15	73.16 ± 2.82	60.38 ± 3.46	67.59 ± 4.18	69.62 ± 2.90	19.20 ± 0.73	74.26 ± 2.56	28.86 ± 9.85	73.00 ± 1.59	67.59 ± 4.18	**75.57 ± 4.64**
GunPoint	85.73 ± 9.32	82.13 ± 8.02	91.73 ± 11.16	68.27 ± 5.09	77.60 ± 4.39	71.78 ± 3.79	64.00 ± 6.57	72.00 ± 1.15	77.73 ± 11.90	76.67 ± 6.67	62.00 ± 16.53	**97.47 ± 2.23**
GunPointAgeSpan	89.56 ± 7.42	72.47 ± 9.28	74.87 ± 11.59	83.54 ± 2.13	82.22 ± 4.04	85.44 ± 1.67	52.22 ± 3.12	85.13 ± 2.39	71.08 ± 9.43	83.54 ± 3.02	53.61 ± 10.43	**92.72 ± 5.80**
GunPointMaleVersusFemale	92.78 ± 8.49	93.48 ± 5.65	92.22 ± 9.65	67.09 ± 4.97	95.57 ± 2.50	91.14 ± 4.91	60.97 ± 6.06	93.57 ± 3.96	74.05 ± 7.11	94.41 ± 2.22	67.22 ± 11.33	**98.54 ± 0.94**
GunPointOldVersusYoung	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	76.19 ± 1.96	79.17 ± 3.15	78.20 ± 2.30	57.46 ± 1.38	77.25 ± 5.77	61.90 ± 5.09	79.26 ± 2.47	85.78 ± 19.21	86.79 ± 4.71
Ham	60.76 ± 5.19	66.48 ± 2.64	65.52 ± 4.83	**71.24 ± 5.65**	62.86 ± 4.86	66.03 ± 7.40	63.49 ± 10.09	68.89 ± 6.69	61.71 ± 8.23	68.89 ± 7.02	61.52 ± 3.79	62.86 ± 3.37
HandOutlines	50.76 ± 12.61	50.92 ± 12.38	57.46 ± 2.17	81.57 ± 4.57	**82.65 ± 3.66**	81.62 ± 2.31	38.65 ± 4.09	78.47 ± 5.74	35.95 ± 0.00	81.62 ± 1.95	76.97 ± 8.74	71.51 ± 5.35
Haptics	29.68 ± 4.36	29.16 ± 4.62	33.31 ± 5.65	37.73 ± 3.74	31.95 ± 2.95	37.66 ± 2.89	23.81 ± 5.22	37.66 ± 2.77	25.45 ± 4.31	38.20 ± 2.76	35.32 ± 3.26	**39.22 ± 4.29**
Herring	55.00 ± 9.20	55.62 ± 5.91	55.00 ± 7.28	50.31 ± 6.57	**59.69 ± 3.39**	52.60 ± 3.93	47.92 ± 2.39	53.65 ± 8.02	49.06 ± 4.50	50.52 ± 1.80	57.19 ± 6.68	55.31 ± 5.01
HouseTwenty	**90.92 ± 1.10**	88.91 ± 1.82	89.58 ± 3.18	65.55 ± 5.48	81.51 ± 3.71	81.23 ± 2.43	60.50 ± 3.36	74.79 ± 1.46	64.20 ± 2.95	81.79 ± 4.63	70.76 ± 6.04	89.24 ± 2.81
InlineSkate	30.04 ± 2.77	24.04 ± 1.78	21.27 ± 3.03	25.16 ± 1.55	21.67 ± 3.27	28.42 ± 1.51	16.91 ± 0.91	28.73 ± 0.83	18.22 ± 1.24	29.09 ± 2.20	22.62 ± 1.07	**33.49 ± 2.77**
InsectEPGRegularTrain	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	54.62 ± 4.82	79.12 ± 3.30	61.31 ± 3.90	42.70 ± 1.23	60.37 ± 1.29	55.18 ± 4.83	64.39 ± 5.02	96.63 ± 7.54	86.59 ± 4.06
InsectEPGSmallTrain	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	60.00 ± 1.05	81.69 ± 6.46	69.08 ± 1.20	55.82 ± 1.06	69.88 ± 0.80	58.71 ± 4.87	66.40 ± 1.81	83.13 ± 0.00	85.06 ± 2.08
InsectWingbeatSound	34.13 ± 0.95	34.52 ± 3.87	30.58 ± 3.83	60.48 ± 0.59	47.36 ± 1.34	60.99 ± 1.05	39.46 ± 3.03	**61.67 ± 1.18**	49.66 ± 3.26	61.20 ± 0.34	46.64 ± 2.91	51.36 ± 2.05
ItalyPowerDemand	92.21 ± 1.02	92.56 ± 1.30	92.75 ± 1.07	53.43 ± 4.80	93.97 ± 0.64	87.66 ± 8.42	94.69 ± 1.03	95.59 ± 0.11	**95.82 ± 0.32**	94.33 ± 0.49	49.23 ± 14.84	92.19 ± 2.62
LargeKitchenAppliances	64.05 ± 4.75	66.13 ± 5.93	66.08 ± 5.44	40.85 ± 3.41	**67.95 ± 4.05**	46.22 ± 6.49	39.82 ± 1.74	43.11 ± 5.30	40.53 ± 2.82	45.33 ± 5.07	41.71 ± 0.52	55.09 ± 5.30
Lightning2	70.49 ± 3.84	64.92 ± 7.48	65.57 ± 4.34	58.03 ± 2.74	69.18 ± 5.84	71.04 ± 1.89	51.37 ± 4.13	63.93 ± 3.28	58.36 ± 5.26	68.85 ± 4.34	60.00 ± 3.77	**77.05 ± 4.18**
Lightning7	64.38 ± 5.97	66.03 ± 3.41	60.00 ± 4.37	60.27 ± 3.36	61.10 ± 9.76	74.43 ± 1.58	20.09 ± 4.81	65.75 ± 1.37	36.71 ± 3.41	**75.34 ± 2.37**	26.30 ± 5.76	66.85 ± 7.34
Mallat	68.22 ± 8.96	69.46 ± 5.32	**95.78 ± 0.57**	90.38 ± 0.87	87.49 ± 9.31	92.79 ± 0.83	66.92 ± 5.03	85.20 ± 1.21	79.12 ± 6.08	89.69 ± 0.98	25.09 ± 0.73	89.72 ± 1.50
Meat	53.00 ± 18.31	59.33 ± 6.08	85.00 ± 4.86	**87.00 ± 7.94**	60.00 ± 5.27	61.11 ± 4.19	68.89 ± 7.52	55.56 ± 19.25	55.33 ± 20.66	86.11 ± 7.88	38.67 ± 15.92	64.00 ± 6.08
MedicalImages	**56.61 ± 3.74**	54.08 ± 3.55	52.39 ± 5.17	34.84 ± 5.66	48.24 ± 3.52	46.27 ± 5.02	40.00 ± 3.41	45.09 ± 2.04	47.45 ± 1.06	47.72 ± 6.48	18.58 ± 8.91	54.68 ± 2.10
MelbournePedestrian	**84.60 ± 1.98**	81.55 ± 1.62	80.47 ± 2.59	30.74 ± 5.50	68.14 ± 1.82	73.08 ± 1.13	75.36 ± 1.75	78.01 ± 1.38	77.48 ± 0.93	76.77 ± 0.49	68.14 ± 1.82	70.88 ± 1.49
MiddlePhalanxOutlineAgeGroup	44.29 ± 12.74	54.68 ± 6.95	55.19 ± 4.97	54.16 ± 10.18	53.51 ± 5.82	**58.66 ± 1.35**	50.43 ± 7.47	54.11 ± 4.79	50.13 ± 8.45	56.49 ± 1.12	41.82 ± 18.66	51.04 ± 7.11
MiddlePhalanxOutlineCorrect	60.00 ± 5.84	**65.02 ± 2.98**	60.62 ± 1.86	54.23 ± 4.23	56.36 ± 2.69	50.86 ± 7.18	64.72 ± 6.49	54.98 ± 7.77	58.97 ± 7.04	52.23 ± 9.34	52.44 ± 10.59	59.18 ± 5.65
MiddlePhalanxTW	**49.74 ± 5.84**	48.05 ± 4.11	49.48 ± 2.12	45.71 ± 2.70	46.23 ± 6.05	46.10 ± 6.40	37.23 ± 5.86	42.64 ± 7.61	35.97 ± 2.62	41.34 ± 8.07	23.64 ± 8.52	44.42 ± 6.27
MixedShapesRegularTrain	67.65 ± 7.57	70.16 ± 5.08	72.53 ± 6.92	70.49 ± 2.59	73.35 ± 4.70	73.87 ± 1.26	24.40 ± 1.33	74.08 ± 2.13	28.76 ± 6.79	73.65 ± 2.23	63.69 ± 2.98	**79.60 ± 1.74**

Continued on next page

TABLE 13: Per-dataset 10-shot classification results on 128 UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	TimeMorph
MixedShapesSmallTrain	70.61 ± 10.30	69.46 ± 11.49	75.93 ± 3.55	73.56 ± 2.63	73.16 ± 8.70	78.08 ± 2.18	23.04 ± 1.80	77.03 ± 1.02	27.06 ± 6.78	77.29 ± 1.92	65.95 ± 4.04	**81.61 ± 2.47**
MoteStrain	89.38 ± 0.88	89.14 ± 3.04	90.18 ± 0.68	86.26 ± 0.20	86.58 ± 1.30	87.19 ± 1.69	77.40 ± 1.04	87.43 ± 0.48	85.32 ± 0.75	**90.65 ± 0.48**	67.41 ± 11.72	89.23 ± 1.53
NonInvasiveFetalECGThorax1	55.91 ± 3.81	50.28 ± 5.45	55.91 ± 3.81	**81.90 ± 0.59**	63.56 ± 4.40	77.13 ± 1.08	36.83 ± 0.46	78.18 ± 0.43	46.96 ± 3.44	75.71 ± 1.08	58.08 ± 1.54	76.84 ± 1.40
NonInvasiveFetalECGThorax2	62.50 ± 1.19	67.13 ± 2.71	62.50 ± 1.19	**86.79 ± 1.25**	75.21 ± 2.50	83.55 ± 0.48	57.79 ± 2.45	83.27 ± 1.00	69.23 ± 2.65	80.68 ± 0.66	70.22 ± 2.79	82.53 ± 1.33
OSULeaf	**72.64 ± 5.62**	69.26 ± 4.17	69.01 ± 3.75	34.88 ± 3.54	39.34 ± 3.85	46.42 ± 2.52	19.83 ± 4.60	40.50 ± 2.19	29.83 ± 2.82	37.19 ± 2.07	28.10 ± 5.12	59.42 ± 3.58
OliveOil	56.00 ± 14.79	40.00 ± 0.00	**66.67 ± 9.72**	40.00 ± 0.00	40.00 ± 0.00	40.00 ± 0.00	45.56 ± 9.62	40.00 ± 0.00	40.00 ± 0.00	40.00 ± 0.00	40.00 ± 0.00	40.00 ± 0.00
PLAID	28.04 ± 4.01	25.14 ± 3.42	22.46 ± 6.07	34.45 ± 1.13	54.53 ± 5.93	50.16 ± 2.34	32.59 ± 0.99	44.82 ± 3.36	37.77 ± 2.12	46.80 ± 0.84	54.53 ± 5.93	**59.89 ± 3.28**
PhalangesOutlinesCorrect	56.88 ± 6.98	56.74 ± 8.68	57.48 ± 7.09	54.27 ± 11.23	49.91 ± 6.62	**58.78 ± 4.01**	55.79 ± 3.74	55.36 ± 17.14	52.42 ± 4.71	54.23 ± 12.22	53.99 ± 12.03	55.73 ± 8.07
Phoneme	**28.10 ± 1.26**	25.96 ± 1.60	**28.10 ± 1.26**	8.58 ± 0.50	11.88 ± 0.54	13.82 ± 0.86	9.55 ± 1.16	11.71 ± 0.45	10.91 ± 1.85	10.88 ± 0.47	10.94 ± 0.56	21.95 ± 1.27
PickupGestureWiimoteZ	50.80 ± 15.01	68.40 ± 2.61	**73.60 ± 2.19**	51.60 ± 1.67	54.40 ± 6.84	56.67 ± 3.06	26.67 ± 1.15	54.67 ± 1.15	42.00 ± 5.10	60.67 ± 1.15	54.40 ± 6.84	70.40 ± 2.61
PigAirwayPressure	26.35 ± 2.05	12.88 ± 0.53	**50.58 ± 0.92**	4.81 ± 0.00	4.13 ± 0.43	4.81 ± 0.48	2.08 ± 1.00	6.25 ± 1.27	4.62 ± 0.87	6.73 ± 0.48	5.00 ± 0.65	9.62 ± 0.48
PigArtPressure	54.71 ± 1.57	25.19 ± 5.85	**69.81 ± 3.29**	8.27 ± 0.40	6.06 ± 2.62	8.97 ± 0.28	6.41 ± 1.00	9.78 ± 0.73	6.35 ± 0.92	8.65 ± 1.27	18.08 ± 2.06	31.73 ± 1.92
PigCVP	28.85 ± 2.68	19.62 ± 1.84	**61.35 ± 2.42**	7.12 ± 0.40	3.27 ± 1.61	7.37 ± 0.73	7.05 ± 1.21	8.33 ± 1.69	4.13 ± 0.80	5.93 ± 1.11	11.63 ± 1.24	19.52 ± 1.34
Plane	**100.00 ± 0.00**	99.43 ± 1.28	**100.00 ± 0.00**	96.57 ± 1.28	96.95 ± 1.24	98.10 ± 0.00	98.73 ± 0.55	97.14 ± 0.95	98.48 ± 0.52	97.14 ± 0.95	49.71 ± 5.36	99.24 ± 1.24
PowerCons	84.00 ± 3.74	85.67 ± 2.68	**88.11 ± 1.91**	73.22 ± 7.34	83.67 ± 3.78	85.93 ± 2.85	53.33 ± 0.56	86.30 ± 2.25	59.67 ± 5.39	84.63 ± 0.85	74.33 ± 14.06	82.33 ± 5.46
ProximalPhalanxOutlineAgeGroup	69.07 ± 5.10	80.49 ± 4.77	80.98 ± 2.76	80.98 ± 2.95	75.90 ± 8.78	81.30 ± 5.55	81.14 ± 3.66	**81.63 ± 4.89**	76.88 ± 9.31	77.56 ± 6.40	68.78 ± 16.36	79.80 ± 3.30
ProximalPhalanxOutlineCorrect	63.71 ± 11.74	64.60 ± 2.91	63.85 ± 3.21	56.01 ± 10.53	61.86 ± 4.47	64.60 ± 0.69	**66.55 ± 3.51**	64.26 ± 0.69	61.86 ± 6.47	65.98 ± 3.31	51.34 ± 15.85	60.07 ± 4.20
ProximalPhalanxTW	60.78 ± 11.91	70.05 ± 3.85	**72.39 ± 1.88**	51.12 ± 7.39	70.73 ± 1.42	64.07 ± 2.20	60.16 ± 2.69	65.53 ± 2.25	57.95 ± 5.18	59.84 ± 3.91	25.07 ± 15.12	69.66 ± 3.47
RefrigerationDevices	42.67 ± 7.00	44.27 ± 6.42	43.41 ± 5.31	34.72 ± 1.59	37.71 ± 6.85	44.00 ± 2.97	34.40 ± 1.33	34.76 ± 2.56	35.31 ± 4.05	**44.62 ± 2.22**	32.37 ± 1.67	41.23 ± 4.90
Rock	41.60 ± 6.84	34.40 ± 4.56	36.00 ± 1.41	81.60 ± 0.89	57.20 ± 9.01	82.67 ± 1.15	84.67 ± 6.11	**88.00 ± 0.00**	62.00 ± 6.16	86.67 ± 3.06	30.80 ± 4.82	76.40 ± 2.97
ScreenType	35.84 ± 5.22	36.48 ± 5.07	37.23 ± 5.64	36.37 ± 2.19	39.15 ± 5.32	**40.53 ± 1.87**	35.73 ± 2.12	36.09 ± 2.74	33.12 ± 2.51	39.82 ± 2.27	36.32 ± 3.91	37.12 ± 3.93
SemgHandGenderCh2	63.47 ± 5.58	**74.93 ± 6.24**	65.43 ± 6.18	67.83 ± 4.45	66.37 ± 2.41	62.89 ± 2.12	51.67 ± 5.43	65.78 ± 6.45	55.43 ± 2.94	60.72 ± 4.24	58.77 ± 7.13	63.80 ± 10.73
SemgHandMovementCh2	37.69 ± 5.41	36.27 ± 1.15	37.82 ± 1.58	37.42 ± 2.10	32.22 ± 2.74	30.67 ± 2.78	25.04 ± 0.56	34.52 ± 3.16	25.69 ± 1.85	33.04 ± 3.33	33.64 ± 4.19	**45.56 ± 2.93**
SemgHandSubjectCh2	43.56 ± 2.78	45.20 ± 5.06	40.22 ± 5.31	62.80 ± 4.54	54.13 ± 4.47	60.37 ± 0.71	34.22 ± 3.27	**68.07 ± 2.65**	39.96 ± 2.48	60.96 ± 1.10	38.18 ± 2.42	66.71 ± 4.45
ShakeGestureWiimoteZ	82.00 ± 6.16	91.20 ± 2.28	88.80 ± 1.10	55.60 ± 2.97	68.40 ± 7.67	60.00 ± 4.00	33.33 ± 3.06	62.67 ± 3.06	45.20 ± 7.29	56.67 ± 1.15	68.40 ± 7.67	**91.60 ± 2.97**
ShapeletSim	86.33 ± 6.80	**100.00 ± 0.00**	**100.00 ± 0.00**	48.78 ± 0.82	95.22 ± 4.26	59.07 ± 5.56	48.70 ± 1.40	50.19 ± 2.80	51.00 ± 1.98	49.81 ± 1.95	52.33 ± 4.64	96.67 ± 1.96
ShapesAll	68.60 ± 2.31	66.63 ± 1.82	68.60 ± 2.31	62.47 ± 0.34	65.47 ± 2.49	72.11 ± 0.69	12.89 ± 4.16	73.28 ± 0.38	31.00 ± 5.80	65.56 ± 1.08	70.00 ± 1.20	**83.13 ± 0.32**
SmallKitchenAppliances	**67.20 ± 8.07**	61.87 ± 4.21	64.69 ± 7.08	36.59 ± 1.57	60.00 ± 3.85	55.11 ± 4.59	41.51 ± 2.40	38.76 ± 2.56	47.25 ± 3.08	51.02 ± 1.54	41.33 ± 5.03	57.60 ± 5.56
SmoothSubspace	**92.00 ± 1.56**	87.07 ± 3.67	**92.00 ± 3.02**	38.13 ± 6.59	67.73 ± 10.31	79.56 ± 2.78	80.67 ± 0.67	60.00 ± 11.02	86.13 ± 3.38	90.67 ± 4.00	43.60 ± 9.96	89.47 ± 2.60
SonyAIBORobotSurface1	95.11 ± 2.42	**96.01 ± 0.87**	92.45 ± 1.40	42.96 ± 0.07	87.95 ± 1.92	64.28 ± 1.83	67.05 ± 1.01	76.43 ± 0.69	64.86 ± 0.43	67.83 ± 1.06	45.09 ± 4.84	86.09 ± 3.97
SonyAIBORobotSurface2	**96.73 ± 3.11**	92.82 ± 3.91	94.67 ± 2.40	77.50 ± 0.71	90.95 ± 2.30	81.25 ± 2.70	83.07 ± 2.21	86.08 ± 2.22	82.71 ± 1.50	85.73 ± 2.18	61.15 ± 12.88	84.11 ± 2.29
StarLightCurves	71.47 ± 14.56	82.56 ± 3.87	75.10 ± 5.06	74.16 ± 3.22	77.51 ± 4.09	75.29 ± 4.53	29.60 ± 9.11	76.31 ± 3.24	30.20 ± 4.95	74.55 ± 5.89	70.71 ± 14.11	**85.26 ± 3.54**
Strawberry	79.73 ± 8.80	76.70 ± 3.26	76.70 ± 4.55	60.81 ± 1.94	79.89 ± 6.30	61.98 ± 1.58	**80.63 ± 1.13**	59.10 ± 1.09	77.51 ± 4.36	68.11 ± 2.15	50.54 ± 10.66	75.84 ± 0.99
SwedishLeaf	**88.03 ± 1.39**	83.68 ± 1.26	**88.03 ± 1.39**	66.11 ± 1.36	73.47 ± 2.54	75.95 ± 1.36	60.69 ± 4.66	81.39 ± 2.49	68.80 ± 1.87	74.72 ± 1.82	28.96 ± 3.20	85.89 ± 1.27
Symbols	**93.83 ± 2.55**	89.89 ± 7.08	92.52 ± 1.13	75.94 ± 0.93	85.87 ± 1.87	83.32 ± 0.52	34.77 ± 0.30	85.96 ± 1.03	61.21 ± 7.54	82.51 ± 0.70	21.11 ± 2.93	89.33 ± 0.82
SyntheticControl	95.60 ± 1.14	**97.87 ± 0.61**	97.60 ± 0.43	59.07 ± 3.98	94.20 ± 1.64	90.78 ± 1.64	59.44 ± 1.95	79.44 ± 4.86	68.40 ± 3.43	90.11 ± 1.35	22.60 ± 7.62	97.33 ± 0.85
ToeSegmentation1	**93.25 ± 1.60**	89.56 ± 3.33	91.58 ± 1.68	55.53 ± 3.14	68.95 ± 9.85	56.73 ± 5.97	50.29 ± 6.71	55.85 ± 3.55	49.39 ± 2.18	55.70 ± 4.98	50.53 ± 1.40	80.53 ± 2.75
ToeSegmentation2	81.38 ± 10.25	**87.08 ± 4.53**	85.85 ± 2.64	50.15 ± 3.62	73.38 ± 4.70	68.21 ± 10.44	56.67 ± 7.86	58.46 ± 6.30	46.77 ± 3.41	61.54 ± 3.35	43.54 ± 14.75	80.31 ± 4.70
Trace	92.20 ± 4.60	**100.00 ± 0.00**	95.20 ± 5.07	55.80 ± 5.72	86.20 ± 10.35	72.67 ± 1.53	55.67 ± 4.51	61.33 ± 4.62	74.80 ± 3.27	78.00 ± 4.36	46.80 ± 8.17	**100.00 ± 0.00**
TwoLeadECG	**99.86 ± 0.08**	98.79 ± 1.25	97.56 ± 2.37	58.05 ± 1.68	91.89 ± 5.00	70.85 ± 3.05	75.62 ± 1.63	65.26 ± 1.91	78.86 ± 1.56	78.99 ± 2.23	54.01 ± 3.73	99.68 ± 0.24
TwoPatterns	70.89 ± 4.77	77.56 ± 2.96	**78.86 ± 3.59**	42.28 ± 1.08	74.73 ± 6.48	42.55 ± 1.37	26.22 ± 0.85	42.35 ± 1.82	29.82 ± 3.35	37.95 ± 1.72	25.60 ± 4.32	70.06 ± 0.70
UMD	92.78 ± 3.35	98.33 ± 1.05	**98.75 ± 0.58**	64.17 ± 3.05	91.53 ± 4.77	76.62 ± 2.44	75.93 ± 7.23	89.58 ± 4.22	93.06 ± 1.47	87.27 ± 3.21	47.50 ± 8.96	98.67 ± 7.42
UWaveGestureLibraryAll	48.45 ± 3.31	48.15 ± 2.07	49.06 ± 3.81	80.57 ± 3.71	68.35 ± 4.39	84.19 ± 2.90	27.54 ± 1.68	**84.63 ± 3.19**	42.77 ± 5.45	83.32 ± 3.39	81.20 ± 2.71	52.41 ± 1.70
UWaveGestureLibraryX	51.98 ± 1.26	57.08 ± 2.26	52.99 ± 2.93	60.99 ± 2.04	57.69 ± 1.76	**62.37 ± 1.85**	17.30 ± 0.19	62.30 ± 0.55	36.29 ± 8.88	61.26 ± 1.95	56.81 ± 2.90	53.50 ± 1.91
UWaveGestureLibraryY	45.58 ± 1.39	47.42 ± 0.67	46.02 ± 2.85	55.44 ± 1.29	53.52 ± 3.23	**56.83 ± 0.99**	23.04 ± 3.42	56.65 ± 1.12	32.16 ± 4.98	55.22 ± 1.56	46.61 ± 1.40	52.03 ± 0.97
UWaveGestureLibraryZ	48.34 ± 2.78	51.60 ± 2.75	44.60 ± 1.67	53.18 ± 1.58	53.45 ± 2.14	54.99 ± 0.25	20.01 ± 2.19	54.36 ± 2.47	34.40 ± 4.42	51.61 ± 2.72	46.80 ± 1.81	**60.65 ± 2.84**
Wafer	84.60 ± 6.19	78.03 ± 11.28	81.53 ± 4.78	69.44 ± 4.07	**84.77 ± 11.10**	82.09 ± 7.52	76.30 ± 12.19	83.57 ± 7.70	81.81 ± 10.33	81.67 ± 9.12	78.96 ± 10.53	82.53 ± 8.58
Wine	**68.15 ± 6.33**	52.59 ± 5.80	64.07 ± 8.03	50.74 ± 1.66	50.00 ± 0.00	58.64 ± 5.35	50.62 ± 1.07	50.00 ± 3.70	50.00 ± 0.00	62.96 ± 7.41	47.78 ± 4.97	49.26 ± 2.11
WordSynonyms	35.14 ± 2.03	30.72 ± 3.50	35.14 ± 2.03	35.86 ± 2.07	39.44 ± 4.12	51.78 ± 3.11	15.15 ± 0.48	51.10 ± 4.21	25.64 ± 4.61	49.01 ± 4.47	33.92 ± 1.82	**58.90 ± 2.04**
Worms	50.39 ± 8.29	**51.95 ± 5.35**	45.71 ± 7.71	23.38 ± 5.11	39.74 ± 4.27	23.38 ± 6.49	26.84 ± 3.97	27.27 ± 9.09	25.45 ± 3.74	25.97 ± 9.09	20.78 ± 5.35	48.57 ± 4.91
WormsTwoClass	**64.94 ± 5.51**	63.12 ± 4.37	61.30 ± 7.20	49.09 ± 4.63	47.27 ± 1.97	50.65 ± 6.49	48.05 ± 4.50	48.05 ± 4.68	48.05 ± 3.56	51.95 ± 4.50	46.49 ± 5.15	56.36 ± 3.85
Yoga	59.91 ± 2.90	**59.97 ± 4.98**	59.88 ± 3.11	55.72 ± 1.71	57.07 ± 1.49	58.79 ± 3.68	53.03 ± 3.90	59.02 ± 1.75	51.43 ± 2.65	59.87 ± 3.99	51.89 ± 2.29	57.73 ± 2.05
Avg. Acc.	66.24	64.93	67.18	55.43	62.72	62.79	46.04	61.94	51.50	63.13	46.88	**69.16**
Avg. Rank	4.83	5.03	4.55	8.00	6.23	5.75	9.87	5.73	8.87	5.64	9.58	**3.91**
#Best	25	15	28	6	5	7	3	5	1	8	1	**40**

TABLE 14
Win/Tie/Loss of TimeMorph against selected baselines on 128 UCR datasets.

Baseline	1-shot	2-shot	5-shot	10-shot
COSCO-ResNet	68 / 0 / 60	68 / 1 / 59	58 / 0 / 70	64 / 0 / 64
PatchTST	97 / 0 / 31	97 / 0 / 31	99 / 0 / 29	94 / 2 / 32
TimesNet	88 / 0 / 40	91 / 1 / 36	86 / 0 / 42	93 / 1 / 34
GPT4TS	106 / 2 / 20	110 / 0 / 18	114 / 1 / 13	121 / 1 / 6
ResNet	64 / 1 / 63	66 / 1 / 61	67 / 0 / 61	69 / 0 / 59
TapNet	75 / 0 / 53	75 / 3 / 50	76 / 1 / 51	72 / 3 / 53

A.1 Win–Tie–Loss Counts

Table 14 reports the complete win/tie/loss counts visualized in Figure 7.

APPENDIX B
ADDITIONAL EXPERIMENTAL DETAILS

B.1 Code Release and Assets

We will release an anonymized code package with the submission. The release includes the TimeMorph implementation, few-shot fine-tuning and evaluation utilities, baseline configurations, table-generation utilities, environment files, and dataset-preparation instructions. The code is released under the MIT license. The package does not redistribute raw third-party datasets or gated pretrained model weights; users should obtain those assets from the original providers and comply with their licenses and terms of use. The release documents which experiments are directly reproducible from the package and which require separately downloaded datasets or pretrained checkpoints.

B.2 Asset and License Table

Table 15 summarizes the main existing assets used in this paper. We list the original source, role, version when available, license or stated terms, and whether the asset is redistributed in our release. For assets whose official source does not state a separate dataset license, we cite the official repository or archive page and do not redistribute the raw data.

TABLE 15: Existing assets used in this paper and their licenses or stated terms. URLs were checked during manuscript preparation.

Asset	Role	Source / version	License or stated terms	Redistribution in our release
UCR Archive 2018	Main few-shot benchmark and unlabeled Stage I source pool	TSC website; Zenodo v1	Zenodo record lists Creative Commons Attribution 4.0 International (CC BY 4.0); archive should be cited via Dau et al.	Not redistributed; users should download it from the original archive or cited mirror.
TSQA / ChatTime	Stage I raw series and Stage II question-answer alignment data	Hugging Face dataset card	Apache-2.0 according to the Hugging Face dataset metadata	Not redistributed; users should obtain it from the original dataset source.
M4 Competition dataset	Stage I raw series and Stage II captioning / attribute-alignment data	Official M4 repository	The official repository publicly provides the dataset and asks users to cite the M4 competition resources; no separate dataset LICENSE file was found in the repository.	Not redistributed; users download from the official repository.
MIT-BIH Arrhythmia Database	External semantic-label evaluation	PhysioNet v1.0.0	Open Data Commons Attribution License v1.0; PhysioNet citation required	Not redistributed; preprocessing requires a local PhysioNet download.
PhysioNet/ CinC 2017 AF dataset	External semantic-label evaluation	PhysioNet v1.0.0	Open Data Commons Attribution License v1.0; PhysioNet and challenge citations required	Not redistributed; preprocessing requires a local PhysioNet download.
Llama 3.2 1B	Pretrained causal LLM backbone	Hugging Face model card	Llama 3.2 Community License and Acceptable Use Policy	Not redistributed; users must obtain access from Meta / Hugging Face.
DINOv2-base	Frozen visual backbone	Hugging Face model card	Apache-2.0	Not redistributed; obtained from the original model repository.
OpenTSLM base code	Starting codebase and time-series language-model components	Project repository and release package	MIT license in the repository license file	Modified code is released in anonymized form under MIT, preserving applicable notices.

TABLE 16
Computational efficiency and fine-tuning cost under the 10-shot UCR protocol. Total parameters count the full deployed model, while updated parameters count only the parameters fine-tuned on the downstream support set. Adapter size estimates the per-dataset trainable-state footprint. Runtime and memory are measured on representative UCR datasets using a single RTX 3090.

Model	Total params	Updated params	Updated %	Adapter MB	Peak GB	Train ms/step	Infer ms/ex.	Infer ex./s
TimeMorph	1.28B	12.2M	0.96	46.7	13.75	1207.8	156.85	6.4
ResNet	629.1K	629.1K	100.00	2.4	0.33	13.2	0.26	4273.3
TapNet	912.1K	912.1K	100.00	3.5	0.22	15.0	0.47	3345.0
COSCO-ResNet	662.1K	662.1K	100.00	2.5	0.58	50.8	0.25	4315.1
PatchTST	665.4K	598.3K	90.80	2.3	1.54	32.6	0.75	2149.8
DLinear	1.2M	1.2M	100.00	4.6	0.04	2.7	0.07	14389.5
TimesNet	2.5M	2.5M	100.00	9.7	0.14	66.7	2.46	614.5
Autoformer	1.2M	1.2M	100.00	4.5	0.40	23.1	11.10	91.3
Crossformer	2.8M	2.8M	100.00	10.8	0.06	24.9	0.59	1709.6
FEDformer	1.8M	1.8M	100.00	6.7	0.26	244.1	5.17	201.9
Informer	1.4M	1.4M	100.00	5.3	0.22	18.0	0.62	1690.1
GPT4TS	82.2M	1.1M	1.34	4.2	0.56	22.6	0.86	1181.2

TABLE 15: Existing assets used in this paper and their licenses or stated terms (continued).

Asset	Role	Source / version	License or stated terms	Redistribution in our release
Python ML libraries, including PyTorch, Transformers, PEFT, NumPy, pandas, scikit-learn, and related dependencies	Training, fine-tuning, evaluation, and reporting	Version constraints are listed in the released environment files	Governed by their respective open-source licenses	Not vendored; installed by the environment files.

B.3 Efficiency Summary

Table 16 reports computational efficiency and fine-tuning cost under the 10-shot UCR protocol. Runtime and memory are measured on representative UCR datasets using a single RTX 3090, while parameter counts are reported separately for total deployed parameters and support-set fine-tuned parameters.

B.4 Baseline Training Recipes and Hyperparameters

Table 17 summarizes the training rules used by the reported baseline families. This table makes the few-shot comparison protocol explicit: all methods share the same sampled support/query split, while checkpoint-selection rules and epoch schedules follow each method family.

B.5 Computational Efficiency Details

The efficiency analysis uses the same model families as the main benchmark and excludes TimeMorph-only ablations. The representative UCR subset contains COFFEE, GUNPOINT, ECG5000, ELECTRICDEVICES, CROP, ACSF1, PHONEME, and NONINVASIVEFETALECGTHORAX1, covering short, medium, and long series as well as different class counts. The external efficiency measurements use the same protocol for MIT-BIH and CinC2017.

APPENDIX C
ADDITIONAL PLOTS

Figure 9 visualizes the shot-scaling trend over the 128 UCR datasets. TimeMorph remains the strongest method on the averaged benchmark curve across the entire few-shot range, with the largest absolute advantage appearing at 10-shot.

C.1 Additional Qualitative Visualizations

Figure 10 extends the representation visualization to the representative datasets used in the qualitative analysis. Although t-SNE layouts can vary with data geometry and projection randomness, the additional panels provide a broader check of whether temporal, morphology, and fused decision states organize query samples in different ways across datasets.

TABLE 17
Few-shot training recipes for TimeMorph and the reported baseline families. All methods use the same sampled support sets within each run and evaluate on all official TEST examples from the target dataset.

Method / family	Optimizer	Epochs	Batch / LR	Model selection rule	Val	ES	Few-shot notes
TimeMorph	AdamW	5 warm-up + 55 joint fine-tuning	8 / 2×10^{-4} (encoder), 1×10^{-4} (projector, LoRA)	Report the last joint fine-tuning checkpoint	No	No	Class-token rows remain trainable; language-side fine-tuning uses LoRA in the joint fine-tuning phase.
ResNet, TapNet	AdamW	60	model-specific / 10^{-4}	Best support-set accuracy, tie-broken by support loss	No	No	Trained only on the sampled support set; the support set doubles as the model-selection set.
COSCO-ResNet	SAM with SGD base optimizer + prototypical loss	100	128 / 10^{-2}	Final trained model; support centroids computed after training	No	No	Query predictions are made by distances to support centroids rather than a learned classifier head.
PatchTST	AdamW + linear warmup	60 total epochs with head warm-up and joint fine-tuning	16 / separate backbone-head LRs	Report the last joint fine-tuning checkpoint	No	No	Uses the same support/query split and full dataset class set as the main protocol.
TSLib family (DLinear, Autoformer, Crossformer, FEDformer, Informer, TimesNet)	RAdam + cosine schedule	Fixed model-specific schedule	usually 16 / 10^{-3}	Report the last checkpoint	No	No	Single-stage training; evaluation masks logits to the target dataset's class set. TimesNet uses a shorter schedule than the other TSLib baselines.
GPT4TS	RAdam	60	64 / 10^{-4}	Report the last checkpoint	No	No	One-Fits-All classification baseline under the same sampled support/query protocol.

TABLE 18
Per-dataset runtime details for the computational-efficiency analysis. Rows with incomplete measurements are omitted.

Dataset	Model	Peak GB	Train ms/step	Infer ms/ex.	Infer ex./s
ACSF1	TimeMorph	13.75	1262.4	173.87	5.8
ACSF1	ResNet	1.01	26.0	0.32	3167.0
ACSF1	TapNet	0.59	33.7	1.45	688.2
ACSF1	COSCO-ResNet	1.64	119.2	0.43	2303.7
ACSF1	PatchTST	6.50	107.6	2.23	447.7
ACSF1	DLinear	0.10	4.5	0.07	13898.9
ACSF1	TimesNet	0.27	121.8	4.54	220.5
ACSF1	Autoformer	1.19	40.5	12.53	79.8
ACSF1	Crossformer	0.12	25.0	0.58	1723.7
ACSF1	FEDformer	0.61	277.0	7.03	142.3
ACSF1	Informer	0.56	23.4	1.09	917.6
ACSF1	GPT4TS	1.00	29.1	0.89	1121.4

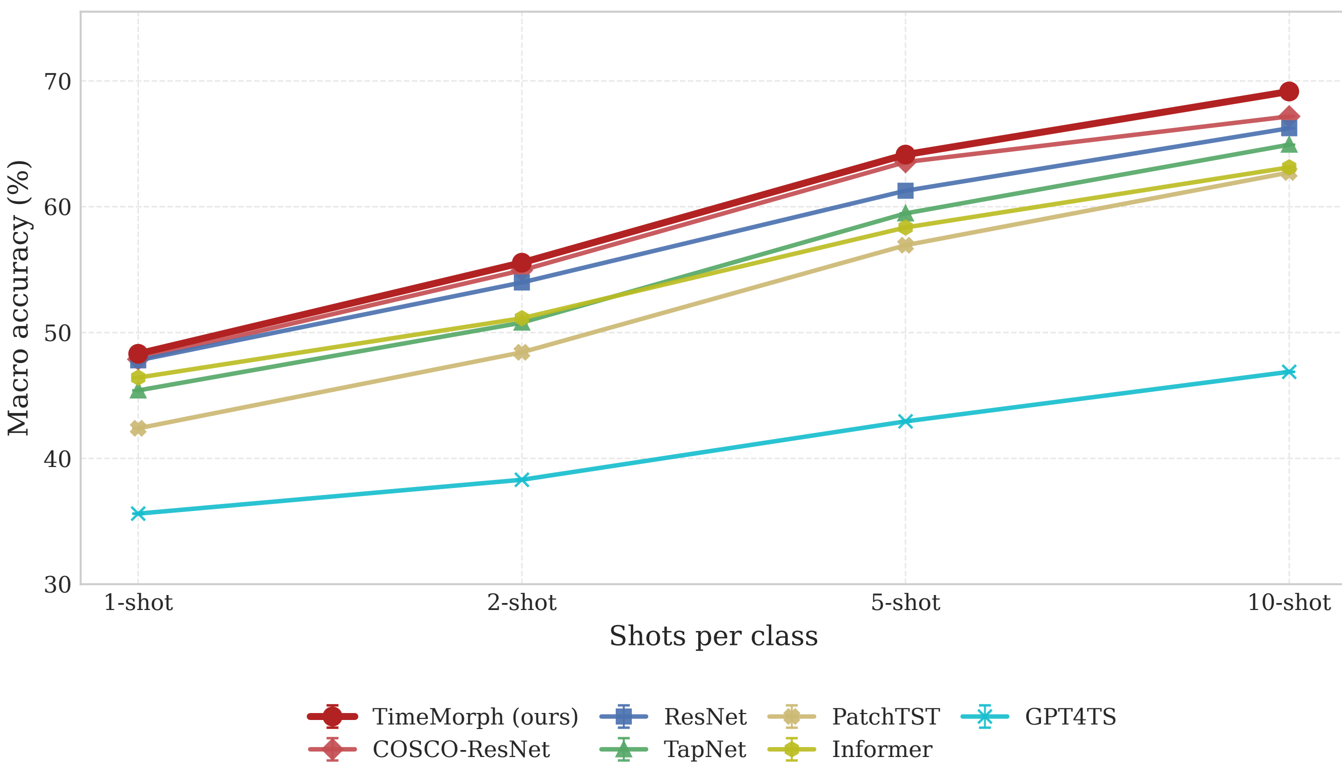

Fig. 9. Shot-scaling trend on the 128 UCR datasets. Each point reports the mean accuracy over five runs. TimeMorph remains the strongest method across all four shot settings.

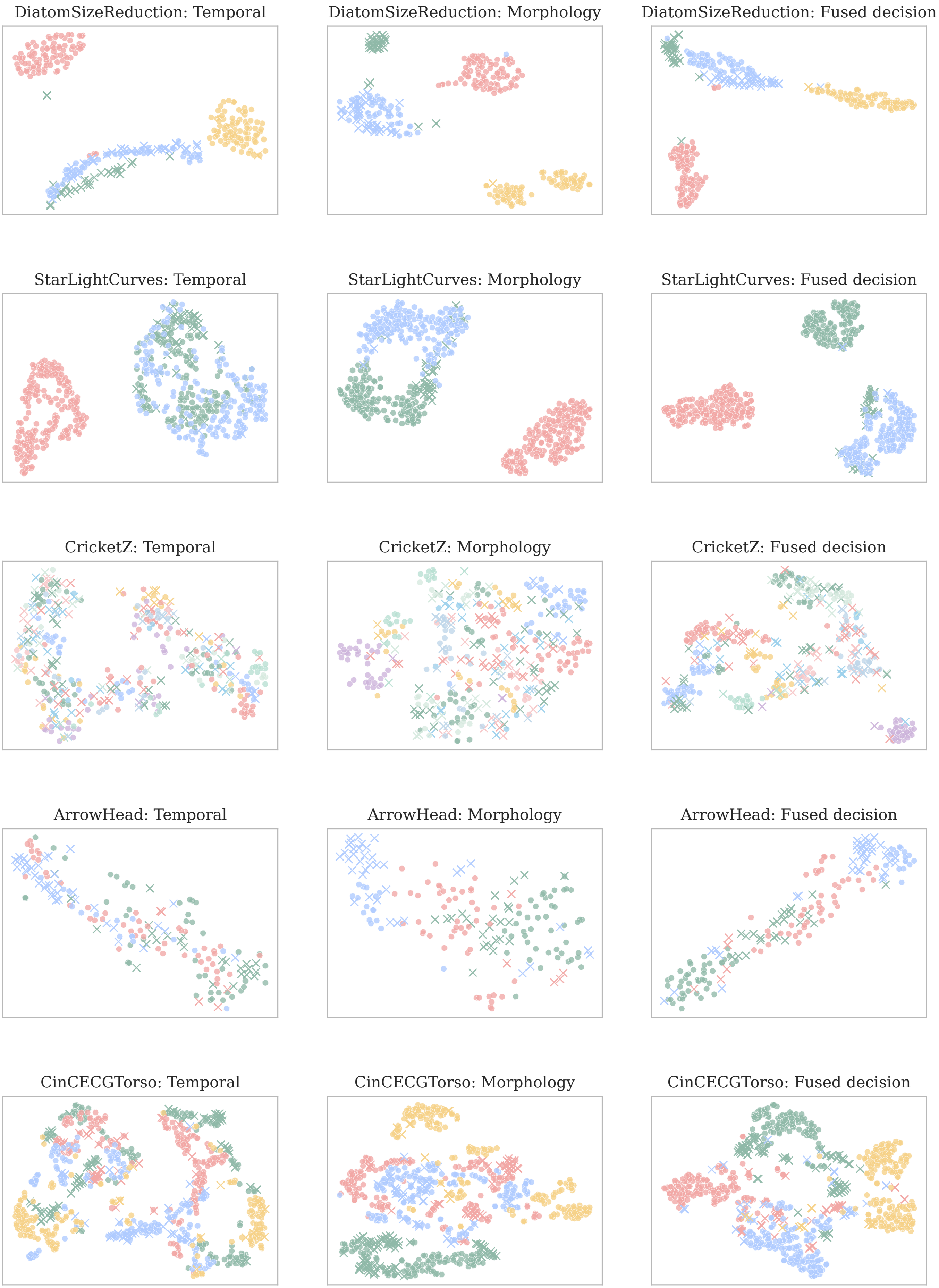

Fig. 10. Additional t-SNE visualizations for temporal, morphology, and fused decision representations on representative UCR datasets. The plots are used as qualitative diagnostics and are not used for model selection.

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